

STAT8561 - Linear Statistical Analysis I

2026 Fall Lecture Notes

Chi-Kuang Yeh

2026-09-22

Table of contents

Preface	7
Office Hours	7
Grade Distribution	7
Assignment	8
Exam	8
Topics and Corresponding Lectures	8
Recommended Textbooks	9
Side Readings	9
Introduction	10
1 Matrix Representation and Notation	13
1.1 Why Linear Statistical Analysis?	13
1.2 A unifying point of view	14
1.3 Basic Notation	14
1.3.1 Inner product and norm	17
1.3.2 Expectation and Covariance Operators	18
1.4 Statistical Models	18
1.4.1 General idea	19
1.4.2 Deterministic part and random part	19
1.5 The Linear Regression Model	19
1.5.1 Geometry Preview	20
1.6 Why projection matters	21
1.7 Discussion	23
1.8 Practice Problems	23
1.9 Take home Practice	24
1.10 Summary	25
2 Least Squares Estimation	26
2.1 The Least Squares Criterion	29
2.2 Derivation of the Least Squares Estimator in a matrix form	33
2.3 Geometric Interpretation of Least Squares	37
2.3.1 Least Squares as Projection	38
2.3.2 Consequence when an Intercept Is Included	44
2.4 The Hat Matrix	45

2.5	The common name to refer to what we have done – Ordinary Least Squares (OLS)	49
2.6	Sum of Squares Decomposition	50
2.6.1	Uncentered Geometric Decomposition	50
2.6.2	Centered Sum of Squares Decomposition	50
2.6.3	Example: Simple Linear Regression	52
2.7	What Happens Without Full Column Rank?	60
2.8	Interpretation of the Geometry	60
2.9	Looking Ahead: Statistical Properties of OLS	62
2.10	Practice at Home	63
3	Distribution Theory of OLS and Inference	66
3.1	Mean and Variance of the OLS Estimator	68
3.1.1	Why Is OLS Unbiased?	69
3.1.2	Variance of OLS	70
3.2	The Normal Linear Model	71
3.3	Distribution of the OLS Estimator	72
3.4	Residual Sum of Squares and Estimation of σ^2	75
3.4.1	Residual Sum of Squares	75
3.4.2	Distribution of SSE	76
3.4.3	Why Are the Degrees of Freedom $n - p$?	77
3.4.4	Estimating σ^2	78
3.5	Independence of OLS and Residual Variation	80
3.5.1	Why Does Independence Occur?	80
3.6	Inference for a Single Regression Coefficient	82
3.6.1	Standard Error	82
3.6.2	Why Does the t Distribution Appear?	83
3.6.3	Confidence Interval for β_j	84
3.6.4	Hypothesis Test for β_j	84
3.7	Inference for Linear Combinations	86
3.8	General Linear Hypotheses	87
3.9	Inference for the Mean Response	88
3.10	Prediction for a New Observation	89
3.11	Worked Example	90
3.12	R Demonstration	94
3.13	Reading Standard Regression Output	98
3.14	Practice at Home	99
4	Model Comparison, Partial F Tests, and ANOVA	102
4.1	Total, Explained, and Unexplained Variation	104
4.2	The Overall F Test	107
4.3	Nested Models	108
4.4	R Demonstration	113
4.5	Summary	120

4.6	Practice Problems	121
5	Multiple Regression and Categorical Predictors	123
5.1	From Simple Regression to Multiple Regression	124
5.2	Categorical Predictors and Indicator Variables	127
5.3	Interactions	129
5.4	R Demonstration With Multiple Regression	131
5.5	Practice Problems	137
6	Contrasts and Estimable Comparisons	139
6.1	Comparisons of group means	139
6.2	Coding and estimability	140
6.3	A complete contrast analysis in R	140
6.4	Practice	143
7	Experimental Design Models: ANOVA and ANCOVA	144
7.1	Design, randomization, and the experimental unit	145
7.2	One-Way ANOVA	147
7.3	Two-Way ANOVA	149
7.4	Analysis of Covariance	151
7.5	R Demonstration	154
7.6	Practice Problems	163
8	Diagnostics and Model Adequacy	165
8.1	Residuals	166
8.2	Diagnostic Plots	168
8.3	Influence	170
8.4	R Demonstration	172
8.5	Practice Problems	179
9	Transformations and Weighted Least Squares	181
9.1	Transformations in Regression	182
9.2	Weighted Least Squares	185
9.3	Response Transformation Versus WLS	188
9.4	R Demonstration	189
9.5	Practice Problems	201
10	Model Assessment and Prediction	203
10.1	Stable estimates and scientific questions	203
10.2	Comparing predictive performance	203
10.3	A small validation example	204
10.4	Practice	206

11 Introduction to Generalized Linear Models	207
11.1 From the normal linear model to GLMs	207
11.2 Logistic regression: probabilities and odds	208
11.3 Poisson regression: counts, rates, and exposure	211
11.4 Inference and model checks	214
11.5 Practice and integrated analysis	217
References	218
 I Assessments	 219
Assignment 1	220
Question 1: Matrix Representation of the Linear Model [8]	220
(a) [2]	221
(b) [2]	222
(c) [2]	222
(d) [2]	223
Question 2: Least Squares and Projection [12]	223
(a) [4]	223
(b) [3]	224
(c) [2]	224
(d) [3]	225
Question 3: Consequences and Geometry of Least Squares [12]	225
(a) Fitted Values and Residuals [3]	225
(b) Consequences of Including an Intercept [3]	226
(c) Residual-Maker Matrix [3]	226
(d) Rank and Uniqueness [3]	226
Question 4: Linear Regression in R [8]	227
(a) [4]	227
(b) [4]	228
Grading Summary	229
 Assignment 2	 230
Question: Distribution and Inference [12]	230
(a) [3]	231
(b) [2]	231
(c) [4]	231
(d) [3]	232
 Appendices	 233
A Matrix Algebra Review	233

B	Linear-Model Extensions (Optional)	234
B.1	Estimability	234
B.2	DFFITS and DFBETAS	236
C	Automatic Model Selection (Optional)	238
C.1	Forward Selection	238
C.2	Additional comparison criteria	239
C.3	An executable selection example	240
D	Generalized Least Squares and Correlated Errors (Optional)	245
D.1	A More General Covariance Model	247
D.2	The Generalized Least Squares Criterion	248
D.3	The Transformed-Model View	249
D.4	Examples of Covariance Structures	251
D.5	R Demonstration With a Hand-Built GLS Computation	256
D.6	Practice Problems	259
E	GLM Likelihood and Computation (Optional)	261
E.1	Exponential-family structure	261
E.2	Why weighted least squares reappears	262
E.3	Likelihood-based inference	262
F	Quantile Regression (Optional)	264
F.1	When the mean is not the whole question	264
F.2	A mean model can miss changing spread	266
F.3	Uncertainty and practical checks	268
F.4	Optional application and self-study	270
G	Mixed-Effects Models (Optional)	271
G.1	Why group membership changes the analysis	271
G.2	Partial pooling and shrinkage	272
G.3	Optional application: repeated measurements and random slopes	276
G.4	Inference, model checks, and further reading	278

Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

The course moves from Gaussian linear-model theory to regression, experimental comparisons, and model checking. A short final unit extends the same design-matrix framework to logistic and Poisson regression. Advanced extensions are marked optional in the appendices.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

Chi-Kuang Yeh, Assistant Professor in the [Department of Mathematics and Statistics, Georgia State University](#).

- Office: Suite 1407, 25 Park Place.
- Email: cych@gsu.edu.

Office Hours

Wednesdays, 9:00 a.m. to 12:00 p.m., or by appointment.

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final – 20%
- Attendance – *7%

Assignment

- ☒ [Assignment 1](#), Due September 11, 2026
- ☐ [Assignment 2](#), Due September 30, 2026

Exam

- ☐ Exam 1, October 14, 2026
- ☐ Exam 2, November 18, 2026

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
		Welcome and Overview	1
	Ch. 1	Math Notation	2–3
	Ch. 2	Least Squares Estimation	3–6
	Ch. 3	Inference on Least Square Estimation	7–
	Ch. 4	Model Comparison, Partial F Tests, and ANOVA	TBA
	Ch. 5	Multiple Regression and Categorical Predictors	TBA
	Ch. 6	Contrasts and Estimable Comparisons	TBA
	Ch. 7	Experimental Design Models: ANOVA and ANCOVA	TBA
	Ch. 8	Diagnostics and Model Adequacy	TBA
	Ch. 9	Transformations and Weighted Least Squares	TBA
	Ch. 10	Model Assessment and Prediction	TBA
	Ch. 11	Introduction to Generalized Linear Models	TBA

Optional self-study readings

Students interested in extensions can read [Quantile Regression](#) and [Mixed-Effects Models](#). These appendices include explanations, graphical examples, R code, and further reading; they are supplementary to the scheduled chapters.

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions*. CRC Press, 2027. [arXiv link](#)
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis*. John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments*. John Wiley & Sons. (Montgomery 2017)

Introduction

This course develops a common language for statistical modelling: specify the research question, construct a design matrix, estimate a mean structure, quantify uncertainty, and check whether the assumptions are credible.

The core sequence covers matrix notation and least squares; normal distributions and quadratic forms; inference and model comparison; categorical predictors, contrasts, and experimental design; diagnostics, transformations, and weighting; and an introduction to generalized linear models for binary and count responses.

The final GLM unit keeps the same linear predictor while extending the response distribution and mean link. Logistic and Poisson regression provide practical examples. The appendices contain optional material on rank-deficient models, automatic selection, correlated errors/GLS, GLM likelihood and computation, quantile regression, and mixed-effects models. The last two appendices are self-study readings for students who want to explore distributional features or grouped observations.

Lectures, homework, two in-class exams, and a final assessment develop both mathematical reasoning and reproducible analysis in R. Core tasks include interpreting estimates and intervals, identifying the experimental unit, and explaining the limits of an analysis.

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal())

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd
= 5)
df <- data.frame(heights, weights)
head(df, 5)
```

```
      heights  weights
1 179.2812  88.01599
2 168.0474  76.10017
3 178.3029  83.46016
4 178.7006  82.01763
5 172.6678  92.84592
```

```
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

Call:

```
lm(formula = weights ~ heights, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.3303	-3.8998	-0.2191	3.2380	12.3478

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.92852	8.22226	-0.235	0.815
heights	0.50907	0.04857	10.481	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

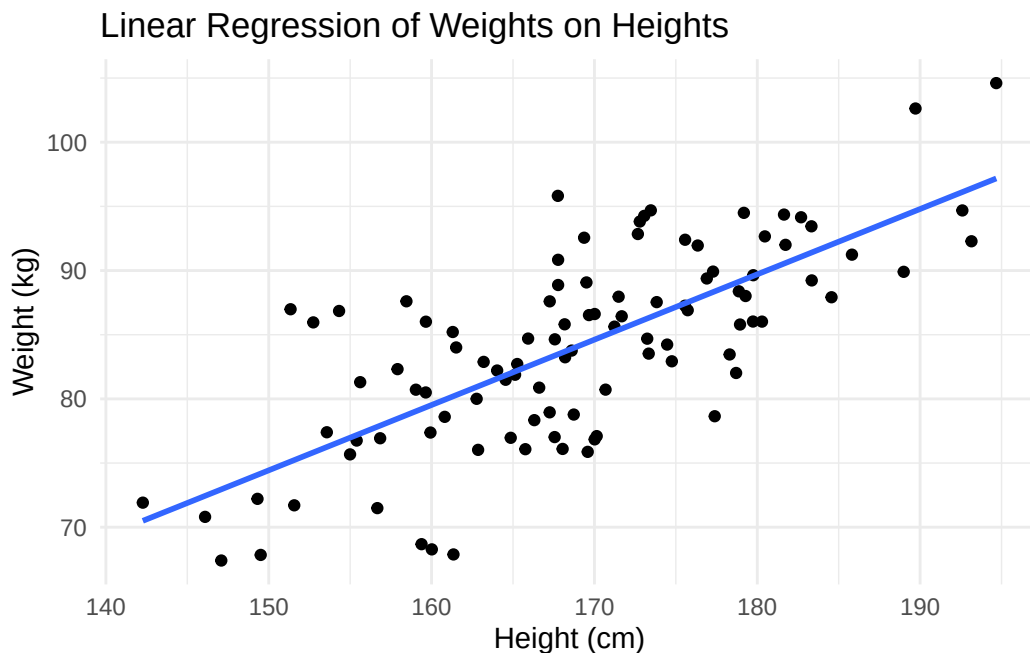
Residual standard error: 5.251 on 98 degrees of freedom

Multiple R-squared: 0.5285, Adjusted R-squared: 0.5237

F-statistic: 109.9 on 1 and 98 DF, p-value: < 2.2e-16

```
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```

`geom\_smooth()` using formula = 'y ~ x'



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

1 Matrix Representation and Notation

In this chapter, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

Learning Objectives

By the end of this chapter, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this chapter:

- Seber and Lee (2003), Chapter 1:
 - 1.1 Notation
 - 1.2 Statistical Models
 - 1.3 Linear Regression Models
 - 1.4 Expectation and Covariance Operators
- Optional preview:
 - Chapter 2: Multivariate Normal Distribution

1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison
- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

1.2 A unifying point of view

A large part of the course can be summarized by the model

$$Y = X\beta + \varepsilon,$$

where

- Y is a response vector,
- X is a design matrix,
- β is an unknown parameter vector,
- ε is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate β , quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in lowercase italic letters, such as a , b , n ;
- vectors are written in bold lowercase letters, such as $\mathbf{x}$, $\mathbf{y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y ;
- realizations of random variables are written in lowercase letters, such as y .

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x .

Some common quantities associated with a univariate random variable X include:

- **Mean**

$$\mu = \mathbb{E}(X).$$

- **Variance**

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

- **k th moment**

$$\mathbb{E}(X^k).$$

- **Cumulative distribution function (CDF)**

$$F(x) = P(X \leq x).$$

- **Probability density function (PDF)**, for a continuous random variable,

$$f(x).$$

- **Probability mass function (PMF)**, for a discrete random variable,

$$p(x) = P(X = x).$$

- **Quantile**

$$q_\alpha = F^{-1}(\alpha),$$

where $0 < \alpha < 1$.

For a random sample of size n , we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \dots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\text{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^n x_i y_i.$$

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n -dimensional random vector.

In linear models, the response is naturally treated as a random vector.

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$.

The covariance matrix of $\mathbf{Y}$ is

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top].$$

Equivalently,

$$\text{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.

Then the corresponding covariance matrix is

$$\text{Var}(\mathbf{Y}) = \begin{pmatrix} \text{Var}(Y_1) & \text{Cov}(Y_1, Y_2) & \text{Cov}(Y_1, Y_3) \\ \text{Cov}(Y_2, Y_1) & \text{Var}(Y_2) & \text{Cov}(Y_2, Y_3) \\ \text{Cov}(Y_3, Y_1) & \text{Cov}(Y_3, Y_2) & \text{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

If $\mathbf{A}$ is a constant matrix, then

$$\text{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A} \text{Var}(\mathbf{Y}) \mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Z}) = \mathbf{A} \text{Cov}(\mathbf{Y}, \mathbf{Z}) \mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

If $Y_1, \dots, Y_n$ are independent and each has variance σ^2 , then

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y . A model introduces assumptions about the distribution of Y , often indexed by an unknown parameter θ .

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2 .

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\varepsilon] = \mathbf{0} \quad \text{and} \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

1.6 Why projection matters

The least squares estimator chooses $\hat{\beta}$ so that

$$\|\mathbf{Y} - \mathbf{X}\beta\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Suppose we observe the following data:

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}$$

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn next chapter that the least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

mod <- lm(y ~ x)
summary(mod)
```

```

Call:
lm(formula = y ~ x)

Residuals:
    1     2     3     4 
-0.2  0.6 -0.6  0.2 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268  0.1515
x              1.2000     0.2828   4.243  0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:  0.9, Adjusted R-squared:  0.85 
F-statistic: 18 on 1 and 2 DF,  p-value: 0.05132

```

```

# Inspecting the design matrix
model.matrix(mod)

```

```

      (Intercept) x
1              1 0
2              1 1
3              1 2
4              1 3
attr("assign")
[1] 0 1

```

```

# Fitted values and residuals
fitted(mod)

```

```

    1     2     3     4 
1.2 2.4 3.6 4.8

```

```

residuals(mod)

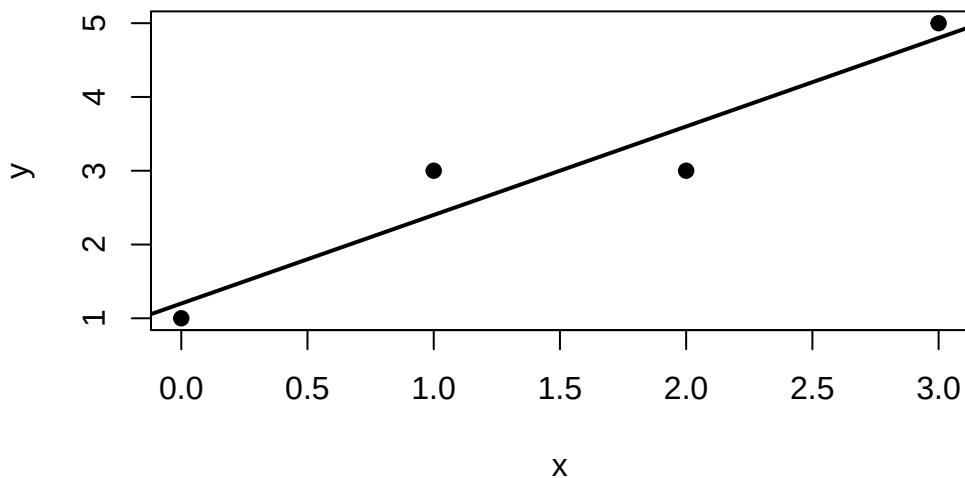
```

```

    1     2     3     4 
-0.2  0.6 -0.6  0.2

```

```
# quick plot
plot(x, y, pch = 19, xlab = "x", ylab = "y")
abline(mod, lwd = 2)
```



1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\text{Var}(\mathbf{Y})$ must be a symmetric matrix.

3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \text{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}]$
2. $\text{Var}(\mathbf{A}\mathbf{Y})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, β , and ε for $n = 5$ observations.

1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}]$ from first principles;
- derive $\text{Var}(\mathbf{A}\mathbf{Y})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,

- fitted values,
- residuals,
- and a scatterplot with the fitted line.

1.10 Summary

This chapter introduced the notation and basic probabilistic tools needed for the rest of the course. We defined random vectors, mean vectors, covariance matrices, and the matrix form of the linear regression model. These ideas will support everything that follows.

Next chapter, we will study least squares estimation and the geometry of projection in more detail.

For some optional review of the matrix algebra, see [Matrix Algebra Review](#).

2 Least Squares Estimation

In this chapter, we study one of the fundamental ideas in linear models: **least squares estimation**.

Recall that the linear model is

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon.$$

The unknown parameter vector β determines the mean structure of the response. Given observed data, our first goal is to estimate β .

The central idea of least squares is simple:

Choose the value of β for which the fitted response $\mathbf{X}\beta$ is as close as possible to the observed response $\mathbf{Y}$.

This optimization problem leads to the **normal equations**, the **least squares estimator**, and a useful geometric interpretation in terms of **orthogonal projection**.

Learning Objectives

By the end of this chapter, students should be able to:

- formulate least squares estimation as an optimization problem;
- define the residual sum of squares;
- derive the normal equations;
- obtain the least squares estimator when $\mathbf{X}$ has full column rank;
- explain the role of the column space $\mathcal{C}(\mathbf{X})$;
- interpret fitted values as an orthogonal projection;
- explain why residuals are orthogonal to the model space;
- define and interpret the hat matrix and residual-maker matrix;
- verify symmetry and idempotence of projection matrices;
- explain the decomposition $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$;
- distinguish the uniqueness of $\hat{\beta}$ from the uniqueness of the fitted values.

Reading

Recommended reading for this week:

- Seber and Lee:
 - Chapter 3: sections on least squares estimation
- Montgomery, Peck, and Vining:
 - introductory sections on estimation in linear regression

i Recap of the Linear Model

Recall from [Week 1](#), the linear model can be written as

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector;
- $\mathbf{X}$ is an $n \times p$ design matrix;
- β is a $p \times 1$ parameter vector;
- ε is an $n \times 1$ error vector.

The design matrix may be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ contains the predictors associated with observation i .
Thus,

$$Y_i = \mathbf{x}_i^\top \beta + \varepsilon_i, \quad i = 1, \dots, n.$$

Under the classical linear model, we often assume

$$\mathbb{E}(\varepsilon) = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Consequently,

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

! Least Squares Does Not Require a Distributional Assumption

The least squares estimator is defined through the optimization problem

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

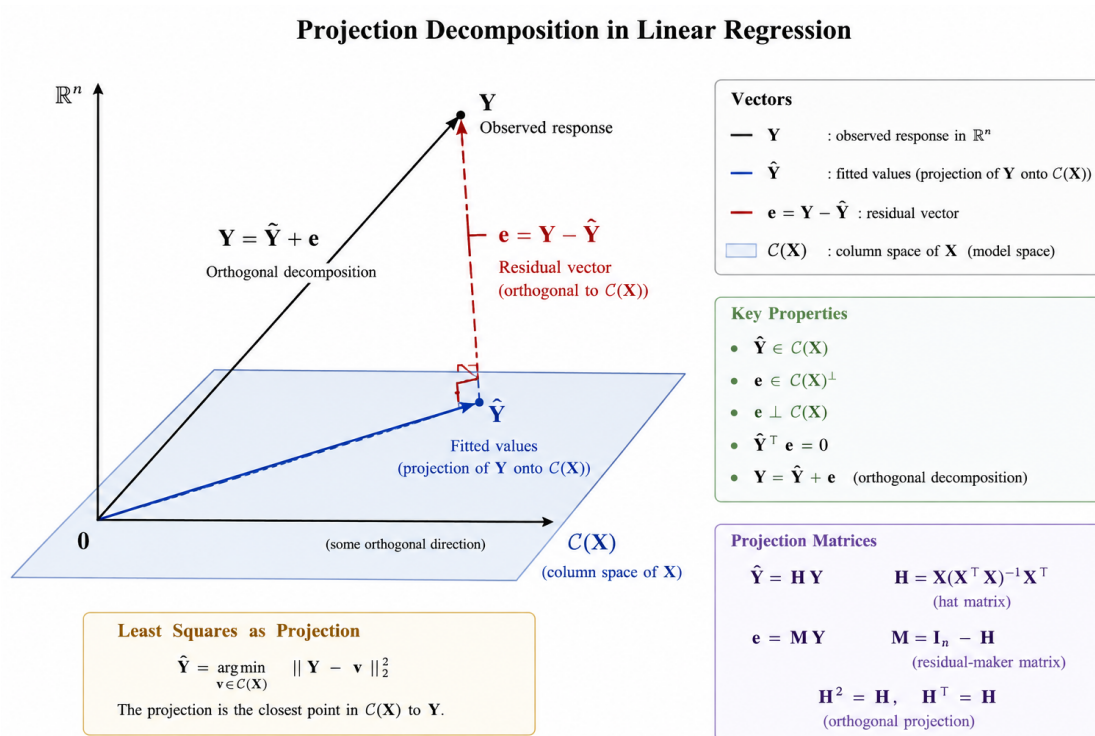


Figure 2.1: Geometric interpretation of least squares as an orthogonal projection.

No distributional assumption on ε is required to define or compute the least squares estimator.

Assumptions such as

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n$$

are needed when we study the **statistical properties** of the estimator.

Normality will be introduced later when we study exact finite-sample inference.

2.1 The Least Squares Criterion

Motivation

For any candidate value β , the corresponding fitted vector is

$$\mathbf{X}\beta.$$

The discrepancy between the observed response $\mathbf{Y}$ and this candidate fit is

$$\mathbf{Y} - \mathbf{X}\beta.$$

A natural idea is to choose β so that this discrepancy is as small as possible.

To measure its size, we use the squared Euclidean norm.

Residual Sum of Squares

The **residual sum of squares** associated with a candidate β is

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

Equivalently,

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta).$$

In scalar notation,

$$S(\beta) = \sum_{i=1}^n (Y_i - \mathbf{x}_i^\top \beta)^2.$$

This last expression explains the name **least squares**: we choose β to make the sum of the squared discrepancies as small as possible.

We therefore define

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} S(\beta).$$

Definition 2.1 (Least Squares Estimator). A **least squares estimator** is any value $\hat{\beta}$ satisfying

$$S(\hat{\beta}) \leq S(\beta)$$

for every $\beta \in \mathbb{R}^p$.

Example 2.1 (LSE for Simple Linear Regression). Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

The fitted value at x_i is

$$\hat{Y}_i = \beta_0 + \beta_1 x_i.$$

The corresponding residual is

$$e_i = Y_i - \beta_0 - \beta_1 x_i.$$

The least squares method chooses β_0 and β_1 to minimize the **sum of squared residuals**

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2.$$

Derivation of the Normal Equations

To minimize $S(\beta_0, \beta_1)$, differentiate with respect to β_0 and β_1 .

For β_0 ,

$$\frac{\partial S}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i).$$

Setting this equal to zero gives

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Therefore,

$$\sum_{i=1}^n Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_i.$$

Dividing by n ,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x},$$

so

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

For β_1 ,

$$\frac{\partial S}{\partial \beta_1} = -2 \sum_{i=1}^n x_i (Y_i - \beta_0 - \beta_1 x_i).$$

Setting this equal to zero gives

$$\sum_{i=1}^n x_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

Hence,

$$\sum_{i=1}^n x_i Y_i = \hat{\beta}_0 \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2.$$

Substituting

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

we obtain

$$\sum_{i=1}^n x_i Y_i = (\bar{Y} - \hat{\beta}_1 \bar{x}) \sum_{i=1}^n x_i + \hat{\beta}_1 \sum_{i=1}^n x_i^2.$$

Since

$$\sum_{i=1}^n x_i = n\bar{x},$$

we have

$$\sum_{i=1}^n x_i Y_i - n\bar{x}\bar{Y} = \hat{\beta}_1 \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right).$$

Using

$$\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^n x_i Y_i - n\bar{x}\bar{Y},$$

and

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2,$$

we obtain

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}.$$

Therefore,

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}}.$$

Alternative Notation

It is common to define

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

and

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}).$$

Then

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

and

$$\hat{\beta}_0 = \bar{Y} - \frac{S_{xy}}{S_{xx}} \bar{x}.$$

Thus, the fitted regression line is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

2.2 Derivation of the Least Squares Estimator in a matrix form

Expansion of the Criterion Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

we expand:

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

Because

$$\mathbf{Y}^\top \mathbf{X}\beta$$

is a scalar,

$$\mathbf{Y}^\top \mathbf{X}\beta = \beta^\top \mathbf{X}^\top \mathbf{Y}.$$

Therefore,

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

Derivation of the Normal Equations

Differentiate $S(\beta)$ with respect to β :

$$\frac{\partial S(\beta)}{\partial \beta} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\beta.$$

At a minimum, the gradient is zero, so

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{0}.$$

Thus,

$$\mathbf{X}^\top \mathbf{X}\hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

These are called the **normal equations**.

i Why Are They Called the Normal Equations?

The normal equations can be written as

$$\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\hat{\beta}) = \mathbf{0}.$$

Later we will define

$$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\beta}.$$

Therefore,

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

This means that $\mathbf{e}$ is perpendicular, or **normal**, to the column space of $\mathbf{X}$.

Thus, the word *normal* is being used in its geometric sense.

Why is the Stationary Point a Minimum?

The Hessian of the least squares criterion is

$$\nabla^2 S(\beta) = 2\mathbf{X}^\top \mathbf{X}.$$

For any $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X} \mathbf{a} = (\mathbf{X} \mathbf{a})^\top (\mathbf{X} \mathbf{a}) = \|\mathbf{X} \mathbf{a}\|_2^2 \geq 0.$$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, and $S(\beta)$ is a convex function.

If $\mathbf{X}$ has full column rank, then for every $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{X} \mathbf{a} \neq \mathbf{0},$$

and hence

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X} \mathbf{a} > 0.$$

Thus, $\mathbf{X}^\top \mathbf{X}$ is positive definite and the least squares minimizer is unique.

A vector $\hat{\beta}$ minimizes

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2$$

if and only if it satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}.$$

If $\mathbf{X}$ has full column rank,

$$\text{rank}(\mathbf{X}) = p,$$

then $\mathbf{X}^\top \mathbf{X}$ is invertible and the least squares estimator is unique:

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.}$$

This is commonly called the **ordinary least squares estimator**, or **OLS estimator**.

i When Is the Closed-Form Formula Valid?

The expression

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

exists if and only if $\mathbf{X}$ has full column rank:

$$\text{rank}(\mathbf{X}) = p.$$

This means that the columns of $\mathbf{X}$ are linearly independent.

If they are linearly dependent, then $\mathbf{X}^\top \mathbf{X}$ is singular and the inverse does not exist.

We will return to this issue later.

Example 2.2 (LSE for Simple Linear Regression continue). As shown in Example 2.1, we know the expression of the LSE. Here, we want to see its connection to the matrix Form.

For simple linear regression,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The general least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For simple linear regression,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix},$$

and

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n x_i Y_i \end{pmatrix}.$$

Therefore,

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

which gives exactly

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

and

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

Hence,

The familiar simple linear regression formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are special cases of the general matrix LSE

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

2.3 Geometric Interpretation of Least Squares

The algebraic derivation gives us the estimator. However, the geometry explains **why** least squares has its important properties.

i Recap: Column Space of the Design Matrix

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\beta : \beta \in \mathbb{R}^p\}.$$

Thus, $\mathcal{C}(\mathbf{X})$ is the set of all response vectors that can be represented exactly by the linear model.

If $\mathbf{X}$ has full column rank p , then

$$\dim \{\mathcal{C}(\mathbf{X})\} = p.$$

Because $\mathbf{X}$ has n rows,

$$\mathcal{C}(\mathbf{X}) \subseteq \mathbb{R}^n.$$

2.3.1 Least Squares as Projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Because

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}),$$

least squares chooses the vector in $\mathcal{C}(\mathbf{X})$ that is closest to the observed vector $\mathbf{Y}$.

That is,

$$\hat{\mathbf{Y}} = \arg \min_{\mathbf{v} \in \mathcal{C}(\mathbf{X})} \|\mathbf{Y} - \mathbf{v}\|_2^2.$$

Therefore, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

So we can say that the fitted vector $\hat{\mathbf{Y}}$ is constrained to live in the model space generated by the predictors. Least squares finds the point in that space closest to the observed data vector $\mathbf{Y}$.

Definition 2.2 (Orthogonality). Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** if their inner product is zero:

$$\mathbf{u}^\top \mathbf{v} = 0.$$

We write

$$\mathbf{u} \perp \mathbf{v}.$$

Geometrically, two nonzero vectors are orthogonal if the angle between them is 90° .

Definition 2.3 (Residual Vector). After obtaining the LSE, define the **residual vector**

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\beta}.$$

The individual residuals are

$$e_i = Y_i - \hat{Y}_i, \quad i = 1, \dots, n.$$

Geometrically, $\mathbf{e}$ is the component of $\mathbf{Y}$ that is orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

Proposition 2.1 (Orthogonality of the Residuals). *Recall that the LSE satisfies the normal equations $\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}$. If we rearrange, we have $\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$.*

Since the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X} \hat{\beta},$$

we obtain the fundamental least squares property

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

To understand what this means, write the design matrix as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \dots & \mathbf{x}^{(p)} \end{pmatrix},$$

where $\mathbf{x}^{(j)}$ is the j th column of $\mathbf{X}$.

Then

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} (\mathbf{x}^{(1)})^\top \mathbf{e} \\ (\mathbf{x}^{(2)})^\top \mathbf{e} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{e} \end{pmatrix} = \mathbf{0}.$$

Therefore,

$$(\mathbf{x}^{(j)})^\top \mathbf{e} = 0, \quad j = 1, \dots, p.$$

Thus, the residual vector is orthogonal to every column of $\mathbf{X}$.

Because the column space $\mathcal{C}(\mathbf{X})$ consists of all linear combinations of the columns of $\mathbf{X}$, it follows that

$$\boxed{\mathbf{e} \perp \mathcal{C}(\mathbf{X}).}$$

Intuition. The columns of $\mathbf{X}$ represent all directions that the linear model is allowed to use. After least squares has chosen the best fitted value, the remaining error $\mathbf{e}$ has no component left in any of these directions. Otherwise, we could move the fitted value within $\mathcal{C}(\mathbf{X})$ and reduce the residual sum of squares further.

Proposition 2.2 (Fundamental Orthogonal Decomposition). *The fitted response vector is*

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

Therefore,

$$\boxed{\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}).}$$

In other words, $\hat{\mathbf{Y}}$ must lie in the space generated by the columns of $\mathbf{X}$.

Since

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

we can write

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.}$$

The two components satisfy

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$$

and

$$\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp.$$

Hence,

$$\boxed{\hat{\mathbf{Y}}^\top \mathbf{e} = 0.}$$

Geometrically, least squares projects $\mathbf{Y}$ onto the column space of $\mathbf{X}$:

$$\mathbf{Y} = \underbrace{\hat{\mathbf{Y}}}_{\text{part explained by the model}} + \underbrace{\mathbf{e}}_{\text{part orthogonal to the model space}}.$$

Because the two components are orthogonal, the Pythagorean theorem gives

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.$$

This geometric decomposition is the foundation of least squares regression and will later lead naturally to the decomposition of sums of squares in ANOVA.

Exercise 2.1 (Computational Problems). Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Compute $\mathbf{X}^\top \mathbf{X}$.
2. Compute $\mathbf{X}^\top \mathbf{Y}$.
3. Find $\hat{\beta}$.
4. Compute $\hat{\mathbf{Y}}$.
5. Compute the residual vector $\mathbf{e}$.
6. Verify that $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

Solution 2.1 (Solution for Exercise 2.1). **Hand calculation**

We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}.$$

Also,

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Therefore,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}.$$

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}.$$

Thus, the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

R calculation

```
# Define X and Y
X <- matrix(
  c(
    1, 0,
    1, 1,
    1, 2
  ),
  nrow = 3,
  byrow = TRUE
)

Y <- matrix(
  c(1, 2, 2),
  ncol = 1
)

# 1. Compute X'X
XtX <- t(X) %*% X
XtX
```

```
      [,1] [,2]
[1,]    3    3
[2,]    3    5
```

```
# 2. Compute X'Y
XtY <- t(X) %*% Y
XtY
```

```
      [,1]
[1,]    5
[2,]    6
```

```
# 3. Compute beta_hat
beta_hat <- solve(XtX) %*% XtY
beta_hat
```

```
      [,1]
[1,] 1.166667
[2,] 0.500000
```

```
# 4. Compute fitted values
Y_hat <- X %*% beta_hat
Y_hat
```

```
      [,1]
[1,] 1.166667
[2,] 1.666667
[3,] 2.166667
```

```
# 5. Compute residuals
e <- Y - Y_hat
e
```

```
      [,1]
[1,] -0.166667
[2,]  0.333333
[3,] -0.166667
```

```
# Optional checks
t(X) %*% e
```

```
      [,1]
[1,] -8.881784e-16
[2,] -8.881784e-16
```

```
t(Y_hat) %*% e
```

```

      [,1]
[1,] -1.474129e-15

```

Exercise 2.2. Explain in words what the column space

$$\mathcal{C}(\mathbf{X})$$

represents in a linear regression model.

Solution 2.2. The column space of $\mathbf{X}$ is the set of all response patterns that can be represented by the linear model.

More intuitively,

$\mathcal{C}(\mathbf{X})$ describes everything the model is capable of fitting. Anything outside this space cannot be explained by the predictors in $\mathbf{X}$.

2.3.2 Consequence when an Intercept Is Included

If the model includes an intercept, then one column of $\mathbf{X}$ is

$$\mathbf{1}_n = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Since the residual vector is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0.$$

Therefore,

$$\boxed{\sum_{i=1}^n e_i = 0.}$$

Since

$$e_i = Y_i - \hat{Y}_i,$$

we also have

$$\sum_{i=1}^n Y_i = \sum_{i=1}^n \hat{Y}_i.$$

Thus,

$$\bar{\hat{Y}} = \bar{Y}.$$

i Intuition for the Mean of the Fitted Values

The intercept allows the fitted model to shift up or down. At the least squares solution, there can be no overall tendency for the fitted values to be systematically too high or too low. Therefore, the residuals sum to zero and the fitted values have the same mean as the observed responses.

2.4 The Hat Matrix

In the full-column-rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Substituting

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

we obtain

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

Then

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the “hat” on $\mathbf{Y}$:

$$\mathbf{Y} \longrightarrow \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Proposition 2.3 (Properties of the Hat Matrix). *When $\mathbf{X}$ has full column rank, the hat matrix satisfies:*

i) Symmetry

$$\mathbf{H}^\top = \mathbf{H}.$$

ii) Idempotence

$$\mathbf{H}^2 = \mathbf{H}.$$

iii) Projection of the design matrix

$$\mathbf{H}\mathbf{X} = \mathbf{X}.$$

iv) Rank

$$\text{rank}(\mathbf{H}) = p.$$

v) Trace

$$\text{tr}(\mathbf{H}) = p.$$

A symmetric and idempotent matrix represents an orthogonal projection.

Therefore, $\mathbf{H}$ is the orthogonal projection matrix onto

$$\mathcal{C}(\mathbf{X}).$$

i Why it is symmetric

We have

$$\begin{aligned} \mathbf{H}^\top &= [\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top]^\top \\ &= \mathbf{X} [(\mathbf{X}^\top \mathbf{X})^{-1}]^\top \mathbf{X}^\top. \end{aligned}$$

Since $\mathbf{X}^\top \mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{H}^\top = \mathbf{H}.$$

i Why it is idempotent

We have

$$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= \mathbf{H}.\end{aligned}$$

Thus,

$$\mathbf{H}^2 = \mathbf{H}.$$

i What Does Idempotence Mean Geometrically?

If we project $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ once, we obtain

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

Projecting the fitted vector again does nothing:

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Once a vector is already inside the model space, projecting it onto the same space leaves it unchanged.

Notice that, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$, we have

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

Definition 2.4 (Residual-Maker Matrix). Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Then

$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

The matrix $\mathbf{M}$ is called the **residual-maker matrix**.

It removes the component of $\mathbf{Y}$ lying in $\mathcal{C}(\mathbf{X})$.

Proposition 2.4 (Properties of the Residual-Maker Matrix). *Since*

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H},$$

we obtain

$$\mathbf{M}^\top = \mathbf{M}$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

Thus, $\mathbf{M}$ is also an orthogonal projection matrix.

While $\mathbf{H}$ projects onto

$$\mathcal{C}(\mathbf{X}),$$

$\mathbf{M}$ projects onto

$$\mathcal{C}(\mathbf{X})^\perp.$$

In addition,

$$\boxed{\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}.$$

Under full column rank,

$$\text{rank}(\mathbf{M}) = n - p$$

and

$$\text{tr}(\mathbf{M}) = n - p.$$

i Two Complementary Projection Matrices

The matrices $\mathbf{H}$ and $\mathbf{M}$ split $\mathbb{R}^n$ into two orthogonal components:

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp.$$

For the observed response,

$$\mathbf{Y} = \underbrace{\mathbf{H}\mathbf{Y}}_{\hat{\mathbf{Y}}} + \underbrace{\mathbf{M}\mathbf{Y}}_{\mathbf{e}}.$$

Thus,

$$\boxed{\mathbf{H} + \mathbf{M} = \mathbf{I}_n.}$$

2.5 The common name to refer to what we have done – Ordinary Least Squares (OLS)

Now that we have seen both the algebraic and geometric views of least squares, we can explain the name **ordinary least squares**, or **OLS**.

- **Squares:** we measure the discrepancy using squared residuals,

$$\sum_{i=1}^n e_i^2 = \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

- **Least:** we choose β to make this quantity as small as possible.
- **Ordinary:** we use the usual unweighted sum of squared residuals, so each observation enters the criterion in the same way.

Thus,

$$\boxed{\text{OLS} = \text{minimize the ordinary sum of squared residuals.}}$$

Geometrically, OLS finds the point in $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{Y}$ using ordinary Euclidean distance.

Later, we will consider extensions such as weighted least squares and generalized least squares.

Exercise 2.3 (Quick Check). What do **ordinary**, **least**, and **squares** each mean in OLS?

2.6 Sum of Squares Decomposition

2.6.1 Uncentered Geometric Decomposition

Since

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

and

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0,$$

we have

$$\begin{aligned}\mathbf{Y}^\top \mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top (\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top \mathbf{e} + \mathbf{e}^\top \mathbf{e} \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}.\end{aligned}$$

Therefore,

$$\boxed{\|\mathbf{Y}\|_2^2 = \|\hat{\mathbf{Y}}\|_2^2 + \|\mathbf{e}\|_2^2.}$$

This is a direct application of the Pythagorean theorem.

2.6.2 Centered Sum of Squares Decomposition

When the regression model contains an intercept,

$$\bar{\hat{Y}} = \bar{Y}.$$

Therefore,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The two components are orthogonal, giving

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

These quantities will later be called

- **total sum of squares**

$$\text{SST} = \sum_{i=1}^n (Y_i - \bar{Y})^2;$$

- **regression sum of squares**

$$\text{SSR} = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2;$$

- **error sum of squares**

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

Thus,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

! Centered vs. Uncentered Decomposition

The geometric identity

$$\mathbf{Y}^\top \mathbf{Y} = \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}$$

is an **uncentered** decomposition.

The familiar regression decomposition

$$\text{SST} = \text{SSR} + \text{SSE}$$

is a **centered** decomposition and relies on the model containing an intercept.

These two decompositions should not be confused.

2.6.3 Example: Simple Linear Regression

Consider the simple regression dataset

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline y_i & 1 & 3 & 3 & 5 \end{array}.$$

We fit the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The response vector is

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix},$$

and the design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

Step 1: Compute $\mathbf{X}^\top \mathbf{X}$

We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

Step 2: Compute $\mathbf{X}^\top \mathbf{Y}$

We obtain

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}.$$

Step 3: Compute $\hat{\beta}$

Since

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix},$$

the least squares estimator is

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix} \begin{pmatrix} 12 \\ 24 \end{pmatrix} \\ &= \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix}. \end{aligned}$$

Thus,

$$\hat{\beta}_0 = 1.2, \quad \hat{\beta}_1 = 1.2,$$

and the fitted regression line is

$$\boxed{\hat{Y} = 1.2 + 1.2x.}$$

Step 4: Compute the Fitted Values

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Step 5: Compute the Residuals

The residual vector is

$$\begin{aligned}
\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\
&= \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} \\
&= \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.
\end{aligned}$$

Notice that

$$\sum_{i=1}^4 e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0.$$

This occurs because the model contains an intercept.

Step 6: Verify Orthogonality

We compute

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus,

$$\boxed{\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

We can also verify that

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0.$$

Step 7: Compute the Hat Matrix

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

For this example,

$$\mathbf{H} = \begin{pmatrix} 0.7 & 0.4 & 0.1 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.1 & 0.4 & 0.7 \end{pmatrix}.$$

Multiplying by $\mathbf{Y}$ gives

$$\mathbf{HY} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix} = \hat{\mathbf{Y}}.$$

Thus, the hat matrix projects the observed response onto the model space.

We can verify the previous calculations in R.

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

df <- data.frame(
  x = x,
  y = y
)

fit <- lm(y ~ x, data = df)

coef(fit)
```

```
(Intercept)      x
          1.2      1.2
```

The estimated coefficients are

$$\hat{\beta}_0 = 1.2, \quad \hat{\beta}_1 = 1.2.$$

We can also compute the estimator directly using matrix operations.

```
X <- model.matrix(fit)
X
```

```

      (Intercept) x
1             1 0
2             1 1
3             1 2
4             1 3
attr(,"assign")
[1] 0 1

```

```

Y <- matrix(y, ncol = 1)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

beta_hat

```

```

      [,1]
(Intercept) 1.2
x           1.2

```

Compute the fitted values and residuals:

```

y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat

```

```

      [,1]
1  1.2
2  2.4
3  3.6
4  4.8

```

```

e

```

```

      [,1]
1 -0.2
2  0.6
3 -0.6
4  0.2

```

Check that the residuals sum to zero:

```
sum(e)
```

```
[1] -3.774758e-15
```

Compute the hat matrix:

```
H <-  
  X %*%  
  solve(t(X) %*% X) %*%  
  t(X)  
  
round(H, 4)
```

```
      1  2  3  4  
1  0.7 0.4 0.1 -0.2  
2  0.4 0.3 0.2  0.1  
3  0.1 0.2 0.3  0.4  
4 -0.2 0.1 0.4  0.7
```

Verify symmetry:

```
round(H - t(H), 10)
```

```
      1 2 3 4  
1 0 0 0 0  
2 0 0 0 0  
3 0 0 0 0  
4 0 0 0 0
```

Verify idempotence:

```
round(H %*% H - H, 10)
```

```
      1 2 3 4  
1 0 0 0 0  
2 0 0 0 0  
3 0 0 0 0  
4 0 0 0 0
```

Verify the normal equations geometrically:

```
round(t(X) %*% e, 10)
```

```
      [,1]
(Intercept) 0
x            0
```

Verify that fitted values and residuals are orthogonal:

```
round(t(y_hat) %*% e, 10)
```

```
      [,1]
[1,]      0
```

Construct the residual-maker matrix:

```
M <- diag(nrow(X)) - H
round(M, 4)
```

```
      1    2    3    4
1  0.3 -0.4 -0.1  0.2
2 -0.4  0.7 -0.2 -0.1
3 -0.1 -0.2  0.7 -0.4
4  0.2 -0.1 -0.4  0.3
```

```
round(M %*% M - M, 10)
```

```
      1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify that

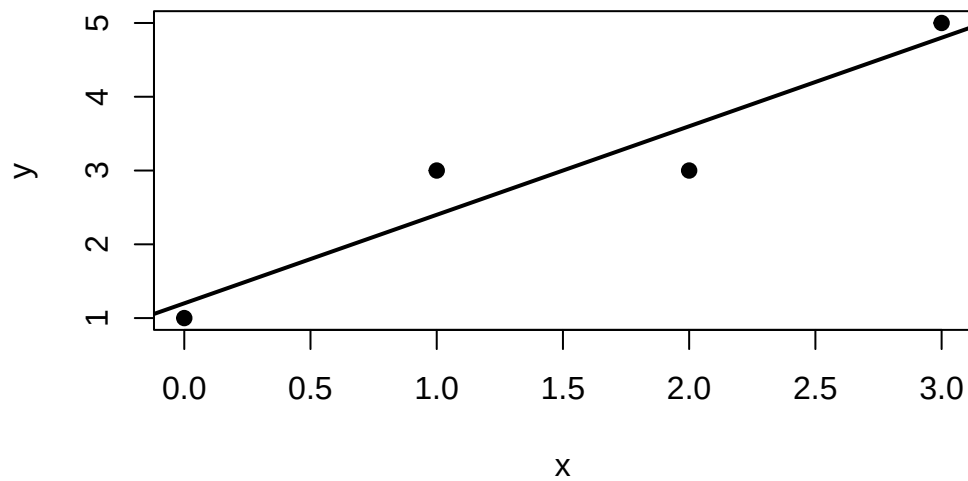
$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

```
round(M %*% Y - e, 10)
```

```
[,1]  
1    0  
2    0  
3    0  
4    0
```

Finally, plot the fitted regression line:

```
plot(  
  x,  
  y,  
  pch = 19,  
  xlab = "x",  
  ylab = "y"  
)  
  
abline(  
  fit,  
  lwd = 2  
)
```



2.7 What Happens Without Full Column Rank?

Suppose

$$\text{rank}(\mathbf{X}) < p.$$

Then the columns of $\mathbf{X}$ are linearly dependent, and

$$\mathbf{X}^\top \mathbf{X}$$

is singular.

Therefore,

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

does not exist.

! A Subtle but Important Point

When $\mathbf{X}$ does not have full column rank,

- a least squares solution still exists;
- $\hat{\beta}$ may not be unique;
- the fitted vector $\hat{\mathbf{Y}}$ is still unique;
- the residual vector $\mathbf{e}$ is still unique.

Thus,

$$\boxed{\hat{\beta} \text{ may not be unique, but } \hat{\mathbf{Y}} \text{ is unique.}}$$

The reason is geometric: even if several coefficient vectors represent the same point in $\mathcal{C}(\mathbf{X})$, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.

We will discuss rank deficiency, generalized inverses, and estimability in more detail later.

2.8 Interpretation of the Geometry

The observed vector

$$\mathbf{Y} \in \mathbb{R}^n.$$

When $\mathbf{X}$ has full column rank p , the model space

$$\mathcal{C}(\mathbf{X})$$

is a p -dimensional subspace of $\mathbb{R}^n$.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

The central relationships are

$$\min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2$$

$$\Downarrow$$

$$\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$$

$$\Downarrow$$

$$\mathbf{e} \perp \mathcal{C}(\mathbf{X})$$

$$\Downarrow$$

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

$$\Downarrow$$

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.$$

Thus,

- fitted values are orthogonal projections;
- residuals are orthogonal to the model space;
- the hat matrix is a projection matrix;
- the residual-maker matrix projects onto the orthogonal complement;
- sum-of-squares decompositions are consequences of the Pythagorean theorem.

2.9 Looking Ahead: Statistical Properties of OLS

i Looking Ahead

Everything we have done so far is primarily **algebraic and geometric**.

We defined $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ by solving an optimization problem.

However, under the statistical model, $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, the response $\mathbf{Y}$ is random. Therefore, $\hat{\beta}$ is also a random vector.

In next chapter, we will study quantities such as

$$\mathbb{E}(\hat{\beta}), \quad \text{and} \quad \text{Var}(\hat{\beta}),$$

followed by estimation of σ^2 , sampling distributions, confidence intervals, and hypothesis testing.

i Summary

In this chapter, we introduced the least squares estimator through the optimization problem

$$\hat{\beta} = \arg \min_{\beta} \|\mathbf{Y} - \mathbf{X}\beta\|_2^2.$$

Differentiating the residual sum of squares leads to the normal equations

$$\boxed{\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y}.}$$

When $\mathbf{X}$ has full column rank,

$$\boxed{\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.}$$

The fitted response is

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

is the hat matrix.

Geometrically,

$$\hat{\mathbf{Y}}$$

is the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

The residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$$

lies in the orthogonal complement of the model space, so

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

Therefore,

$$\boxed{\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}}$$

is an orthogonal decomposition.

In next chapter, we will treat $\hat{\beta}$ as a random vector and study its expectation, variance, sampling distribution, and the foundations of statistical inference for linear models.

2.10 Practice at Home

1. Explain why

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}$$

means that the residual vector is orthogonal to the column space $\mathcal{C}(\mathbf{X})$.

2. Explain why

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

is an orthogonal projection, and why

$$\mathbf{H}^2 = \mathbf{H}.$$

3. If the model contains an intercept, explain why

$$\sum_{i=1}^n e_i = 0$$

and hence

$$\bar{\hat{Y}} = \bar{Y}.$$

4. Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

Compute

$$\hat{\beta}, \quad \hat{\mathbf{Y}}, \quad \mathbf{e},$$

and verify that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

5. Show that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

is symmetric and idempotent.

Side Fact: Matrix Calculus Facts Used in this chapter

Let β be a column vector.

For a constant vector $\mathbf{a}$,

$$\frac{\partial}{\partial \beta} (\mathbf{a}^\top \beta) = \mathbf{a}.$$

For a constant matrix $\mathbf{A}$,

$$\frac{\partial}{\partial \beta} (\beta^\top \mathbf{A} \beta) = (\mathbf{A} + \mathbf{A}^\top) \beta.$$

If $\mathbf{A}$ is symmetric,

$$\mathbf{A}^\top = \mathbf{A},$$

and therefore

$$\boxed{\frac{\partial}{\partial \beta} (\beta^\top \mathbf{A} \beta) = 2\mathbf{A}\beta.}$$

These identities justify the derivative used in the least squares criterion.

3 Distribution Theory of OLS and Inference

In the previous chapter, we studied least squares primarily from an **algebraic and geometric** point of view.

We obtained

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and showed that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

We now move from **estimation** to **statistical inference**.

Since $\mathbf{Y}$ is random, quantities such as

$$\hat{\beta}, \quad \hat{\mathbf{Y}}, \quad \mathbf{e}, \quad \text{SSE}$$

are also random.

Our goal is to understand their sampling distributions and use them to quantify uncertainty.

Learning Objectives

By the end of this chapter, students should be able to:

- derive the expectation and variance of the OLS estimator;
- state the normal linear model;

- derive the distribution of the OLS estimator under normality;
- explain the distribution of the residual sum of squares;
- explain why the residual degrees of freedom are $n - p$;
- estimate σ^2 using the mean squared error;
- explain why $\hat{\beta}$ and SSE are independent;
- construct confidence intervals for regression coefficients;
- perform hypothesis tests for regression coefficients;
- perform inference for linear combinations of regression coefficients;
- distinguish inference for a mean response from prediction of a future observation;
- interpret standard regression output from R.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on distribution theory of least squares estimators;
 - estimation of the error variance;
 - inference for regression coefficients.
- Montgomery, Peck, and Vining:
 - sections on confidence intervals;
 - hypothesis testing;
 - estimation and prediction in linear regression.

i Recap from Least Squares

From Chapter 2, when $\mathbf{X}$ has full column rank,

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

We also have

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

with

$$\mathbf{H} + \mathbf{M} = \mathbf{I}_n$$

and

$$\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}.$$

Geometrically,

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

is an orthogonal decomposition.

In this chapter, we will see that this geometric decomposition also provides the foundation for regression inference.

! Throughout This Chapter

We treat the design matrix $\mathbf{X}$ as fixed.

If the predictors are random, the results can instead be interpreted conditionally on the observed design matrix $\mathbf{X}$.

3.1 Mean and Variance of the OLS Estimator

Before introducing normality, we can already study the expectation and variance of the OLS estimator.

Assume

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Recall that

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon.$$

Substituting this into the OLS estimator gives

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \varepsilon) \\ &= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon. \end{aligned}$$

Thus,

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon.$$

This representation will be useful throughout the chapter.

Proposition 3.1 (Mean and Variance of OLS). *Under*

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

the OLS estimator satisfies

$$\mathbb{E}(\hat{\beta}) = \beta$$

and

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

3.1.1 Why Is OLS Unbiased?

Using

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon,$$

we obtain

$$\begin{aligned} \mathbb{E}(\hat{\beta}) &= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\varepsilon) \\ &= \beta. \end{aligned}$$

Thus,

$$\mathbb{E}(\hat{\beta}) = \beta.$$

OLS is therefore an **unbiased estimator** of β .

Intuitively, if we repeated the experiment many times using the same design matrix, the OLS estimates would fluctuate around the true parameter vector.

3.1.2 Variance of OLS

Similarly,

$$\begin{aligned}\text{Var}(\hat{\beta}) &= \text{Var}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon] \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{Var}(\varepsilon) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.\end{aligned}$$

! Normality Is Not Needed Yet

The results

$$\mathbb{E}(\hat{\beta}) = \beta$$

and

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

do **not** require normally distributed errors.

They follow from the first two moment assumptions

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Normality becomes important when we want exact finite sample distributions and inference.

Exercise 3.1 (Which Assumptions Are Needed?). Suppose

$$\mathbb{E}(\varepsilon) = \mathbf{0}$$

and

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Do we need to assume that the errors are normally distributed to conclude that

$$\mathbb{E}(\hat{\beta}) = \beta?$$

Solution 3.1. No.

Normality is **not** required for unbiasedness.

The result follows from

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon$$

and

$$\mathbb{E}(\varepsilon) = \mathbf{0}.$$

Normality will be needed later for exact finite sample inference.

3.2 The Normal Linear Model

To obtain exact sampling distributions, we now strengthen the assumptions.

Definition 3.1 (Normal Linear Model). The **normal linear model** assumes

$$\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n).$$

Equivalently,

$$\varepsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

Under this assumption:

- each response has a normal distribution;
- the errors have common variance σ^2 ;
- the errors are independent;
- exact finite sample distributions can be obtained for many regression statistics.

i Why Does Normality Matter?

Least squares itself does not require normality.

Normality is introduced because it gives us exact distributions for

$$\hat{\beta}, \quad \text{SSE}, \quad T, \quad F.$$

These distributions allow us to construct exact finite sample confidence intervals and hypothesis tests.

Exercise 3.2 (What Changes When We Add Normality?). Suppose the errors satisfy

$$\mathbb{E}(\varepsilon) = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n,$$

but are not normally distributed.

Which of the following can still hold?

1. OLS can still be computed.
2. OLS can still be unbiased.
3. We can still have

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

4. Exact finite sample t and F distributions automatically hold.

Solution 3.2. Statements 1, 2, and 3 can still hold.

Statement 4 does not generally hold.

Normality is not required to define OLS or derive its mean and variance. Its main role here is to provide exact finite sample distributions for inference.

3.3 Distribution of the OLS Estimator

Recall that

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Thus, $\hat{\beta}$ is a **linear transformation** of $\mathbf{Y}$.

A fundamental property of the multivariate normal distribution is that a linear transformation of a multivariate normal vector is again multivariate normal.

Theorem 3.1 (Distribution of the OLS Estimator). *Under the normal linear model,*

$$\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}).$$

Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Then

$$\text{Var}(\hat{\beta}) = \sigma^2 \mathbf{C}.$$

If c_{jj} denotes the j th diagonal element of $\mathbf{C}$, then

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 c_{jj}).$$

Therefore,

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}} \sim N(0, 1).$$

Exercise 3.3 (Reading the Covariance Matrix). Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

If c_{jj} is the j th diagonal element of $\mathbf{C}$:

1. What is $\text{Var}(\hat{\beta}_j)$?
2. What happens to the uncertainty of $\hat{\beta}_j$ if c_{jj} becomes larger?

Solution 3.3. The variance is

$$\boxed{\text{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}.$$

Therefore, a larger c_{jj} means greater sampling variability and less precise estimation of β_j .

Thus, the precision of OLS depends not only on σ^2 , but also on the design matrix $\mathbf{X}$.

There is still one difficulty.

The quantity

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}}$$

depends on the unknown parameter σ .

Before performing inference, we need to estimate σ^2 .

Exercise 3.4 (Why Can We Not Use the Normal Statistic Directly?). We have

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}} \sim N(0, 1).$$

Why can we usually not use this statistic directly for inference?

Solution 3.4. Because the error standard deviation σ is unknown.

We therefore need to estimate σ from the data.

Replacing σ with an estimator will lead naturally to the t distribution.

3.4 Residual Sum of Squares and Estimation of σ^2

Recall from Chapter 2 that

$$\mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Since

$$\mathbf{M}\mathbf{X} = \mathbf{0},$$

we have

$$\begin{aligned}\mathbf{e} &= \mathbf{M}\mathbf{Y} \\ &= \mathbf{M}(\mathbf{X}\beta + \varepsilon) \\ &= \mathbf{M}\varepsilon.\end{aligned}$$

Thus,

$$\boxed{\mathbf{e} = \mathbf{M}\varepsilon.}$$

The residual vector therefore contains the part of the random error that lies outside the model space.

3.4.1 Residual Sum of Squares

The residual sum of squares is

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}.$$

Since

$$\mathbf{e} = \mathbf{M}\varepsilon,$$

and $\mathbf{M}$ is symmetric and idempotent,

$$\begin{aligned}\text{SSE} &= \varepsilon^\top \mathbf{M}^\top \mathbf{M} \varepsilon \\ &= \varepsilon^\top \mathbf{M} \varepsilon.\end{aligned}$$

Therefore,

$$\boxed{\text{SSE} = \varepsilon^\top \mathbf{M} \varepsilon.}$$

3.4.2 Distribution of SSE

Under the normal linear model,

$$\frac{\varepsilon}{\sigma} \sim N_n(\mathbf{0}, \mathbf{I}_n).$$

Let

$$\mathbf{Z} = \frac{\varepsilon}{\sigma}.$$

Then

$$\frac{\text{SSE}}{\sigma^2} = \mathbf{Z}^\top \mathbf{M} \mathbf{Z}.$$

Recall that $\mathbf{M}$ is symmetric and idempotent and

$$\text{rank}(\mathbf{M}) = n - p.$$

Therefore,

Theorem 3.2 (Distribution of the Residual Sum of Squares). *Under the normal linear model,*

$$\boxed{\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2.}$$

3.4.3 Why Are the Degrees of Freedom $n - p$?

Recall that

$$\text{rank}(\mathbf{H}) = p$$

and

$$\text{rank}(\mathbf{M}) = n - p.$$

The fitted vector lies in

$$\mathcal{C}(\mathbf{X}),$$

which has dimension p .

The residual vector lies in

$$\mathcal{C}(\mathbf{X})^\perp,$$

which has dimension

$$n - p.$$

Therefore, the residual variation has

$$\boxed{n - p}$$

degrees of freedom.

i Geometric Interpretation of Degrees of Freedom

The response space can be decomposed as

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp.$$

The fitted component occupies p dimensions.

The remaining

$$n - p$$

dimensions are available for residual variation.

This provides a geometric explanation for why estimating p regression parameters costs p degrees of freedom.

Exercise 3.5 (Degrees of Freedom). Suppose a regression model has

$$n = 20$$

observations and

$$p = 3$$

regression parameters.

1. What is the residual degrees of freedom?
2. Why is it not 20?

Solution 3.5. The residual degrees of freedom are

$$n - p = 20 - 3 = 17.$$

It is not 20 because fitting the regression model uses $p = 3$ dimensions of the response space.

The residual vector therefore lies in a space of dimension

$$20 - 3 = 17.$$

3.4.4 Estimating σ^2

Since a chi-square random variable with ν degrees of freedom has expectation ν ,

$$\mathbb{E}\left(\frac{\text{SSE}}{\sigma^2}\right) = n - p.$$

Therefore,

$$\mathbb{E}(\text{SSE}) = (n - p)\sigma^2.$$

This gives us an unbiased estimator of the error variance.

Definition 3.2 (Mean Squared Error). The **mean squared error** is defined as

$$\text{MSE} = \hat{\sigma}^2 = \frac{\text{SSE}}{n - p}.$$

It satisfies

$$\mathbb{E}(\hat{\sigma}^2) = \sigma^2.$$

The estimated error standard deviation is

$$\hat{\sigma} = \sqrt{\text{MSE}}.$$

Exercise 3.6 (Estimate the Error Variance). Suppose

$$n = 20, \quad p = 3, \quad \text{SSE} = 34.$$

Compute

$$\hat{\sigma}^2$$

and

$$\hat{\sigma}.$$

Solution 3.6. The residual degrees of freedom are

$$n - p = 17.$$

Therefore,

$$\hat{\sigma}^2 = \frac{34}{17} = 2.$$

Hence,

$$\hat{\sigma} = \sqrt{2} \approx 1.414.$$

3.5 Independence of OLS and Residual Variation

A crucial result under the normal linear model is

$$\hat{\beta} \text{ and } \text{SSE} \text{ are independent.}$$

This result is closely connected to the projection geometry from the previous chapter.

3.5.1 Why Does Independence Occur?

Recall that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

and

$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

We also know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Thus, the fitted component and the residual component come from orthogonal parts of the response vector.

i Geometry Becomes Probability

In the previous chapter, we learned that

$$\hat{\mathbf{Y}} \perp \mathbf{e}.$$

This is a geometric statement.

Under the normal linear model, these components are jointly normal.

For jointly normal random vectors, zero covariance implies independence.

Thus, under normality, geometric orthogonality leads to probabilistic independence.

More formally,

$$\hat{\beta} - \beta = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$$

and

$$\mathbf{e} = \mathbf{M}\varepsilon.$$

Therefore,

$$\begin{aligned}\text{Cov}(\hat{\beta}, \mathbf{e}) &= \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M} \\ &= \mathbf{0},\end{aligned}$$

because

$$\mathbf{X}^\top \mathbf{M} = \mathbf{0}.$$

Since the two random vectors are jointly normal,

$$\hat{\beta} \perp\!\!\!\perp \mathbf{e}.$$

Since

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}$$

is a function of $\mathbf{e}$,

$$\boxed{\hat{\beta} \perp\!\!\!\perp \text{SSE}.}$$

Exercise 3.7 (Does Orthogonal Always Mean Independent?). We know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Does the orthogonality between the fitted and residual components automatically imply that they are independent?

Why is normality important?

Solution 3.7. No.

In general, zero covariance or orthogonality does **not** imply independence.

Under the normal linear model, the fitted and residual components are jointly normal.

For jointly normal random vectors, zero covariance implies independence.

Thus, normality is what allows us to move from orthogonality to independence.

3.6 Inference for a Single Regression Coefficient

For the j th regression coefficient,

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 c_{jj}).$$

If σ were known, we could use

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}} \sim N(0, 1).$$

In practice, σ is unknown.

We replace it with

$$\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n - p}}.$$

3.6.1 Standard Error

The estimated standard deviation of $\hat{\beta}_j$ is

Definition 3.3 (Standard Error of an OLS Coefficient). The standard error of $\hat{\beta}_j$ is

$$\boxed{\text{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{c_{jj}}.}$$

3.6.2 Why Does the t Distribution Appear?

We know that

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}} \sim N(0, 1).$$

We also know that

$$U = \frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2.$$

In addition,

$$Z \quad \text{and} \quad U$$

are independent.

Therefore,

$$\frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}.$$

Substituting the regression quantities gives

Theorem 3.3 (t Distribution for an OLS Coefficient). *Under the normal linear model,*

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma} \sqrt{c_{jj}}} \sim t_{n-p}.$$

i Normal vs. t Distribution

If σ were known, we would use a normal distribution.

Because σ is unknown and must be estimated from the data, we introduce additional uncertainty.

The t distribution accounts for this additional uncertainty.

As the degrees of freedom increase,

$$t_\nu$$

approaches

$$N(0, 1).$$

Exercise 3.8 (Why Do We Use a t Distribution?). Suppose

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 c_{jj}).$$

Why do we use a t distribution rather than a standard normal distribution when performing inference for β_j ?

Solution 3.8. Because σ is unknown.

We replace σ with the estimate $\hat{\sigma}$ obtained from the residual sum of squares.

This additional estimation uncertainty changes the standardized statistic from a normal distribution to a t distribution.

3.6.3 Confidence Interval for β_j

A $100(1 - \alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{1-\alpha/2, n-p} \text{SE}(\hat{\beta}_j).$$

Equivalently,

$$\hat{\beta}_j \pm t_{1-\alpha/2, n-p} \hat{\sigma} \sqrt{c_{jj}}.$$

3.6.4 Hypothesis Test for β_j

Suppose we want to test

$$H_0 : \beta_j = \beta_{j,0}$$

against

$$H_1 : \beta_j \neq \beta_{j,0}.$$

The test statistic is

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{SE}(\hat{\beta}_j)}.$$

Under H_0 ,

$$T \sim t_{n-p}.$$

For a two sided test at significance level α , reject H_0 if

$$|T| > t_{1-\alpha/2, n-p}.$$

Exercise 3.9 (Confidence Interval and Hypothesis Test). Suppose we test

$$H_0 : \beta_j = 0$$

against

$$H_1 : \beta_j \neq 0$$

at significance level

$$\alpha = 0.05.$$

What should happen if the corresponding 95% confidence interval for β_j does not contain zero?

Solution 3.9. We reject

$$H_0 : \beta_j = 0$$

at the 5% significance level.

A two sided hypothesis test at level α and a $100(1 - \alpha)\%$ confidence interval give equivalent conclusions.

3.7 Inference for Linear Combinations

Sometimes the quantity of interest is not a single coefficient. Suppose we are interested in $\mathbf{a}^\top \beta$, where $\mathbf{a}$ is a fixed $p \times 1$ vector. Since

$$\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}),$$

we have

$$\mathbf{a}^\top \hat{\beta} \sim N(\mathbf{a}^\top \beta, \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}).$$

Therefore,

$$\frac{\mathbf{a}^\top \hat{\beta} - \mathbf{a}^\top \beta}{\hat{\sigma} \sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}} \sim t_{n-p}.$$

A $100(1 - \alpha)\%$ confidence interval for $\mathbf{a}^\top \beta$ is

$$\mathbf{a}^\top \hat{\beta} \pm t_{1-\alpha/2, n-p} \hat{\sigma} \sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}.$$

i Why Is This Useful?

Many regression questions can be expressed as linear combinations of the regression coefficients. Examples include:

- a single coefficient;
- the difference between two coefficients;
- the mean response at a particular predictor value.

Thus, inference for a single regression coefficient is a special case of inference for a linear combination.

Exercise 3.10 (A Single Coefficient as a Linear Combination). Suppose

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

What vector $\mathbf{a}$ would you choose so that $\mathbf{a}^\top \beta = \beta_1$?

Solution 3.10. Choose

$$\mathbf{a} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then $\mathbf{a}^\top \beta = \beta_1$. Thus, inference for an individual coefficient is a special case of inference for a linear combination.

3.8 General Linear Hypotheses

Sometimes we want to test several linear restrictions simultaneously. Suppose $H_0 : \mathbf{C}\beta = \mathbf{d}$, where $\mathbf{C}$ is an $r \times p$ matrix of rank r and $\mathbf{d}$ is an $r \times 1$ vector.

Definition 3.4 (General Linear Hypothesis). A hypothesis of the form

$$H_0 : \mathbf{C}\beta = \mathbf{d}$$

is called a **general linear hypothesis**. It allows several linear restrictions on the regression coefficients to be tested simultaneously.

The corresponding test statistic is

$$F = \frac{(\mathbf{C}\hat{\beta} - \mathbf{d})^\top [\mathbf{C}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{C}^\top]^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d})}{r\hat{\sigma}^2}.$$

Under H_0 ,

$$F \sim F_{r, n-p}.$$

Relationship Between t and F

When only one restriction is being tested, $r = 1$. In this case,

$$F = T^2.$$

Thus, the familiar t test for one coefficient is a special case of the general F test. We will study F tests more thoroughly in the next chapter when we develop the ANOVA decomposition and compare nested regression models.

Exercise 3.11 (*t* Test or *F* Test?). Suppose we want to test only $H_0 : \beta_2 = 0$. Could we use either a *t* test or an *F* test? How are the two test statistics related?

Solution 3.11. Yes. Because there is only one linear restriction, either test can be used. The corresponding statistics satisfy

$$\boxed{F = T^2.}$$

Therefore, the two tests produce the same conclusion.

3.9 Inference for the Mean Response

Suppose we want to estimate the mean response at a predictor vector $\mathbf{x}_0$. The true mean response is

$$\mu(\mathbf{x}_0) = \mathbf{x}_0^\top \beta.$$

Its estimator is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\beta}.$$

Since this is a linear combination of $\hat{\beta}$,

$$\text{Var}(\hat{\mu}(\mathbf{x}_0)) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0.$$

Thus,

$$\frac{\mathbf{x}_0^\top \hat{\beta} - \mathbf{x}_0^\top \beta}{\hat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-p}.$$

Therefore, a $100(1 - \alpha)\%$ confidence interval for the mean response is

$$\boxed{\mathbf{x}_0^\top \hat{\beta} \pm t_{1-\alpha/2, n-p} \hat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

3.10 Prediction for a New Observation

Now suppose we want to predict one new observation at the same predictor value $\mathbf{x}_0$:

$$Y_{\text{new}} = \mathbf{x}_0^\top \beta + \varepsilon_{\text{new}},$$

where

$$\varepsilon_{\text{new}} \sim N(0, \sigma^2)$$

and is independent of the original data. The predictor is

$$\hat{Y}_{\text{new}} = \mathbf{x}_0^\top \hat{\beta}.$$

The prediction error is

$$Y_{\text{new}} - \mathbf{x}_0^\top \hat{\beta}.$$

Its variance is

$$\text{Var} \left(Y_{\text{new}} - \mathbf{x}_0^\top \hat{\beta} \right) = \sigma^2 \left[1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 \right].$$

Therefore, a $100(1 - \alpha)\%$ prediction interval is

$$\mathbf{x}_0^\top \hat{\beta} \pm t_{1-\alpha/2, n-p} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

! Mean Response vs. New Observation

The two intervals answer different questions.

Goal	Source of uncertainty
Estimate a mean response	Uncertainty in estimating the regression mean
Predict a new observation	Uncertainty in the regression mean plus new random variation

The additional 1 in $1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ represents the variability of the new observation itself. Therefore, at the same $\mathbf{x}_0$ and confidence level, the prediction interval is wider than the confidence interval for the mean response.

Exercise 3.12 (Mean Response or Prediction?). Consider the following questions.

1. What is the **average stopping distance** of all cars travelling at 50 mph?
2. What stopping distance should we expect for **one particular future car** travelling at 50 mph?

Which question requires a confidence interval for the mean response, and which requires a prediction interval?

Solution 3.12. The first question concerns the **mean response**, so we use a confidence interval for the regression mean. The second question concerns **one future observation**, so we use a prediction interval. The prediction interval is wider because it also includes the random variation associated with the future observation.

3.11 Worked Example

Consider again the simple regression dataset from Chapter 2:

$$\begin{array}{c|cccc} x_i & 0 & 1 & 2 & 3 \\ \hline Y_i & 1 & 3 & 3 & 5 \end{array}.$$

The design matrix and response vector are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}.$$

From the previous chapter,

$$\hat{\beta} = \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix},$$

and

$$\hat{\mathbf{Y}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Therefore,

$$\mathbf{e} = \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.$$

Estimate the Error Variance

The residual sum of squares is

$$\begin{aligned} \text{SSE} &= \mathbf{e}^\top \mathbf{e} \\ &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 \\ &= 0.8. \end{aligned}$$

Since $n = 4$ and $p = 2$, the residual degrees of freedom are $n - p = 2$. Thus,

$$\hat{\sigma}^2 = \frac{0.8}{2} = 0.4.$$

Therefore,

$$\hat{\sigma} = \sqrt{0.4} \approx 0.632.$$

Standard Error of the Slope

We have

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

For the slope,

$$c_{22} = \frac{4}{20} = 0.2.$$

Therefore,

$$\text{SE}(\hat{\beta}_1) = \sqrt{0.4(0.2)} = \sqrt{0.08} \approx 0.283.$$

Test the Slope

Consider $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. The test statistic is

$$T = \frac{1.2}{0.283} \approx 4.243.$$

The residual degrees of freedom are 2. For a 5% two sided test, $t_{0.975,2} \approx 4.303$. Since $|4.243| < 4.303$, we fail to reject H_0 at the 5% significance level.

i What Does This Example Show?

The estimated slope $\hat{\beta}_1 = 1.2$ is relatively large. However, there are only four observations, leaving only two residual degrees of freedom. Therefore, there is substantial uncertainty in the estimate. A large estimated effect does not automatically imply strong evidence against the null hypothesis.

Confidence Interval for the Slope

A 95% confidence interval is $1.2 \pm 4.303(0.283)$, which gives approximately

$$(-0.017, 2.417).$$

The interval contains zero, which agrees with the hypothesis test.

Exercise 3.13 (Check the Connection). Without performing another hypothesis test, what conclusion can you draw from the fact that the 95% confidence interval $(-0.017, 2.417)$ contains zero?

Solution 3.13. We fail to reject $H_0 : \beta_1 = 0$ at the 5% significance level. This agrees with the previous t test.

Mean Response at $x_0 = 2$

For simple linear regression with an intercept,

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The estimated mean response is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\beta} = 3.6.$$

Also,

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = 0.3.$$

Therefore,

$$\text{SE}(\hat{\mu}(\mathbf{x}_0)) = \sqrt{0.4(0.3)} \approx 0.346.$$

The 95% confidence interval for the mean response is approximately

$$\boxed{(2.110, 5.090)}.$$

Prediction at $x_0 = 2$

For one new observation,

$$\text{SE}_{\text{pred}} = \sqrt{0.4(1 + 0.3)} \approx 0.721.$$

Therefore, the 95% prediction interval is approximately

$$\boxed{(0.497, 6.703)}.$$

Notice that the prediction interval is substantially wider than the confidence interval for the mean response.

3.12 R Demonstration

We can reproduce the previous calculations using R.

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)

summary(fit)
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

```
      1      2      3      4
-0.2   0.6 -0.6   0.2
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.2000	0.5292	2.268	0.1515
x	1.2000	0.2828	4.243	0.0513 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom

Multiple R-squared: 0.9, Adjusted R-squared: 0.85

F-statistic: 18 on 1 and 2 DF, p-value: 0.05132

Estimated Coefficients

```
coef(fit)
```

(Intercept)	x
1.2	1.2

Residual Degrees of Freedom and Error Variance

```
df.residual(fit)
```

```
[1] 2
```

```
sigma(fit)
```

```
[1] 0.6324555
```

```
sigma(fit)^2
```

```
[1] 0.4
```

Covariance Matrix and Standard Errors

```
vcov(fit)
```

	(Intercept)	x
(Intercept)	0.28	-0.12
x	-0.12	0.08

```
sqrt(diag(vcov(fit)))
```

(Intercept)	x
0.5291503	0.2828427

Verify the Formulas Using Matrix Operations

```
X <- model.matrix(fit)
Y <- matrix(y, ncol = 1)

n <- nrow(X)
p <- ncol(X)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y
```

```
e <- Y - X %*% beta_hat

SSE <- as.numeric(t(e) %*% e)

sigma2_hat <- SSE / (n - p)

V_beta_hat <-
  sigma2_hat *
  solve(t(X) %*% X)

SE_beta_hat <-
  sqrt(diag(V_beta_hat))

beta_hat
```

```
      [,1]
(Intercept) 1.2
x           1.2
```

```
SSE
```

```
[1] 0.8
```

```
sigma2_hat
```

```
[1] 0.4
```

```
V_beta_hat
```

```
      (Intercept)      x
(Intercept)    0.28 -0.12
x             -0.12  0.08
```

```
SE_beta_hat
```

```
      (Intercept)      x
0.5291503    0.2828427
```

Hypothesis Tests and Confidence Intervals

R automatically reports coefficient level t tests:

```
summary(fit)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.2	0.5291503	2.267787	0.1514719
x	1.2	0.2828427	4.242641	0.0513167

Confidence intervals for the regression coefficients can be obtained using

```
confint(fit)
```

		2.5 %	97.5 %
(Intercept)	-1.07674982	3.476750	
x	-0.01697397	2.416974	

Mean Response and Prediction

At $x_0 = 2$, use

```
newdat <- data.frame(x = 2)
```

```
predict(
  fit,
  newdata = newdat,
  interval = "confidence"
)
```

	fit	lwr	upr
1	3.6	2.109517	5.090483

```
predict(
  fit,
  newdata = newdat,
  interval = "prediction"
)
```

	fit	lwr	upr
1	3.6	0.497313	6.702687

The first interval estimates the **mean response**. The second interval predicts **one future observation**.

3.13 Reading Standard Regression Output

For

```
summary(fit)
```

the coefficient table contains four important columns:

Column	Interpretation
Estimate	Estimated regression coefficient
Std. Error	Estimated standard deviation of the coefficient estimator
t value	t statistic for testing whether the coefficient equals zero
Pr(> t)	Two sided p value for the test

The output also includes:

- the residual standard error;
- the residual degrees of freedom;
- R^2 ;
- adjusted R^2 ;
- an overall F test.

We will study R^2 , the ANOVA decomposition, and the overall F test more carefully in the next chapter.

Exercise 3.14 (Reading an R Regression Table). Suppose R reports

	Estimate	Std. Error	t value	Pr(> t)
x_1	2.40	0.60	4.00	0.001

Answer the following.

1. What is $\hat{\beta}_1$?
2. What is $SE(\hat{\beta}_1)$?
3. What null hypothesis is being tested by default?

Solution 3.14. The estimate is $\hat{\beta}_1 = 2.40$. The standard error is $\text{SE}(\hat{\beta}_1) = 0.60$. The default coefficient level test is $H_0 : \beta_1 = 0$. The reported t statistic is

$$\frac{2.40}{0.60} = 4.$$

3.14 Practice at Home

1. Starting from

$$\hat{\beta} = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon,$$

derive

$$\mathbb{E}(\hat{\beta}) = \beta$$

and

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

2. Explain why normality is not required for OLS to be unbiased but is useful for exact finite sample inference.
3. Explain geometrically why the residual degrees of freedom are $n - p$.
4. Explain why $\hat{\beta}$ and SSE are independent under the normal linear model.
5. Derive the t statistic for testing $H_0 : \beta_j = \beta_{j,0}$.
6. Explain the connection between a two sided hypothesis test and the corresponding confidence interval.
7. Explain the difference between a confidence interval for $\mathbf{x}_0^\top \beta$ and a prediction interval for a future observation at $\mathbf{x}_0$.
8. Fit a simple regression model in R and identify:
 - the coefficient estimates;
 - their standard errors;
 - the t statistics;
 - the p values;
 - the residual standard error;
 - the residual degrees of freedom.

Summary

This chapter moved from the **geometry of least squares** to the **probability of least squares**. Under the first two moment assumptions,

$$\mathbb{E}(\hat{\beta}) = \beta$$

and

$$\text{Var}(\hat{\beta}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}.$$

Under the additional normality assumption,

$$\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}).$$

The residual sum of squares satisfies

$$\frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2.$$

This leads to the unbiased estimator

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n-p}.$$

The independence of $\hat{\beta}$ and SSE allows us to replace the unknown σ with $\hat{\sigma}$ and obtain exact t inference. The same framework extends naturally to:

- linear combinations of regression coefficients;
- general linear hypotheses;
- confidence intervals for the mean response;
- prediction intervals for future observations.

In the next chapter, we will use these results to develop the ANOVA decomposition, the overall F test, and comparisons of nested regression models.

Side Fact: Distribution Results Used in This Chapter

If

$$\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$$

and $\mathbf{A}$ is symmetric and idempotent with $\text{rank}(\mathbf{A}) = r$, then

$$\boxed{\mathbf{Z}^\top \mathbf{A} \mathbf{Z} \sim \chi_r^2.}$$

If

$$Z \sim N(0, 1), \quad U \sim \chi_\nu^2,$$

and Z and U are independent, then

$$\boxed{\frac{Z}{\sqrt{U/\nu}} \sim t_\nu.}$$

If

$$U_1 \sim \chi_r^2, \quad U_2 \sim \chi_\nu^2,$$

and U_1 and U_2 are independent, then

$$\boxed{\frac{U_1/r}{U_2/\nu} \sim F_{r,\nu}.}$$

These results provide the distributional foundation for regression inference.

4 Model Comparison, Partial F Tests, and ANOVA

In this chapter, we study the ANOVA decomposition for linear regression, the overall significance test, and the comparison of nested models. These ideas connect the geometry of least squares with the inferential tools developed in earlier chapters.

Learning Objectives

By the end of this chapter, students should be able to:

- define the total, regression, and error sums of squares;
- explain the ANOVA decomposition in regression with an intercept;
- interpret the degrees of freedom associated with SST, SSR, and SSE;
- perform the overall F test for regression;
- compare nested linear models using extra sums of squares;
- interpret ANOVA tables produced by statistical software.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on analysis of variance in regression
 - sums of squares and decomposition of variability
 - tests for nested models
- Montgomery, Peck, and Vining:
 - sections on the ANOVA table
 - overall model significance
 - partial and sequential sums of squares

i Motivation: does measuring tree height add useful information?

Suppose a forestry team wants to estimate timber volume before felling a tree. Diameter is already available; measuring height requires additional effort. Does height contribute information beyond diameter?

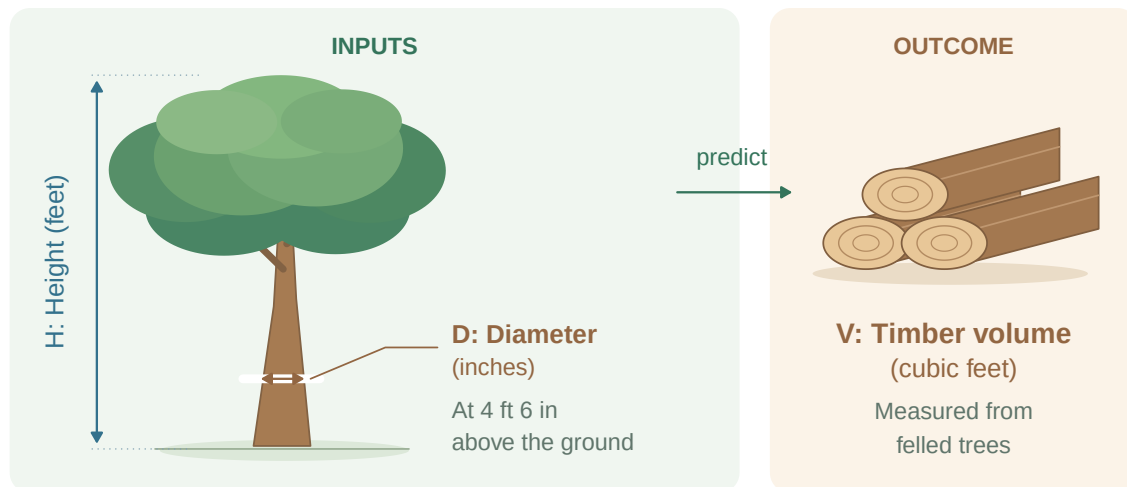
The real `trees` dataset in R lets us explore this question with measurements on 31 felled black cherry trees. Consider three candidate mean models:

Model	Information used	Mean volume
M_0	No tree measurements	β_0
M_1	Diameter	$\beta_0 + \beta_1 \text{ Diameter}$
M_2	Diameter and height	$\beta_0 + \beta_1 \text{ Diameter} + \beta_2 \text{ Height}$

Discuss before calculating: Which comparison answers whether height adds information? Why is choosing the smallest training SSE insufficient? Keep these questions in mind; the optional data example below returns to them with fitted models.

From a tree to a regression model

Predict timber volume before felling



At the same diameter, does height help predict volume?

Conceptual sketch, not to scale or a formula for volume.

Figure 4.1: From measurements to a prediction question. Diameter and height are predictors; measured timber volume is the response. The tree and timber drawings are conceptual, not to scale.

Review of the Linear Model

Recall the normal linear model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

When $\mathbf{X}$ has full column rank, the ordinary least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}.$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

From Chapter 2, we know that $\hat{\mathbf{Y}}$ and $\mathbf{e}$ are orthogonal. From Chapter 3, we know that this leads to useful distributional results for inference. In this chapter, we organize these ideas into the ANOVA framework.

Vector notation. Following the convention used in this course, $\mathbf{a} = (a_1, a_2, a_3)$ denotes a column vector, while $\mathbf{a}^\top = (a_1 \ a_2 \ a_3)$ denotes a row vector. Commas distinguish the column-vector tuple; spaces separate entries of the row vector.

4.1 Total, Explained, and Unexplained Variation

A central question in regression is: How much of the variation in the response can be explained by the model? To answer this, we decompose the total variation in $\mathbf{Y}$ into:

- variation explained by regression;
- variation left unexplained by the model.

When the model includes an intercept, this decomposition takes a particularly simple and important form.

Definition 4.1 (Regression Sums of Squares). For an OLS fit with an intercept, let $\bar{Y} = n^{-1} \sum_{i=1}^n Y_i$, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta}$, and $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$. Define the **total**, **error**, and **regression sums of squares** by

$$\begin{aligned}\text{SST} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2, \\ \text{SSE} &= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \mathbf{e}^\top \mathbf{e}, \\ \text{SSR} &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2.\end{aligned}$$

SST measures variation around the sample mean; SSE measures deviations from the fitted model; SSR measures how much the fitted values vary around the sample mean. These are definitions. Their additive relationship below follows from OLS orthogonality and the inclusion of an intercept.

Theorem 4.1 (ANOVA Decomposition with an Intercept). *For an ordinary least squares fit whose design matrix contains an intercept,*

$$\text{SST} = \text{SSR} + \text{SSE}.$$

The centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to the residual vector $\mathbf{e}$. Thus total variation around the sample mean splits into explained variation and residual variation. This geometric identity does not require normal errors.

Why the Decomposition Holds

The key reason is orthogonality. We can write

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + (\mathbf{Y} - \hat{\mathbf{Y}}).$$

That is,

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + \mathbf{e}.$$

Because the model contains an intercept, the vector $\bar{Y}\mathbf{1}$ lies in the column space of $\mathbf{X}$. Hence both $\hat{\mathbf{Y}}$ and $\bar{Y}\mathbf{1}$ lie in the model space, so their difference also lies in the model space. Since the residual vector $\mathbf{e}$ is orthogonal to the model space, we have

$$(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})^\top \mathbf{e} = 0.$$

Therefore, by the Pythagorean theorem,

$$\|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2 + \|\mathbf{e}\|^2,$$

which is exactly

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Exercise 4.1 (In-Class Discussion: Why Does the Intercept Matter?). Why does the usual centered decomposition $\text{SST} = \text{SSR} + \text{SSE}$ require the model space to contain the constant vector? Which step could fail without an intercept? Does this identity require normal errors?

Solution 4.1. With an intercept, $\mathbf{1}$ and therefore $\bar{Y}\mathbf{1}$ belong to the model space. Thus the centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to $\mathbf{e}$, and the cross term vanishes when its squared norm is expanded. Without a constant vector in the model space, $\sum_i e_i$ need not be zero, so the cross term $\mathbf{e}^\top(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = -\bar{Y}\sum_i e_i$ need not vanish. The identity follows from OLS geometry, not normality.

Exercise 4.2 (In Class: One Observation, Three Deviations). For one observation, $y_i = 1$, $\hat{y}_i = 1.2$, and $\bar{y} = 3$. In one minute, identify the total, explained, and residual deviations. Does the squared total deviation equal the sum of the other two squared deviations for this observation?

Solution 4.2. The deviations are $y_i - \bar{y} = -2$, $\hat{y}_i - \bar{y} = -1.8$, and $y_i - \hat{y}_i = -0.2$. They add as signed deviations: $-2 = -1.8 - 0.2$. Their squares do not add: $4 \neq 3.24 + 0.04$. The ANOVA identity holds after summing over all observations because the fitted and residual vectors are orthogonal; it is not a point-by-point identity for squared deviations.

Degrees of Freedom

The ANOVA decomposition is accompanied by a decomposition of degrees of freedom. When the model includes an intercept and $\mathbf{X}$ has rank p , we have:

- total degrees of freedom: $n - 1$;
- regression degrees of freedom: $p - 1$;
- error degrees of freedom: $n - p$.

Thus,

$$n - 1 = (p - 1) + (n - p).$$

These match the sum of squares decomposition:

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Definition 4.2 (Regression and Error Mean Squares). A **mean square** is a sum of squares divided by its associated degrees of freedom. For a full-rank regression design with an intercept and $1 < p < n$, define

$$\text{MSR} = \frac{\text{SSR}}{p-1}, \quad \text{MSE} = \frac{\text{SSE}}{n-p}.$$

Here p includes the intercept coefficient.

Dividing by degrees of freedom accounts for the different dimensions contributing to each sum of squares. Under the linear-model moment assumptions from Chapter 3, MSE is an unbiased estimator of σ^2 . MSR estimates the same noise level under the overall null hypothesis, which motivates their ratio.

4.2 The Overall F Test

A major inferential question is whether the regression model provides any explanatory power beyond the intercept-only model. Suppose the model includes an intercept and $p-1$ additional predictors. The null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0,$$

where β_0 is the intercept and p counts all regression coefficients. Equivalently, under H_0 , the mean response does not depend on the predictors. The alternative is that at least one non-intercept coefficient is nonzero.

Theorem 4.2 (Overall F Test). *Under the normal linear model with a fixed, full-rank design containing an intercept and $1 < p < n$, the null hypothesis that all non-intercept coefficients are zero gives*

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{p-1, n-p}.$$

Large values provide evidence against the intercept-only mean model. Exact finite-sample inference requires the normal, common-variance error assumptions; the sum-of-squares identity alone does not supply this distribution.

Interpretation of the Overall F Test

The numerator measures explained variation per regression degree of freedom. The denominator measures unexplained variation per residual degree of freedom. So the F statistic compares:

- how much signal the model explains;
- how much noise remains in the residuals.

If the predictors have no effect, then both quantities should be of similar size, and the ratio should not be unusually large. If the predictors explain substantial variation, then the numerator should be much larger than the denominator.

Exercise 4.3 (In Class: What Does an Overall Rejection Mean?). An intercept model with two predictors is fitted to $n = 25$ observations using a full-rank design. Write the overall null hypothesis and the F reference degrees of freedom under the normal linear model. How does this joint test differ from the individual t tests? If it rejects, must both individual slope tests also reject?

Solution 4.3. Here $p = 3$, so $H_0 : \beta_1 = \beta_2 = 0$ and the reference distribution is $F_{2,22}$. Rejection gives evidence that the two slopes are not jointly zero. It does not show that both are nonzero, and it does not require either individual test to reject at the same level. Each individual t test asks whether one slope is zero while retaining the other predictor; the overall F test asks whether both slopes are jointly zero. These are different questions, especially when predictors are correlated.

Relationship to the Intercept-Only Model

The overall F test compares two models:

- the reduced model: intercept only;
- the full model: intercept plus predictors.

Thus the ANOVA decomposition provides the basis for formal model comparison. This leads naturally to the idea of nested models.

4.3 Nested Models

Definition 4.3 (Nested Linear Models). For the same response vector and observations, let the reduced and full models have mean spaces $\mathcal{C}(\mathbf{X}_R)$ and $\mathcal{C}(\mathbf{X}_F)$. The models are **nested** when

$$\mathcal{C}(\mathbf{X}_R) \subseteq \mathcal{C}(\mathbf{X}_F).$$

Thus every mean vector allowed by the reduced model is also allowed by the full model. A strict comparison requires ranks $p_R < p_F$; adding columns that are linear combinations of existing columns does not enlarge the mean space.

Example 4.1 (Adding a Predictor). Consider the reduced model $Y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$ and the full model $Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$. Setting $\beta_2 = 0$ recovers the reduced model. If the x_2 column is not in the span of the intercept and x_1 , the full model adds one degree of freedom.

Definition 4.4 (Extra Sum of Squares). For nested models fitted by OLS to the same response and observations, define the **extra sum of squares** for the full model relative to the reduced model as

$$SS_{\text{extra}} = SSE_R - SSE_F.$$

Its associated degrees of freedom are $p_F - p_R$, where p_R and p_F are the design-matrix ranks.

Exercise 4.4 (In-Class Discussion: Can the Larger Model Fit Worse?). Why must $SSE_F \leq SSE_R$ when nested models are fitted by OLS to the same observations? Must the inequality be strict whenever a genuinely new predictor direction is added?

Solution 4.4. Every fitted mean vector allowed by the reduced model is also available to the full model. Minimizing the same residual sum of squares over this larger set cannot give a larger minimum. Equality is possible even if the model space grows: for the observed response, the added directions may provide no further reduction in residual variation. Thus greater flexibility guarantees a weak improvement in training fit, not a strictly positive improvement or a useful predictor.

A positive extra sum of squares alone is not evidence against the reduced model: the F test scales that improvement by its degrees of freedom and the estimated noise variance.

Theorem 4.3 (Partial F Test for Nested Normal Linear Models). *Fit both models to the **same response and the same observations**. Suppose their fixed design matrices have nested column spaces, with ranks $p_R < p_F < n$, and the full model has independent $N(0, \sigma^2)$ errors. Under the null hypothesis that the mean belongs to the reduced model,*

$$F = \frac{(SSE_R - SSE_F)/(p_F - p_R)}{SSE_F/(n - p_F)} \sim F_{p_F - p_R, n - p_F}.$$

Large values give evidence against the reduced mean model. The numerator measures improvement per added degree of freedom; the denominator estimates the unexplained variance under the full model.

! Compare the models, not just the printed p-values

Different missing-value patterns can cause two R fits to use different observations. Construct a common analysis dataset before comparing them. For one added coefficient, the partial F statistic equals the square of its two-sided t-test statistic from the full model.

Exercise 4.5 (In Class: A Better Fit, but Enough Improvement?). Two nested models use the same $n = 30$ observations. The reduced model has rank $p_R = 2$ and $\text{SSE}_R = 90$; the full model has rank $p_F = 4$ and $\text{SSE}_F = 78$. Compute the extra sum of squares, its degrees of freedom, the full-model MSE, and the partial F statistic. Does a reduction in SSE by itself justify rejecting the reduced model?

Solution 4.5. The improvement is $90 - 78 = 12$ on $4 - 2 = 2$ degrees of freedom. The full-model MSE is $78/(30 - 4) = 3$, so $F = (12/2)/3 = 2$. Under the normal-model null, compare this with $F_{2,26}$; the upper-tail p-value is about 0.156. The improvement is insufficient for rejection at the 5% level. More flexibility always weakly reduces training SSE, including when the added coefficients are zero.

Connection with General Linear Hypotheses

Nested column-space models correspond to linear restrictions on the full-model coefficients. More generally, Chapter 3 allows a specified affine restriction of the form

$$H_0 : \mathbf{C}\beta = \mathbf{d}.$$

So the ANOVA comparison of nested models is another way of expressing the general F test from Chapter 3. In practice, this is one of the most common uses of regression ANOVA tables.

Definition 4.5 (Sequential and Partial Sums of Squares). For a term in a specified regression model, both quantities are extra sums of squares from a nested comparison on the same observations:

- **Sequential sum of squares:** the reduction in SSE when the term is added after the preceding terms in a specified order.
- **Partial sum of squares:** the increase in SSE when the specified term is removed from a full model while the other specified terms are retained.

When predictors are correlated, sequential sums of squares can depend on their order. Partial sums of squares answer a comparison conditional on the terms retained in the reduced model. With interactions or rank deficiency, state the exact restriction and respect model hierarchy rather than relying only on a Type I, II, or III label.

Definition 4.6 (Coefficient of Determination). For an OLS fit with an intercept and $SST > 0$, the **coefficient of determination** is

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

By the ANOVA decomposition, R^2 lies between 0 and 1 and measures the proportion of centered variation explained by the fitted model. If every observed response is identical, $SST = 0$ and this ratio is undefined.

Interpretation of R Squared

- If R^2 is close to 1, the model explains a large proportion of the variation in the response.
- If R^2 is close to 0, the model explains little of the variation.

However, R^2 alone does not guarantee that the model is appropriate. A high R^2 does not ensure that assumptions are satisfied, and a low R^2 does not necessarily imply the model is useless.

Definition 4.7 (Adjusted Coefficient of Determination). For an OLS fit with an intercept, $SST > 0$, and design rank $p < n$, define the **adjusted coefficient of determination** as

$$R_{\text{adj}}^2 = 1 - \frac{SSE/(n-p)}{SST/(n-1)}.$$

Ordinary R^2 cannot decrease when the model space is enlarged on the same data. Adjusted R^2 accounts for the loss of residual degrees of freedom and can decrease or become negative. It compares residual mean square with the sample variance of the response; it is not a literal proportion of explained variation or an estimate of out-of-sample predictive performance.

Example 4.2 (A Small ANOVA Calculation). For the column vectors $\mathbf{x} = (0, 1, 2, 3)$ and $\mathbf{y} = (1, 3, 3, 5)$, the fitted line is $\hat{y} = 1.2 + 1.2x$. Therefore $\hat{\mathbf{y}} = (1.2, 2.4, 3.6, 4.8)$, $\mathbf{e} = (-0.2, 0.6, -0.6, 0.2)$, and $\bar{y} = 3$.

$$SST = (1-3)^2 + (3-3)^2 + (3-3)^2 + (5-3)^2 = 8,$$

$$SSE = (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 = 0.8,$$

$$SSR = SST - SSE = 8 - 0.8 = 7.2.$$

Here $n = 4$ and $p = 2$, so the regression, residual, and total degrees of freedom are 1, 2, and 3, respectively.

Source	Sum of squares	df	Mean square	F
Regression	7.2	1	7.2	18
Error	0.8	2	0.4	
Total	8.0	3		

The model has $R^2 = 7.2/8 = 0.9$, yet the F-test p-value is $P(F_{1,2} \geq 18) \approx 0.0513$. With only two residual degrees of freedom, a high training R^2 does not automatically give evidence at the 5% significance level.

Exercise 4.6 (In Class: Can Two Fit Summaries Disagree?). For the same $n = 10$ observations with $SST = 100$, an intercept model has rank 2 and SSE 20. Adding a predictor whose column is linearly independent of the existing columns increases the rank to 3 and reduces SSE to 19. Compute R^2 and adjusted R^2 for both fits. Did the one-unit reduction in SSE improve both summaries? Why can relying on R^2 alone be misleading?

Solution 4.6. Ordinary R^2 increases from $1 - 20/100 = 0.80$ to $1 - 19/100 = 0.81$. Adjusted R^2 decreases from $1 - (20/8)/(100/9) = 0.775$ to $1 - (19/7)/(100/9) \approx 0.756$. The extra predictor reduces SSE but increases residual mean square, from 2.5 to about 2.714. Training R^2 cannot decrease when the model space grows, even when the new term only fits noise. A high R^2 also does not establish a correct mean function, appropriate error assumptions, or causality. Neither summary alone establishes better prediction on new observations.

ANOVA Table Structure

A standard regression ANOVA table has the following structure:

Source	Sum of Squares	Degrees of Freedom	Mean Square	F
Regression	SSR	$p - 1$	MSR	MSR/MSE
Error	SSE	$n - p$	MSE	
Total	SST	$n - 1$		

Students should learn to move fluently between:

- formulas;
- geometric interpretation;
- software output.

4.4 R Demonstration

Fit a simple regression model

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)
summary(fit)
```

Call:

```
lm(formula = y ~ x)
```

Residuals:

```
      1      2      3      4
-0.2   0.6 -0.6   0.2
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.2000	0.5292	2.268	0.1515
x	1.2000	0.2828	4.243	0.0513 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom

Multiple R-squared: 0.9, Adjusted R-squared: 0.85

F-statistic: 18 on 1 and 2 DF, p-value: 0.05132

Obtain the ANOVA table

```
anova(fit)
```

Analysis of Variance Table

Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
x	1	7.2	7.2	18	0.05132 .
Residuals	2	0.8	0.4		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Verify sums of squares manually

```
ybar <- mean(y)
yhat <- fitted(fit)
e <- resid(fit)

SST <- sum((y - ybar)^2)
SSE <- sum(e^2)
SSR <- sum((yhat - ybar)^2)

c(SST = SST, SSR = SSR, SSE = SSE)
```

```
SST SSR SSE
8.0 7.2 0.8
```

```
SST - SSR - SSE
```

```
[1] 3.330669e-16
```

See the three sources of variation. Each vertical segment below represents one deviation. Squaring and adding the four segment lengths gives the corresponding sum of squares.

```
par(mfrow = c(1, 3), mar = c(4, 4, 2.4, 0.6),
    mgp = c(2.4, 0.7, 0))
for (component in c("Total", "Explained", "Residual")) {
  plot(x, y, pch = 16, ylim = c(0.5, 5.5), xlab = "x", ylab = "y",
       main = component, cex.main = 1)
  abline(h = ybar, col = "gray60", lty = 2)
  abline(fit, col = "steelblue4")
  if (component == "Total") {
    segments(x, ybar, x, y, col = "gray35", lwd = 2)
  } else if (component == "Explained") {
    segments(x, ybar, x, yhat, col = "steelblue4", lwd = 2)
    points(x, yhat, pch = 1, col = "steelblue4")
  } else {
    segments(x, yhat, x, y, col = "darkorange3", lwd = 2)
  }
}
par(mfrow = c(1, 1))
```

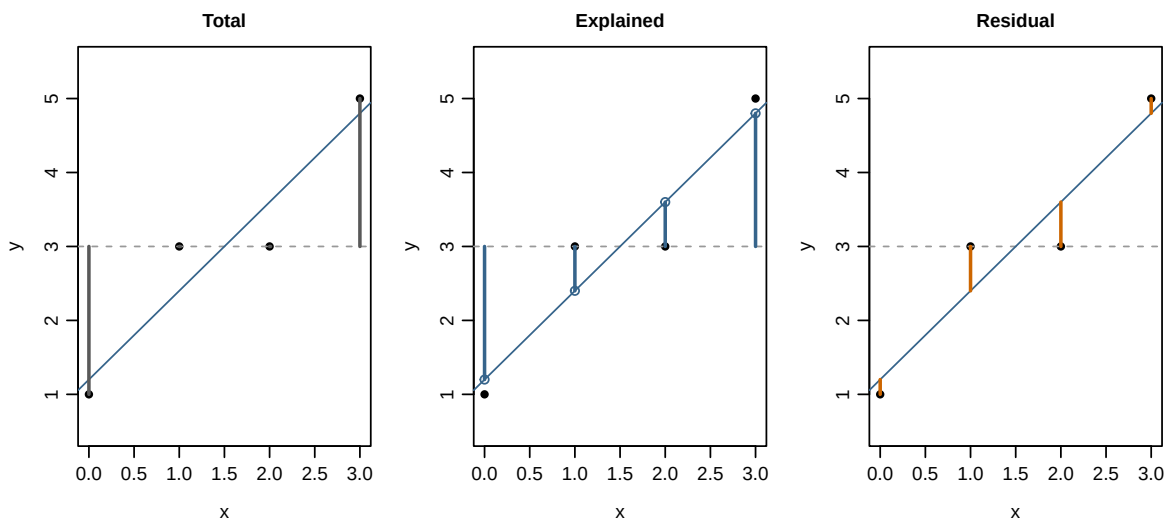


Figure 4.2: Total deviations from the sample mean, explained deviations of fitted values from the mean, and residual deviations from the fitted line. $SST = SSR + SSE$ holds after summing across observations; it is not a point-by-point identity between squared deviations.

Compute R squared manually

```
R2 <- SSR / SST
R2
```

```
[1] 0.9
```

```
summary(fit)$r.squared
```

```
[1] 0.9
```

```
summary(fit)$adj.r.squared
```

```
[1] 0.85
```

```
pf(18, df1 = 1, df2 = 2, lower.tail = FALSE)
```

```
[1] 0.0513167
```

Compare nested models

```
dat <- data.frame(
  y = c(4, 5, 7, 10, 8, 12, 13, 14),
  x1 = c(1, 2, 3, 4, 5, 6, 7, 8),
  x2 = c(2, 1, 3, 2, 5, 4, 6, 5)
)

fit_reduced <- lm(y ~ x1, data = dat)
fit_full <- lm(y ~ x1 + x2, data = dat)

anova(fit_reduced, fit_full)
```

Analysis of Variance Table

```
Model 1: y ~ x1
Model 2: y ~ x1 + x2
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	6	6.8214				
2	5	4.6613	1	2.1601	2.3171	0.1884

Inspect the two fitted models

```
summary(fit_reduced)
```

Call:

```
lm(formula = y ~ x1, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.85714	-0.30357	0.03571	0.33036	1.60714

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.5357	0.8308	3.052	0.022453 *
x1	1.4643	0.1645	8.900	0.000112 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.066 on 6 degrees of freedom

Multiple R-squared: 0.9296, Adjusted R-squared: 0.9178
 F-statistic: 79.21 on 1 and 6 DF, p-value: 0.0001121

```
summary(fit_full)
```

Call:

```
lm(formula = y ~ x1 + x2, data = dat)
```

Residuals:

1	2	3	4	5	6	7	8
0.4032	-1.0403	0.3306	0.8871	-1.1371	0.4194	0.7903	-0.6532

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.9677	0.8041	3.691	0.01413 *
x1	1.8387	0.2876	6.394	0.00139 **
x2	-0.6048	0.3973	-1.522	0.18845

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9655 on 5 degrees of freedom

Multiple R-squared: 0.9519, Adjusted R-squared: 0.9326

F-statistic: 49.46 on 2 and 5 DF, p-value: 0.0005079

Interpretation of Software Output

For a fitted model in R:

- `anova(fit)` gives sequential sums of squares in the order terms enter a single model;
- `anova(fit_reduced, fit_full)` compares nested models;
- `summary(fit)` reports the overall F statistic, R^2 , and adjusted R^2 .

Students should understand that these outputs are not separate topics. They are all built from the same least squares geometry and distribution theory.

Example 4.3 (Optional Real-Data Example: Is Tree Height Useful Beyond Diameter?). Return to the forestry question. R's `trees` data contain timber volume in cubic feet, height in feet, and a variable named `Girth` that is actually **diameter in inches**, measured 4 feet 6 inches above the ground. We rename it `Diameter` to avoid that labeling trap. Each row represents one of 31 felled black cherry trees. See the [official dataset documentation](#) for the source and measurement definitions.

State the question first. Comparing M_0 with M_2 asks whether diameter and height together improve on a constant mean. Comparing M_1 with M_2 asks whether height contributes after adjusting for diameter. For the second question, $H_0 : \beta_{\text{Height}} = 0$. These are distinct questions even though both use an F test.

```
tree_dat <- datasets::trees
tree_dat$Diameter <- tree_dat$Girth
tree_dat <- tree_dat[, c("Volume", "Diameter", "Height")]
stopifnot(!anyNA(tree_dat))
fit_tree0 <- lm(Volume ~ 1, data = tree_dat)
fit_tree_d <- lm(Volume ~ Diameter, data = tree_dat)
fit_tree_dh <- lm(Volume ~ Diameter + Height, data = tree_dat)
knitr::kable(data.frame(
  Model = c("Constant mean", "Diameter", "Diameter + height"),
  Rank = c(fit_tree0$rank, fit_tree_d$rank, fit_tree_dh$rank),
  SSE = c(deviance(fit_tree0), deviance(fit_tree_d),
    deviance(fit_tree_dh))), digits = 2)
```

Model	Rank	SSE
Constant mean	1	8106.08
Diameter	2	524.30
Diameter + height	3	421.92

```
anova(fit_tree_d, fit_tree_dh)
```

Analysis of Variance Table

Model 1: Volume ~ Diameter

Model 2: Volume ~ Diameter + Height

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	29	524.30				
2	28	421.92	1	102.38	6.7943	0.01449 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Read the comparison. The SSE drops from about 524.30 to 421.92. The added term uses one degree of freedom, and the full model has $31 - 3 = 28$ residual degrees of freedom. Thus

$$F = \frac{(524.30 - 421.92)/1}{421.92/28} \approx 6.79, \quad p \approx 0.0145.$$

Under the stated normal, common-variance error model, this provides evidence against a zero height coefficient after accounting for diameter. It is not the overall test of both slopes.

```
op <- par(mfrow = c(1, 2), mar = c(4, 4, 2.2, 0.6),
          mgp = c(2.4, 0.7, 0))
tree_models <- list("Diameter only" = fit_tree_d,
                    "Diameter + height" = fit_tree_dh)
volume_limits <- range(c(tree_dat$Volume, fitted(fit_tree_d),
                        fitted(fit_tree_dh)))
for (label in names(tree_models)) {
  prediction <- fitted(tree_models[[label]])
  plot(tree_dat$Diameter, tree_dat$Volume, pch = 16, cex = 0.65,
        ylim = volume_limits, xlab = "Diameter (inches)",
        ylab = "Timber volume (cubic feet)", main = label)
  segments(tree_dat$Diameter, tree_dat$Volume,
            tree_dat$Diameter, prediction, col = "darkorange3")
  points(tree_dat$Diameter, prediction, pch = 1,
          col = "steelblue4", cex = 0.8)
  legend("topleft", c("Observed", "Fitted"), pch = c(16, 1),
        col = c("black", "steelblue4"), bty = "n", cex = 0.8)
}
par(op)
```

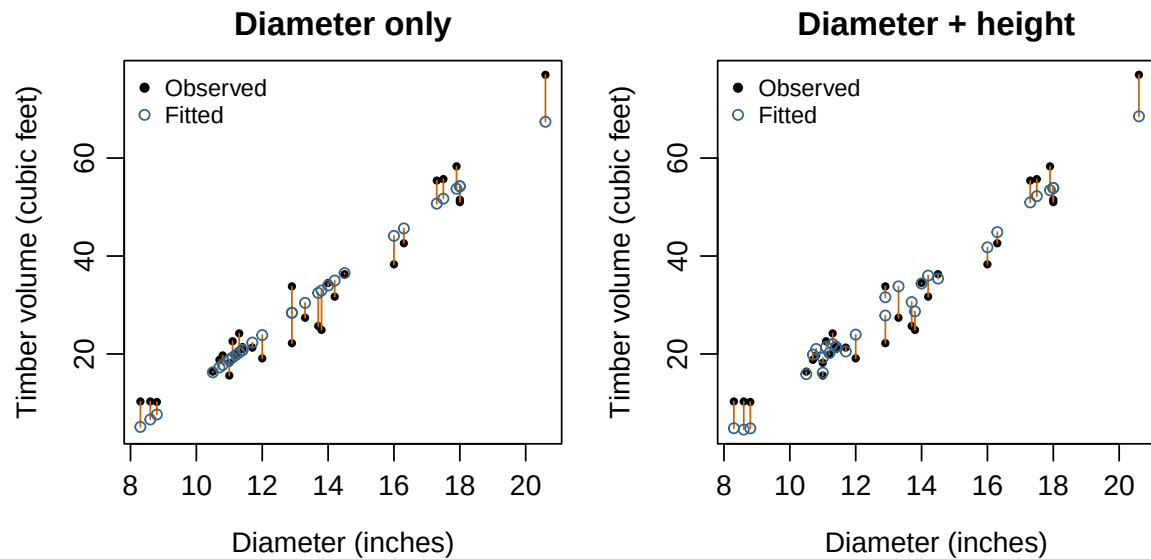


Figure 4.3: The same 31 trees under two mean models. Black dots are measured volumes; blue circles are fitted volumes. Orange segments show residuals. Adding height reduces the sum of squared residual lengths, but the F test asks whether that improvement is large relative to the estimated noise.

Check your understanding. Why use 28 rather than 29 in the variance estimate? What would change if some trees had missing heights? The denominator comes from the full model. With missing heights, first form one common dataset and refit both models on it; comparing fits on different observations confounds the added predictor with a changed sample.

What this example does not settle. These are observational measurements, so the comparison is not a causal effect of making a tree taller. A statistically significant height term also does not decide whether measuring height is worth its cost or how much it helps predict new trees. Those decisions require practical error tolerances and validation. Inspect residuals and the mean function before using the raw-scale linear model operationally; transformations and model assessment appear later in the course.

4.5 Summary

In this chapter, we developed the ANOVA framework for linear regression. We defined:

$$\text{SST}, \quad \text{SSR}, \quad \text{SSE},$$

and showed that, with an intercept,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

This decomposition led to:

- the ANOVA table;
- the overall F test for regression;
- the comparison of nested models through extra sums of squares;
- the interpretation of R^2 and adjusted R^2 .

4.6 Practice Problems

Conceptual

1. Explain the meaning of SST, SSR, and SSE in words.
2. Explain why the regression degrees of freedom are $p - 1$ when the model includes an intercept.
3. Explain the difference between the overall F test and a test for a single coefficient.

Computational

Suppose a regression model with intercept has:

- $n = 20$,
- $p = 4$,
- $\text{SST} = 100$,
- $\text{SSE} = 40$.

Compute:

1. SSR,
2. the degrees of freedom for regression and error,
3. MSR,
4. MSE,
5. the overall F statistic,
6. R^2 .

Nested Model Problem

A reduced model has $\text{SSE}_R = 120$ with $p_R = 3$, and a full model has $\text{SSE}_F = 90$ with $p_F = 5$. If $n = 30$, compute the nested-model F statistic.

Suggested Homework

Complete the following tasks:

- prove the decomposition $SST = SSR + SSE$ when the model includes an intercept;
- derive the overall F statistic from the ANOVA decomposition;
- fit a regression model in R and reproduce the ANOVA table by hand;
- compare two nested models using an extra sum of squares test;
- interpret both R^2 and adjusted R^2 for a chosen dataset.

5 Multiple Regression and Categorical Predictors

In this chapter, we move from simple regression ideas to multiple regression. We study how regression coefficients are interpreted when several predictors are included in the model, how categorical predictors enter through indicator variables, and how interactions change the meaning of coefficients. The main goal is to help students read, build, and interpret regression models in realistic settings.

Learning Objectives

By the end of this chapter, students should be able to:

- interpret coefficients in a multiple regression model;
- explain the meaning of a partial regression coefficient;
- distinguish between marginal association and adjusted association;
- incorporate categorical predictors using indicator variables;
- interpret regression models with interactions;
- use software output to explain fitted multiple regression models.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on multiple linear regression
 - interpretation of regression coefficients
 - indicator variables and model formulation
- Montgomery, Peck, and Vining:
 - sections on multiple regression
 - qualitative predictors
 - interaction terms and interpretation

Review of the Regression Framework

Recall the linear model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

When the design matrix $\mathbf{X}$ has full column rank, the ordinary least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

In earlier weeks, we focused on estimation, inference, ANOVA decomposition, and the comparison of nested models. In this chapter, the emphasis shifts toward interpretation and modelling structure.

5.1 From Simple Regression to Multiple Regression

In simple linear regression, we write

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here, the slope β_1 measures the expected change in the response associated with a one-unit increase in x . In multiple regression, the model becomes

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i.$$

Here there are k predictors and $p = k + 1$ regression parameters, including the intercept. Throughout the course, p counts all columns of the full-rank design matrix. Now each coefficient must be interpreted while holding the other predictors fixed. This is the key conceptual shift.

Why Multiple Regression Matters

Multiple regression is important because real data usually involve several explanatory variables. Reasons for including multiple predictors include:

- improving prediction;
- adjusting for confounding variables;
- estimating the effect of one variable while controlling for others;
- allowing more realistic scientific interpretation.

A coefficient in multiple regression is therefore usually an adjusted effect, not a purely marginal one.

Interpreting the Intercept

In the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i,$$

the intercept β_0 is the expected value of the response when all predictors equal zero:

$$\mathbb{E}[Y_i \mid x_{i1}, \dots, x_{ip}] = \beta_0 \quad \text{when } x_{i1} = \cdots = x_{ip} = 0.$$

This interpretation may or may not be scientifically meaningful. Sometimes zero is a natural baseline. Sometimes it is outside the observed range, in which case the intercept is mainly a mathematical anchor for the model.

Interpreting Partial Regression Coefficients

Consider the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i.$$

Then:

- β_1 is the expected change in Y associated with a one-unit increase in x_1 , holding x_2 fixed;
- β_2 is the expected change in Y associated with a one-unit increase in x_2 , holding x_1 fixed.

This is called a **partial effect** or **adjusted effect**. The phrase “holding other variables fixed” is essential and should always be stated clearly.

Marginal Association Versus Adjusted Association

Suppose x_1 and x_2 are correlated. Then the relationship between Y and x_1 in a simple regression of Y on x_1 alone may differ from the coefficient of x_1 in a multiple regression including both x_1 and x_2 . This happens because:

- the simple regression coefficient describes a marginal association;
- the multiple regression coefficient describes an adjusted association.

These can differ substantially when predictors are related to each other.

Example of Adjusted Interpretation

Suppose we fit

$$\hat{Y} = 12.5 + 0.8x_1 - 1.2x_2.$$

Then:

- for each one-unit increase in x_1 , the fitted mean response increases by 0.8 units, holding x_2 fixed;
- for each one-unit increase in x_2 , the fitted mean response decreases by 1.2 units, holding x_1 fixed.

This interpretation is valid only within the modelling assumptions and over the range of data where the model is reasonable.

Matrix View of Multiple Regression

In multiple regression, the design matrix has the form

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}.$$

Each column corresponds to a predictor or model term. This allows the same least squares and inference framework to handle:

- continuous predictors;
- categorical predictors represented by indicator variables;
- interactions;
- polynomial terms.

So the general linear model framework is highly flexible.

5.2 Categorical Predictors and Indicator Variables

Many real predictors are categorical rather than numeric. Examples include:

- treatment group;
- sex;
- school type;
- region;
- machine type.

These enter a regression model through indicator variables, also called dummy variables.

Binary Predictor

Suppose z_i is a binary variable:

$$z_i = \begin{cases} 1, & \text{if observation } i \text{ is in group A,} \\ 0, & \text{if observation } i \text{ is in group B.} \end{cases}$$

Then the model

$$Y_i = \beta_0 + \beta_1 z_i + \varepsilon_i$$

has the following interpretation:

- when $z_i = 0$, the expected response is β_0 ;
- when $z_i = 1$, the expected response is $\beta_0 + \beta_1$.

Thus, β_1 measures the difference in group means.

More Than Two Categories

Suppose a categorical predictor has three levels:

- A,
- B,
- C.

We usually represent it with two indicator variables, for example:

$$z_{1i} = \begin{cases} 1, & \text{if level B,} \\ 0, & \text{otherwise,} \end{cases} \quad z_{2i} = \begin{cases} 1, & \text{if level C,} \\ 0, & \text{otherwise.} \end{cases}$$

Then level A is the reference group, and the model is

$$Y_i = \beta_0 + \beta_1 z_{1i} + \beta_2 z_{2i} + \varepsilon_i.$$

Interpretation:

- level A mean: β_0 ;
- level B mean: $\beta_0 + \beta_1$;
- level C mean: $\beta_0 + \beta_2$.

So coefficients are interpreted relative to the reference category.

Why We Do Not Include All Indicators with an Intercept

If a categorical variable has k levels, then with an intercept we include only $k - 1$ indicator variables. If we include all k indicators and also include an intercept, then the columns of the design matrix become linearly dependent. This causes rank deficiency. This is sometimes called the dummy variable trap.

Continuous and Categorical Predictors Together

Regression models often mix continuous and categorical predictors. For example,

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i$$

may contain:

- a continuous predictor x_i ;
- a binary indicator z_i .

Interpretation:

- β_1 is the slope in x holding group fixed;
- β_2 is the group difference holding x fixed.

This is one of the simplest forms of ANCOVA-style modelling.

5.3 Interactions

An interaction allows the effect of one predictor to depend on the value of another predictor. For two continuous predictors, we may write

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \varepsilon_i.$$

Now the effect of x_1 is no longer constant. Instead,

$$\frac{\partial \mathbb{E}[Y_i | x_{i1}, x_{i2}]}{\partial x_{i1}} = \beta_1 + \beta_3 x_{i2}.$$

So the slope for x_1 depends on the value of x_2 . Likewise, the slope for x_2 depends on x_1 .

Interaction Between a Continuous and a Binary Predictor

Suppose we fit

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i,$$

where $z_i \in \{0, 1\}$. Then:

- when $z_i = 0$,

$$\mathbb{E}[Y_i | x_i, z_i = 0] = \beta_0 + \beta_1 x_i;$$

- when $z_i = 1$,

$$\mathbb{E}[Y_i | x_i, z_i = 1] = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)x_i.$$

So:

- β_2 changes the intercept;
- β_3 changes the slope.

This allows the two groups to have different regression lines.

Main Effects in the Presence of Interaction

When an interaction is present, the meaning of the main effects changes. For example, in

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i,$$

the coefficient β_1 is the slope for x only when $z = 0$, the reference group. Similarly, β_2 is the group difference only when $x = 0$. So students should be careful not to interpret main effects in isolation when interactions are included.

Centering Predictors for Interpretation

Sometimes the value zero is not meaningful for a continuous predictor. In that case, it can be helpful to centre the predictor: $x_i^* = x_i - \bar{x}$. Then the intercept becomes the expected response at the average value of x , which may be much easier to interpret. Centering can also improve interpretability in models with interactions.

Comparing Models With and Without Interaction

To assess whether an interaction is needed, we can compare nested models. For example:

- reduced model:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i;$$

- full model:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i.$$

The reduced model is nested within the full model by setting $\beta_3 = 0$. So the interaction can be tested by a standard extra sum of squares F test.

Collinearity and Interpretation

In multiple regression, predictors may be strongly related to each other. When this happens:

- coefficient estimates can become unstable;
- standard errors can become large;
- interpretation becomes more delicate.

Even when the overall model seems useful, individual coefficients may be hard to estimate precisely. This issue is called **multicollinearity**. We will study diagnostics for it more formally later, but students should already know that “holding other variables fixed” may become practically difficult if predictors tend to move together.

! Adjusted association and causal effects

An adjusted regression coefficient describes a conditional association under the chosen mean model. A causal interpretation additionally requires a defensible study design and identification assumptions. Check the overlap of predictor values across groups before interpreting an adjusted comparison.

Multiple Regression as Conditional Mean Modelling

A good way to summarize multiple regression is: $\mathbb{E}[Y \mid X_1, \dots, X_k]$ is being modelled as a linear function of predictors and model terms. This viewpoint helps unify:

- continuous predictors;
- factors;

- interactions;
- transformed predictors.

It also helps students distinguish between the response itself and its conditional mean.

Worked Example With a Continuous and a Binary Predictor

Suppose we observe a response Y , a study-hours variable x , and a tutoring indicator z , where

- $z = 0$ means no tutoring;
- $z = 1$ means tutoring.

Consider the model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i.$$

Suppose the fitted equation is

$$\hat{Y} = 50 + 3x + 8z.$$

Then:

- among students with the same tutoring status, one extra hour of study is associated with an increase of 3 points in the fitted mean score;
- among students with the same number of study hours, the tutoring group has a fitted mean score 8 points higher than the non-tutoring group.

If we add an interaction and obtain

$$\hat{Y} = 48 + 4x + 10z - 1.5xz,$$

then:

- for students without tutoring, the slope in study hours is 4;
- for students with tutoring, the slope in study hours is $4 - 1.5 = 2.5$.

Thus, the effect of study time depends on tutoring status.

5.4 R Demonstration With Multiple Regression

Fit a model with two continuous predictors

```

dat1 <- data.frame(
  y = c(12, 15, 14, 18, 20, 19, 23, 25),
  x1 = c(1, 2, 2, 3, 4, 4, 5, 6),
  x2 = c(5, 4, 6, 4, 3, 5, 2, 1)
)

fit1 <- lm(y ~ x1 + x2, data = dat1)
summary(fit1)

```

Call:

```
lm(formula = y ~ x1 + x2, data = dat1)
```

Residuals:

```

      1      2      3      4      5      6      7
-3.158e-01 -2.632e-02 -1.316e-01  7.105e-01  3.300e-17 -1.053e-01  2.895e-01
      8
-4.211e-01

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.2895	1.1598	10.596	0.000129 ***
x1	2.2632	0.1681	13.464	4.05e-05 ***
x2	-0.4474	0.1697	-2.636	0.046186 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.423 on 5 degrees of freedom

Multiple R-squared: 0.9936, Adjusted R-squared: 0.991

F-statistic: 387.3 on 2 and 5 DF, p-value: 3.295e-06

Compare with a simple regression

```

fit1_simple <- lm(y ~ x1, data = dat1)
summary(fit1_simple)

```

Call:

```
lm(formula = y ~ x1, data = dat1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.89308	-0.27201	0.05031	0.39308	0.73585

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.3774	0.4988	18.80	1.46e-06 ***
x1	2.6289	0.1339	19.63	1.13e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.597 on 6 degrees of freedom

Multiple R-squared: 0.9847, Adjusted R-squared: 0.9821

F-statistic: 385.4 on 1 and 6 DF, p-value: 1.132e-06

```
summary(fit1)
```

Call:

```
lm(formula = y ~ x1 + x2, data = dat1)
```

Residuals:

	1	2	3	4	5	6	7
	-3.158e-01	-2.632e-02	-1.316e-01	7.105e-01	3.300e-17	-1.053e-01	2.895e-01
	8						
	-4.211e-01						

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.2895	1.1598	10.596	0.000129 ***
x1	2.2632	0.1681	13.464	4.05e-05 ***
x2	-0.4474	0.1697	-2.636	0.046186 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.423 on 5 degrees of freedom

Multiple R-squared: 0.9936, Adjusted R-squared: 0.991

F-statistic: 387.3 on 2 and 5 DF, p-value: 3.295e-06

Fit a model with a categorical predictor

```

dat2 <- data.frame(
  y = c(60, 62, 58, 71, 73, 70, 66, 68),
  hours = c(4, 5, 3, 4, 5, 6, 4, 5),
  group = factor(c("A", "A", "A", "B", "B", "B", "A", "B"))
)

fit2 <- lm(y ~ hours + group, data = dat2)
summary(fit2)

```

Call:

```
lm(formula = y ~ hours + group, data = dat2)
```

Residuals:

1	2	3	4	5	6	7	8
-1.50	-0.25	-2.75	1.25	2.50	-1.25	4.50	-2.50

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	58.500	6.236	9.381	0.000232 ***
hours	0.750	1.512	0.496	0.641001
groupB	8.250	2.620	3.149	0.025399 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.025 on 5 degrees of freedom

Multiple R-squared: 0.7821, Adjusted R-squared: 0.695

F-statistic: 8.975 on 2 and 5 DF, p-value: 0.02215

```
model.matrix(fit2)
```

	(Intercept)	hours	groupB
1	1	4	0
2	1	5	0
3	1	3	0
4	1	4	1
5	1	5	1
6	1	6	1
7	1	4	0
8	1	5	1

attr("assign")

```
[1] 0 1 2
attr("contrasts")
attr("contrasts")$group
[1] "contr.treatment"
```

Fit a model with interaction

```
fit3 <- lm(y ~ hours * group, data = dat2)
summary(fit3)
```

Call:

```
lm(formula = y ~ hours * group, data = dat2)
```

Residuals:

```
      1      2      3      4      5      6      7
-1.500e+00 -1.500e+00 -1.500e+00 -6.771e-16  2.500e+00 -1.343e-15  4.500e+00
      8
-2.500e+00
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	53.500	9.026	5.927	0.00406 **
hours	2.000	2.222	0.900	0.41897
groupB	19.500	14.401	1.354	0.24715
hours:groupB	-2.500	3.142	-0.796	0.47083

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.142 on 4 degrees of freedom

Multiple R-squared: 0.8119, Adjusted R-squared: 0.6708

F-statistic: 5.755 on 3 and 4 DF, p-value: 0.06202

```
anova(fit2, fit3)
```

Analysis of Variance Table

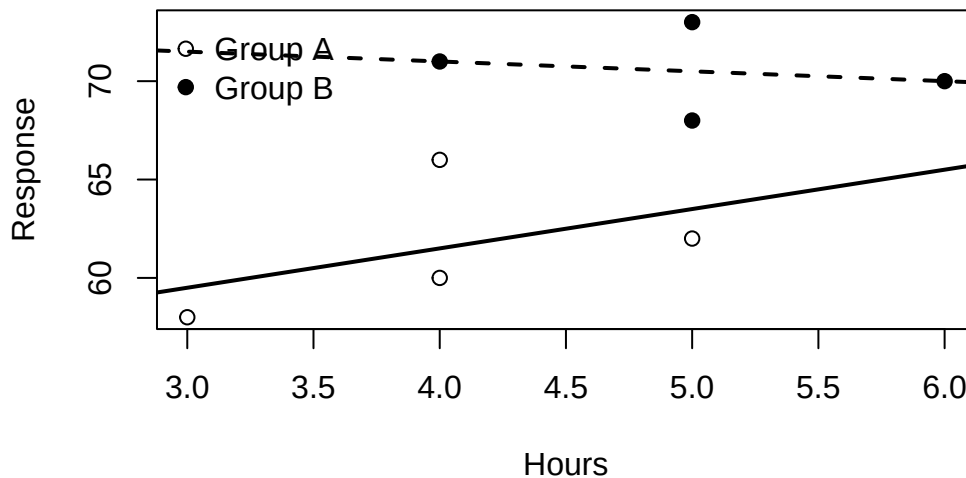
Model 1: y ~ hours + group

Model 2: y ~ hours * group

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	5	45.75				
2	4	39.50	1	6.25	0.6329	0.4708

Plot group-specific regression lines

```
plot(dat2$hours, dat2$y,
     pch = ifelse(dat2$group == "A", 1, 19),
     xlab = "Hours",
     ylab = "Response")
abline(a = coef(fit3)[1], b = coef(fit3)[2], lwd = 2)
abline(a = coef(fit3)[1] + coef(fit3)[3],
      b = coef(fit3)[2] + coef(fit3)[4],
      lwd = 2, lty = 2)
legend("topleft", legend = c("Group A", "Group B"),
      pch = c(1, 19), bty = "n")
```



Interpreting Software Output

In `summary(lm(...))`, each coefficient estimate answers a question about the conditional mean given the model terms included. Students should always ask:

- what variables are being held fixed;
- what is the reference category;
- whether an interaction changes the meaning of the main effects;
- whether zero is a meaningful baseline for interpretation.

These questions matter more than memorizing formulas.

In-Class Discussion Questions

1. Why can the coefficient of a predictor change when a second predictor is added to the model?
2. Why do we need a reference category for categorical predictors?
3. How does an interaction change the interpretation of a main effect?
4. When might centring a predictor improve interpretation?

5.5 Practice Problems

Conceptual

1. Explain the meaning of a partial regression coefficient in your own words.
2. Explain the difference between a marginal association and an adjusted association.
3. Explain why a model with an interaction requires more careful interpretation than an additive model.

Computational

Suppose the fitted model is

$$\hat{Y} = 10 + 2x_1 - 3x_2.$$

1. Interpret the coefficient of x_1 .
2. Interpret the coefficient of x_2 .
3. Compute the fitted mean response when $x_1 = 4$ and $x_2 = 1$.

Now suppose the fitted model is

$$\hat{Y} = 20 + 5x + 7z - 2xz,$$

where z is binary.

1. Write the fitted mean function when $z = 0$.
2. Write the fitted mean function when $z = 1$.
3. Interpret the interaction coefficient.

Indicator Variable Problem

A factor has four levels: A, B, C, and D.

1. How many indicator variables are needed if the model includes an intercept?
2. If A is the reference group, write a regression model using indicators for B, C, and D.
3. State the expected response for each group.

Suggested Homework

Complete the following tasks:

- fit a multiple regression model with at least two continuous predictors and interpret all coefficients;
- fit a model with one continuous predictor and one categorical predictor, then identify the reference category and explain all coefficients;
- add an interaction term and compare the additive and interaction models;
- use `model.matrix()` in R to inspect the design matrix for a model with factors;
- write a short explanation of why coefficient interpretation changes when additional predictors are added.

Summary

In this chapter, we developed the interpretation of multiple regression models. We emphasized that:

- regression coefficients in multiple regression are partial effects;
- categorical predictors are incorporated through indicator variables;
- interactions allow the effect of one predictor to depend on another;
- the meaning of a coefficient depends on the full model specification.

These ideas are essential for moving from formal least squares theory to practical statistical modelling.

6 Contrasts and Estimable Comparisons

A scientific question often concerns a difference between means rather than one coefficient printed by software. This chapter uses the linear-combination inference from Chapter 3 to express those questions. The next chapter connects these comparisons to experimental designs.

Learning objectives. Express a comparison of group means as a linear function of regression coefficients, obtain its estimate and interval, and distinguish a meaningful comparison from a coding-dependent coefficient.

6.1 Comparisons of group means

Suppose the conditional means of three groups are μ_A , μ_B , and μ_C . A question such as “Does A differ from the average of B and C?” concerns $\theta = \mu_A - (\mu_B + \mu_C)/2$.

Definition 6.1 (Contrast of Group Means). A contrast is a linear combination $\theta = \sum_{j=1}^g c_j \mu_j$ whose weights satisfy $\sum_{j=1}^g c_j = 0$. The zero-sum condition makes the comparison unchanged when the same constant is added to every group mean.

From means to coefficients. With A as the reference group, write $\beta = (\beta_0, \beta_B, \beta_C)^\top$. Then $\mu_A = \beta_0$, $\mu_B = \beta_0 + \beta_B$, and $\mu_C = \beta_0 + \beta_C$, so

$$\theta = -\frac{1}{2}\beta_B - \frac{1}{2}\beta_C = \mathbf{a}^\top \beta, \quad \mathbf{a} = (0, -\frac{1}{2}, -\frac{1}{2})^\top.$$

! The zero-sum condition applies to mean weights

The weights on (μ_A, μ_B, μ_C) sum to zero. The equivalent weights on treatment-coded regression coefficients need not sum to zero. Always translate the scientific comparison through the chosen parameterization.

Under the full-rank normal linear model, use

$$\hat{\theta} = \mathbf{a}^\top \hat{\beta}, \quad \text{SE}(\hat{\theta}) = \hat{\sigma} \sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}.$$

A confidence interval is $\hat{\theta} \pm t_{1-\alpha/2, n-p} \text{SE}(\hat{\theta})$. This is the same procedure as in Chapter 3, applied to a different target. Several independent restrictions can be collected into $C\beta = d$, with $\text{rank}(C) = r$, and tested jointly using the general F test.

Interpretation. Report the contrast estimate in the units of the response, its interval, and which population means it compares. Equality of two slopes is only a meaningful scientific question when their units and scales make that comparison defensible.

6.2 Coding and estimability

Changing the reference group changes the coefficient labels and values, but not the fitted group means or a fixed comparison of those means. The vector **a** must change with the coding so that it represents the same target.

Example 6.1 (An Intercept and All Group Indicators). In $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, adding a constant to every τ_i and subtracting it from μ leaves every group mean unchanged. The separate parameters are not identified without a constraint, but $\tau_A - \tau_B$ is unchanged and is estimable.

The practical lesson is to state the target in terms of observable mean responses before reading coefficients. For the row-space criterion and generalized-inverse formulation, see [Linear-Model Extensions \(Optional\)](#).

i One comparison versus a family of comparisons

A single prespecified contrast and a search through many contrasts are different inferential procedures. Define the family being reported. For all pairwise group comparisons, Tukey intervals are useful; for a specified family of tests, Holm adjustment is another option. An omnibus F test does not automatically justify every unadjusted pairwise test.

6.3 A complete contrast analysis in R

This simulated example separates the known data-generating means from their noisy estimates. Changing the seed changes the sample, not the target comparison.

```
set.seed(8561)
group <- factor(rep(c("A", "B", "C"), each = 12))
mu <- c(A = 10, B = 13, C = 15)
dat <- data.frame(group,
                  y = unname(mu[group]) + rnorm(length(group), sd = 2))
fit <- lm(y ~ group, data = dat)
```

```
# A minus the average of B and C, under treatment coding.
a <- c(0, -1/2, -1/2)
estimate <- sum(a * coef(fit))
se <- sqrt(drop(t(a) %*% vcov(fit) %*% a))
df <- df.residual(fit)
ci <- estimate + c(-1, 1) * qt(0.975, df) * se
c(estimate = estimate, SE = se, lower = ci[1], upper = ci[2],
  p_value = 2 * pt(-abs(estimate / se), df))
```

estimate	SE	lower	upper	p_value
-3.3388312952	0.9030487544	-5.1760978004	-1.5015647901	0.0007874472

Verify the same question under another coding. A cell-means model directly estimates (μ_A, μ_B, μ_C) , so its contrast weights are $(1, -1/2, -1/2)$.

```
fit_means <- lm(y ~ 0 + group, data = dat)
c_mean <- c(1, -1/2, -1/2)
estimate_means <- sum(c_mean * coef(fit_means))
se_means <- sqrt(drop(t(c_mean) %*% vcov(fit_means) %*% c_mean))
rbind(treatment_coding = c(estimate, se),
      cell_means_coding = c(estimate_means, se_means))
```

	[,1]	[,2]
treatment_coding	-3.338831	0.9030488
cell_means_coding	-3.338831	0.9030488

```
plot(estimate, 1, xlim = range(c(ci, 0)) + c(-0.5, 0.5), ylim = c(0.5, 1.5),
     yaxt = "n", ylab = "", xlab = "A - (B + C) / 2", pch = 19)
segments(ci[1], 1, ci[2], 1, lwd = 2)
abline(v = 0, lty = 2, col = "gray50")
```

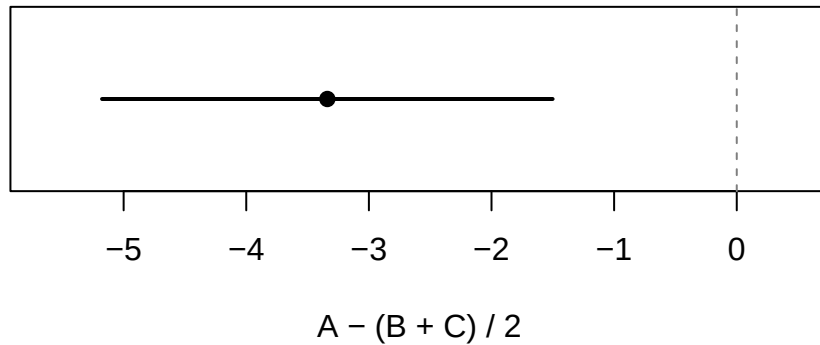


Figure 6.1: The estimated A-minus-average-of-B-and-C contrast and its 95% confidence interval. Zero represents no difference for this particular comparison.

A family of comparisons. For all three pairwise group differences, fit the one-way ANOVA and request simultaneous Tukey intervals:

```
TukeyHSD(aov(y ~ group, data = dat))
```

```
Tukey multiple comparisons of means
 95% family-wise confidence level
```

```
Fit: aov(formula = y ~ group, data = dat)
```

```
$group
      diff      lwr      upr    p adj
B-A 2.8862540 0.3275576 5.444950 0.0242654
C-A 3.7914086 1.2327122 6.350105 0.0026175
C-B 0.9051546 -1.6535418 3.463851 0.6639004
```

These intervals answer a different collection of questions from the single contrast above. Do not infer a difference between group means by inspecting whether two separate mean-response intervals overlap.

6.4 Practice

1. Express $\mu_B - \mu_C$ under treatment coding with A as reference, then under cell-means coding.
2. Verify that the estimate and standard error agree under the two codings.
3. Explain why the coefficients can change after changing the reference group while the scientific comparison remains unchanged.
4. Specify the comparison family before reporting all pairwise differences, and explain the role of multiplicity adjustment.

7 Experimental Design Models: ANOVA and ANCOVA

In this chapter, we study one-way ANOVA, two-way ANOVA, and analysis of covariance as special cases of the general linear model. The main goal is to show that these topics are not separate from regression, but are natural modelling frameworks within the same matrix-based system. This helps students unify mean comparisons, factor effects, covariate adjustment, and interaction analysis under one common language.

Learning Objectives

By the end of this chapter, students should be able to:

- explain how one-way ANOVA is a special case of the linear model;
- formulate and interpret two-way ANOVA models with and without interaction;
- explain the role of a covariate in ANCOVA;
- distinguish between main effects and interaction effects in factorial models;
- interpret ANOVA and ANCOVA models using regression notation and software output;
- compare group means appropriately after adjusting for covariates.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on analysis of variance
 - analysis of covariance
 - linear model treatment of factor and covariate effects
- Montgomery, Peck, and Vining:
 - sections on qualitative predictors
 - ANOVA-style linear models
 - ANCOVA and adjusted comparisons

Why These Topics Belong Together

Students often first encounter ANOVA and regression as if they were separate techniques. But in fact, they are part of the same linear modelling framework. For example:

- one-way ANOVA compares group means;
- two-way ANOVA studies the effects of two factors and possibly their interaction;
- ANCOVA compares groups while adjusting for a continuous covariate.

All of these can be written as linear models using an appropriate design matrix. This week emphasizes that unification.

Review of the General Linear Model

Recall the linear model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

with

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta, \quad \text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

The content of the model depends on how we construct $\mathbf{X}$. If $\mathbf{X}$ contains continuous predictors, we obtain regression models. If $\mathbf{X}$ contains indicator variables for factor levels, we obtain ANOVA-type models. If $\mathbf{X}$ contains both factor indicators and continuous covariates, we obtain ANCOVA-type models.

7.1 Design, randomization, and the experimental unit

The **experimental unit** is the smallest unit independently assigned a treatment. Replication means independently treated units; repeated measurements or multiple subsamples from one unit do not create additional treatment replicates. Identify the unit before constructing the data table or choosing error degrees of freedom.

Randomization allocates treatments by a random mechanism, helping separate treatment assignment from potential outcomes. **Blocking** groups similar units using information available before treatment and randomizes within those groups. Random sampling supports generalization to a population; random treatment assignment supports a causal comparison in the experiment. Neither alone guarantees both claims.

! Avoid pseudoreplication

If one classroom receives A and another receives B, measuring 30 students per classroom does not produce 30 independently randomized classrooms per treatment. A model that treats all 60 measurements as independent treatment replicates can severely understate uncertainty. The design determines the relevant replication.

Example 7.1 (A Randomized Complete Block Design). Suppose five production batches each provide three comparable units. Within every batch, randomly assign one unit to each of three treatments. There are 15 experimental units and five replications per treatment. An additive fixed-block model is

$$Y_{bj} = \mu + \gamma_b + \tau_j + \varepsilon_{bj}, \quad \varepsilon_{bj} \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

Comparing $Y \sim \text{block}$ with $Y \sim \text{block} + \text{treatment}$ tests treatment differences after accounting for batch variation. There are $15 - (1 + 4 + 2) = 8$ residual degrees of freedom. With one unit in each block-by-treatment cell, treatment-by-block interaction cannot be separated from error; the additive interpretation is an assumption.

```
set.seed(8561)
blocked <- data.frame(block = factor(rep(1:5, each = 3)),
                      unit = rep(1:3, 5))
# Randomize treatment labels among the three units in each block.
blocked$treatment <- factor(
  unlist(lapply(1:5, function(b) sample(c("A", "B", "C")))),
  levels = c("A", "B", "C"))
block_effect <- c(-3, -1, 0, 2, 4)
treatment_effect <- c(A = 0, B = 1.5, C = 3)
blocked$y <- 10 + block_effect[as.integer(blocked$block)] +
  treatment_effect[blocked$treatment] + rnorm(nrow(blocked), sd = 0.8)
fit_block_only <- lm(y ~ block, data = blocked)
fit_block_treatment <- lm(y ~ block + treatment, data = blocked)
anova(fit_block_only, fit_block_treatment)
```

Analysis of Variance Table

Model 1: $y \sim \text{block}$

Model 2: $y \sim \text{block} + \text{treatment}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	10	35.101				
2	8	2.873	2	32.227	44.869	4.489e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
with(blocked, interaction.plot(treatment, block, y, type = "b", pch = 16,
                              col = hcl.colors(5, "Dark 3"), lwd = 1.5,
                              xlab = "Treatment", ylab = "Response",
                              trace.label = "Block"))
```

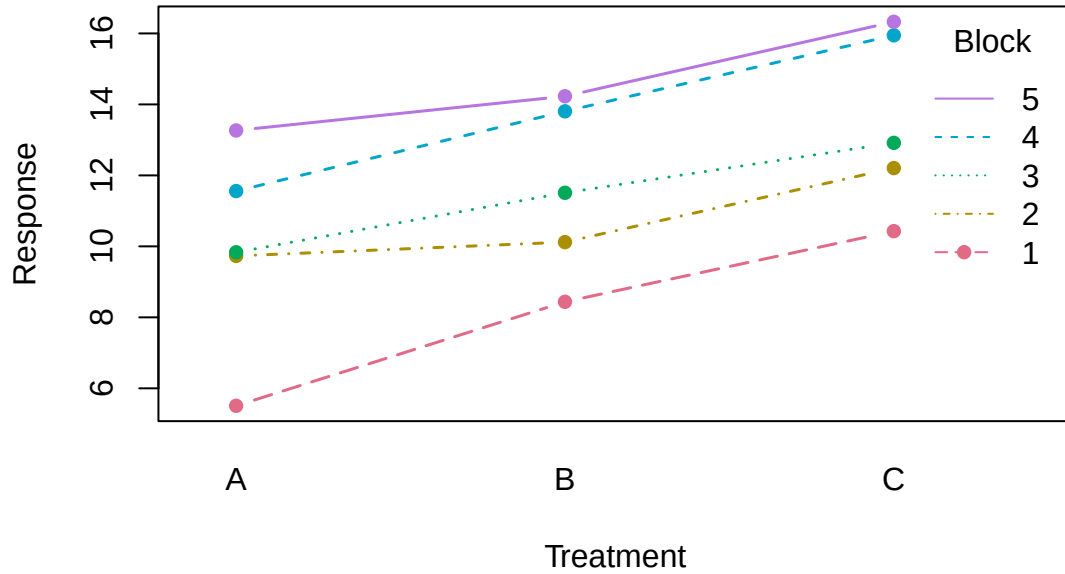


Figure 7.1: Simulated responses within five blocks. Comparing treatments within the same block avoids attributing the large between-batch shifts to treatment. The connecting lines guide comparisons; the model does not assert a linear treatment scale.

A randomized complete design without blocks uses treatment alone. The same regression machinery handles either design, but the design and scientific question decide which comparison is meaningful.

7.2 One-Way ANOVA

Suppose observations are divided into g groups, and we want to compare their means. One-way ANOVA can be written as

$$Y_{ij} = \mu_i + \varepsilon_{ij}, \quad i = 1, \dots, g,$$

where:

- μ_i is the mean of group i ;
- ε_{ij} are independent errors with mean zero and common variance σ^2 .

This is sometimes called the cell-means model.

Alternative Parameterization of One-Way ANOVA

Another common parameterization is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, \dots, g,$$

where:

- μ is an overall baseline level;
- τ_i is the effect of group i .

To identify the model, we usually impose a constraint such as

$$\sum_{i=1}^g \tau_i = 0.$$

Different coding schemes lead to different coefficient interpretations, but the fitted group means are the same.

Null Hypothesis in One-Way ANOVA

The usual one-way ANOVA null hypothesis is

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_g.$$

Equivalently, in the effect parameterization, this is

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_g = 0$$

subject to the model constraints. This is a general linear hypothesis and can be tested by an F test.

One-Way ANOVA as Regression With Indicators

Suppose there are three groups: A, B, and C. Using treatment coding with group A as the reference group, we may write

$$Y_i = \beta_0 + \beta_1 z_{1i} + \beta_2 z_{2i} + \varepsilon_i,$$

where:

- $z_{1i} = 1$ if observation i is in group B and 0 otherwise;

- $z_{2i} = 1$ if observation i is in group C and 0 otherwise.

Then:

- mean of group A: β_0 ;
- mean of group B: $\beta_0 + \beta_1$;
- mean of group C: $\beta_0 + \beta_2$.

Thus one-way ANOVA is simply a regression model with categorical predictors.

ANOVA Table Interpretation

For one-way ANOVA, the total variation is decomposed into:

- variation between groups;
- variation within groups.

The standard ANOVA table compares the mean square for groups to the mean square for error. This is an F test for whether the group means are all equal. Within the linear model framework, this is just a comparison of a reduced intercept-only model and a fuller model that includes group indicators.

7.3 Two-Way ANOVA

Now suppose there are two factors:

- factor A with levels $i = 1, \dots, a$;
- factor B with levels $j = 1, \dots, b$.

A general two-way ANOVA model with interaction is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}.$$

Here:

- α_i is the effect of level i of factor A;
- β_j is the effect of level j of factor B;
- $(\alpha\beta)_{ij}$ is the interaction effect for the combination of levels (i, j) .

Main Effects in Two-Way ANOVA

A **main effect** describes how the mean response changes across levels of one factor, averaging over the levels of the other factor, when the interaction is absent or appropriately interpreted. For example:

- factor A main effect asks whether changing the level of A shifts the mean response;
- factor B main effect asks the analogous question for B.

However, if interaction is strong, main effects must be interpreted with care.

Interaction in Two-Way ANOVA

Interaction means that the effect of one factor depends on the level of the other factor. If there is no interaction, the mean structure is additive:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}.$$

If interaction is present, the difference among levels of factor A changes across levels of factor B, or vice versa. Graphically, in an interaction plot, no interaction corresponds roughly to parallel profiles.

Null Hypotheses in Two-Way ANOVA

Common hypotheses include:

- no main effect of factor A;
- no main effect of factor B;
- no interaction between A and B.

Each of these is a general linear hypothesis and can be tested with an F statistic by comparing appropriate reduced and full models.

Importance of Testing Interaction First

In practice, it is often wise to assess the interaction term before making strong statements about main effects. Why? Because if interaction is present, the effect of factor A is not the same at all levels of factor B, and vice versa. So a marginal main-effect interpretation may be misleading.

Balanced and Unbalanced Designs

A design is **balanced** if each factor combination has the same number of observations. Balanced designs have many nice algebraic properties:

- sums of squares decompose cleanly;
- effects are more orthogonal;
- interpretations are often simpler.

In unbalanced designs, sequential sums of squares can depend on term order, and software defaults may test different hypotheses. A fixed estimable hypothesis has the same inference under equivalent full-rank recodings. State the comparison of means first, then choose the model comparison that tests it. Students should be aware that real data are often unbalanced.

Two-Way ANOVA as Regression

Just as in one-way ANOVA, two-way ANOVA can be represented using indicator variables. The design matrix can include:

- indicators for levels of factor A;
- indicators for levels of factor B;
- products of indicators for interaction terms.

So two-way ANOVA is also just a regression model with categorical predictors and possible interactions.

7.4 Analysis of Covariance

Analysis of covariance, or **ANCOVA**, combines:

- one or more categorical predictors;
- one or more continuous covariates.

A simple ANCOVA model is

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i,$$

where:

- x_i is a continuous covariate;
- z_i is a group indicator.

This model compares groups while adjusting for the covariate.

Why ANCOVA Is Useful

Suppose two groups differ in an outcome, but also differ in a background variable related to the outcome. If we compare the raw group means, the comparison may be confounded by the background variable. ANCOVA adjusts for the covariate so that the group comparison is made at a common covariate level. This often improves both fairness of comparison and precision of inference.

Interpreting an ANCOVA Model

Suppose

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i,$$

with $z_i \in \{0, 1\}$. Then:

- β_1 is the change in the mean response associated with a one-unit increase in x , holding group fixed;
- β_2 is the difference between groups, holding the covariate fixed.

Thus ANCOVA is a regression model in which one of the predictors is a factor.

Parallel Slopes Assumption

A standard ANCOVA model without interaction assumes the relationship between the response and the covariate has the same slope in all groups. That is, if

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \varepsilon_i,$$

then both groups share the same slope β_1 . This is often called the **parallel slopes assumption**.

ANCOVA With Interaction

If we want to allow the covariate effect to differ by group, we add an interaction:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 x_i z_i + \varepsilon_i.$$

Now the slope in x depends on the group. This model should be considered when the relationship between the response and the covariate appears different across groups.

Why the Parallel Slopes Assumption Matters

If the slopes are not truly parallel but we fit a common-slope ANCOVA model, then the adjusted group comparison may be misleading. So a practical workflow is often:

- fit the interaction model first;
- assess whether the interaction is needed;
- if the interaction is not important, use the simpler common-slope ANCOVA model.

Adjusted Means

A major idea in ANCOVA is the comparison of **adjusted means**. These are group means adjusted to a common covariate value, often the overall mean of the covariate. Adjusted means are useful because they allow group comparison after accounting for systematic covariate differences.

One-Way ANOVA, Two-Way ANOVA, and ANCOVA as Nested Models

These topics can all be understood through model comparison. Examples:

- one-way ANOVA compares the intercept-only model to a model with group effects;
- two-way ANOVA compares models with and without main effects or interaction;
- ANCOVA compares models with and without the covariate, group effect, or interaction.

This nested-model view links directly back to the general linear hypothesis framework.

Post Hoc Comparisons

Plan scientifically meaningful comparisons before inspecting the results when possible. An omnibus F test and a comparison of particular groups answer different questions; omnibus significance does not justify an arbitrary family of unadjusted pairwise tests. Examples include:

- all pairwise comparisons;
- treatment versus control;
- preplanned contrasts.

These are again linear functions of model parameters. Use a stated comparison family: for example, Tukey intervals for all pairwise mean differences, or Holm-adjusted p-values for a specified family of tests. Planned comparisons also require multiplicity control when the inferential claim concerns a family. These comparisons use the contrast framework from the preceding chapter.

Interpretation and Caution

In factor models, coefficient interpretation depends on coding. For example, treatment coding, sum-to-zero coding, and cell-means coding all yield different raw coefficients. Therefore students should focus on:

- fitted means;
- differences among means;
- interactions;
- clearly defined hypotheses.

These are more stable and meaningful than memorizing one coding-specific coefficient interpretation.

Worked Example With One-Way ANOVA

Suppose three teaching methods are compared. A one-way ANOVA model asks whether the mean outcome differs across the three methods. This can be written either as a group-means model or as a regression with two indicators and an intercept. The overall F test asks whether all three group means are equal. If the null is rejected, further contrasts can be used to compare specific methods.

Worked Example With Two-Way ANOVA

Suppose yield is measured under:

- three fertilizer types;
- two irrigation levels.

A two-way ANOVA model can assess:

- whether fertilizer matters;
- whether irrigation matters;
- whether the effect of fertilizer depends on irrigation.

This is a direct example of main effects and interaction in a factorial design.

Worked Example With ANCOVA

Suppose two treatment groups are compared on a final score, and baseline score is available as a covariate. A simple ANCOVA model adjusts the treatment comparison for baseline. This often provides a more precise and fairer comparison than comparing final means alone.

7.5 R Demonstration

Simulate one-way ANOVA data

```
set.seed(8561)
group <- factor(rep(c("A", "B", "C"), each = 12))
mu <- c(A = 10, B = 13, C = 15)
y <- mu[group] + rnorm(length(group), sd = 2)

dat1 <- data.frame(y = y, group = group)
fit1 <- lm(y ~ group, data = dat1)

summary(fit1)
```

```

Call:
lm(formula = y ~ group, data = dat1)

Residuals:
    Min       1Q   Median       3Q      Max
-4.8920 -1.1980  0.1736  1.3460  4.7288

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.4231     0.7373   14.136 1.48e-15 ***
groupB        2.8863     1.0428    2.768 0.009179 **
groupC        3.7914     1.0428    3.636 0.000933 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.554 on 33 degrees of freedom
Multiple R-squared:  0.3041,    Adjusted R-squared:  0.262
F-statistic: 7.212 on 2 and 33 DF,  p-value: 0.002521

```

```
anova(fit1)
```

Analysis of Variance Table

```

Response: y
      Df Sum Sq Mean Sq F value    Pr(>F)
group    2  94.098   47.049   7.2117 0.002521 **
Residuals 33 215.291    6.524
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Group means and model matrix

```
tapply(dat1$y, dat1$group, mean)
```

```

      A      B      C
10.42315 13.30940 14.21455

```

```
model.matrix(fit1)[1:10, ]
```

	(Intercept)	groupB	groupC
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
5	1	0	0
6	1	0	0
7	1	0	0
8	1	0	0
9	1	0	0
10	1	0	0

Pairwise comparisons through linear contrasts

```

b <- coef(fit1)
V <- vcov(fit1)

# Under treatment coding with A as reference:
# mean_A = beta0
# mean_B = beta0 + beta_groupB
# mean_C = beta0 + beta_groupC

# Compare B and C
a <- c(0, 1, -1)
est <- sum(a * b)
se <- sqrt(t(a) %*% V %*% a)
t_stat <- est / se
df <- df.residual(fit1)
p_val <- 2 * pt(abs(t_stat), df = df, lower.tail = FALSE)

c(estimate = est, se = se, t = t_stat, p_value = p_val)

```

estimate	se	t	p_value
-0.9051546	1.0427509	-0.8680449	0.3916388

R Demonstration With Two-Way ANOVA

Simulate factorial data

```

set.seed(8561)
A <- factor(rep(c("Low", "Medium", "High"), each = 16),
            levels = c("Low", "Medium", "High"))

```

```

B <- factor(rep(rep(c("I1", "I2"), each = 8), times = 3))

mean_table <- matrix(c(
  10, 12,
  13, 15,
  16, 20
), nrow = 3, byrow = TRUE)

mu2 <- numeric(length(A))
for (i in seq_along(mu2)) {
  mu2[i] <- mean_table[which(levels(A) == A[i]), which(levels(B) == B[i])]
}

y2 <- mu2 + rnorm(length(mu2), sd = 1.8)
dat2 <- data.frame(y = y2, A = A, B = B)

```

Fit additive and interaction models

```

fit2_add <- lm(y ~ A + B, data = dat2)
fit2_int <- lm(y ~ A * B, data = dat2)

summary(fit2_add)

```

Call:

```
lm(formula = y ~ A + B, data = dat2)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.0540	-0.9970	0.0499	1.2429	4.4947

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.3493	0.6189	16.722	< 2e-16 ***
AMedium	2.3237	0.7580	3.066	0.003705 **
AHigh	5.7129	0.7580	7.537	1.88e-09 ***
BI2	2.2215	0.6189	3.589	0.000829 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.144 on 44 degrees of freedom

Multiple R-squared: 0.6152, Adjusted R-squared: 0.589
F-statistic: 23.45 on 3 and 44 DF, p-value: 3.208e-09

```
summary(fit2_int)
```

Call:

```
lm(formula = y ~ A * B, data = dat2)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4.6795	-1.0816	0.0961	1.2392	4.0989

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.3280	0.7666	13.472	< 2e-16 ***
AMedium	2.7409	1.0842	2.528	0.0153 *
AHigh	5.3598	1.0842	4.944	1.28e-05 ***
BI2	2.2641	1.0842	2.088	0.0429 *
AMedium:BI2	-0.8343	1.5333	-0.544	0.5892
AHigh:BI2	0.7063	1.5333	0.461	0.6474

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.168 on 42 degrees of freedom

Multiple R-squared: 0.6243, Adjusted R-squared: 0.5795

F-statistic: 13.96 on 5 and 42 DF, p-value: 4.793e-08

```
anova(fit2_add, fit2_int)
```

Analysis of Variance Table

Model 1: y ~ A + B

Model 2: y ~ A * B

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	44	202.24				
2	42	197.48	2	4.758	0.506	0.6066

```
anova(fit2_int)
```

Analysis of Variance Table

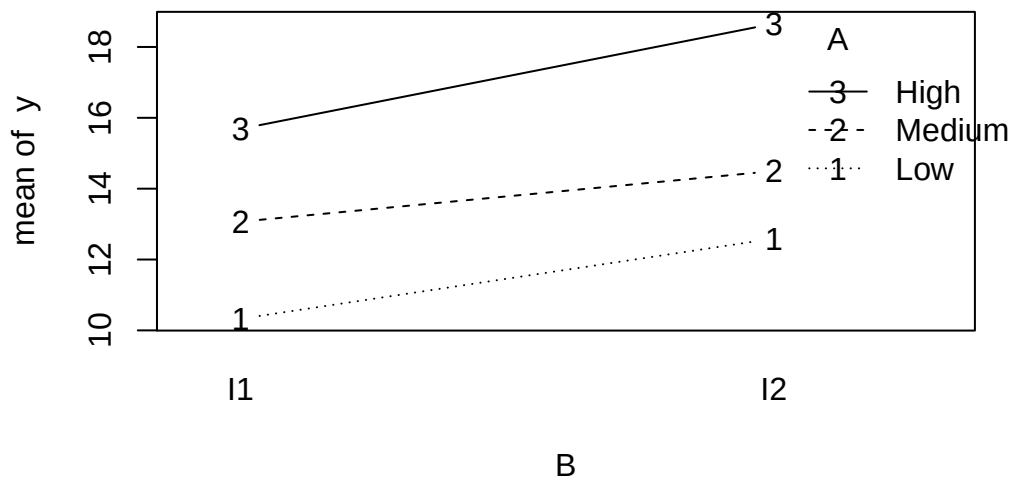
Response: y

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
A	2	264.129	132.065	28.087	1.804e-08 ***
B	1	59.220	59.220	12.595	0.0009681 ***
A:B	2	4.758	2.379	0.506	0.6065586
Residuals	42	197.484	4.702		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Interaction plot

```
with(dat2, interaction.plot(x.factor = B, trace.factor = A, response = y,  
                           fun = mean, type = "b", legend = TRUE))
```



R Demonstration With ANCOVA

Simulate ANCOVA-style data

```
set.seed(8561)  
n <- 80  
group3 <- factor(rep(c("Control", "Treatment"), each = n / 2))
```

```
x <- rnorm(n, mean = 50, sd = 10)
y3 <- 20 + 0.7 * x + ifelse(group3 == "Treatment", 4, 0) + rnorm(n, sd = 4)

dat3 <- data.frame(y = y3, x = x, group = group3)
fit3 <- lm(y ~ x + group, data = dat3)
fit3_int <- lm(y ~ x * group, data = dat3)

summary(fit3)
```

Call:

```
lm(formula = y ~ x + group, data = dat3)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.1507	-2.5151	-0.1405	2.1190	9.7267

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	20.6057	2.0562	10.021	1.32e-15 ***
x	0.6784	0.0390	17.395	< 2e-16 ***
groupTreatment	4.1699	0.8934	4.667	1.27e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.975 on 77 degrees of freedom

Multiple R-squared: 0.8016, Adjusted R-squared: 0.7964

F-statistic: 155.5 on 2 and 77 DF, p-value: < 2.2e-16

```
summary(fit3_int)
```

Call:

```
lm(formula = y ~ x * group, data = dat3)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.8391	-2.7615	-0.1328	2.4626	9.8774

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```

(Intercept)      22.06211      2.66126      8.290 3.06e-12 ***
x                 0.64943      0.05151     12.607 < 2e-16 ***
groupTreatment    0.83332      3.96347      0.210      0.834
x:groupTreatment  0.06830      0.07903      0.864      0.390
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.981 on 76 degrees of freedom
Multiple R-squared:  0.8035,    Adjusted R-squared:  0.7958
F-statistic: 103.6 on 3 and 76 DF,  p-value: < 2.2e-16

```

```
anova(fit3, fit3_int)
```

Analysis of Variance Table

```

Model 1: y ~ x + group
Model 2: y ~ x * group
      Res.Df  RSS Df Sum of Sq    F Pr(>F)
1         77 1216.6
2         76 1204.7  1    11.837 0.7468 0.3902

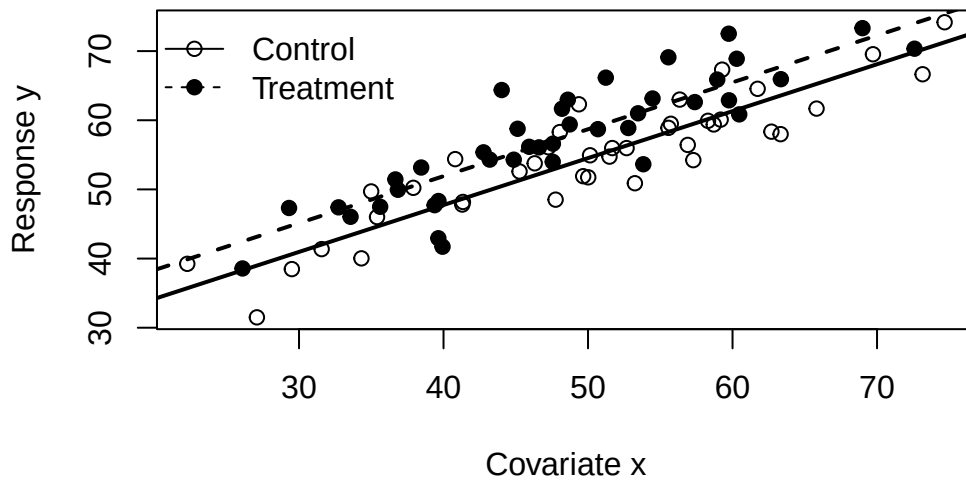
```

Plot ANCOVA fit

```

plot(dat3$x, dat3$y,
     pch = ifelse(dat3$group == "Control", 1, 19),
     xlab = "Covariate x",
     ylab = "Response y")
abline(a = coef(fit3)[1], b = coef(fit3)[2], lwd = 2)
abline(a = coef(fit3)[1] + coef(fit3)[3], b = coef(fit3)[2],
      lwd = 2, lty = 2)
legend("topleft",
      legend = c("Control", "Treatment"),
      pch = c(1, 19),
      lty = c(1, 2),
      bty = "n")

```



Interpreting Software Output

In R:

- `lm(y ~ group)` fits a one-way ANOVA model;
- `lm(y ~ A * B)` fits a two-way ANOVA model with interaction;
- `lm(y ~ x + group)` fits a common-slope ANCOVA model;
- `lm(y ~ x * group)` fits an ANCOVA model with group-specific slopes.

Students should recognize that these are all regression models differing only in the structure of the design matrix.

A Practical Workflow

A useful modelling workflow is:

- identify whether predictors are factors, covariates, or both;
- decide whether interactions are scientifically plausible;
- fit the more complete model when appropriate;
- test whether interaction is needed;
- interpret fitted means or adjusted means rather than raw coding-specific coefficients;
- use contrasts for focused follow-up questions.

In-Class Discussion Questions

1. Why is one-way ANOVA just a regression model with indicator variables?

2. Why should interaction be considered before interpreting main effects in two-way ANOVA?
3. What does ANCOVA adjust for that ordinary group-mean comparison does not?
4. Why can coefficient interpretation depend on coding while fitted means do not?

7.6 Practice Problems

Conceptual

1. Explain the difference between a one-way ANOVA model and an ANCOVA model.
2. Explain what interaction means in a two-way ANOVA setting.
3. Explain why adjusted means are useful in ANCOVA.

Computational

Suppose there are three groups A, B, and C, and treatment coding is used with A as the reference group. The fitted model is

$$\hat{Y} = 8 + 2z_1 + 5z_2.$$

1. What is the fitted mean for group A?
2. What is the fitted mean for group B?
3. What is the fitted mean for group C?
4. What is the estimated difference between groups B and C?

Now consider the ANCOVA model

$$\hat{Y} = 12 + 0.6x + 3z,$$

where $z = 1$ for treatment and $z = 0$ for control.

1. Interpret the coefficient of x .
2. Interpret the coefficient of z .
3. Compute the fitted mean when $x = 10$ for control and treatment.

Model-Comparison Problem

You fit the following two models:

- Model 1: $y \sim x + \text{group}$
- Model 2: $y \sim x * \text{group}$

1. What extra term is in Model 2?
2. What scientific question does the comparison test?
3. Why would a significant interaction affect the interpretation of the group effect?

Suggested Homework

Complete the following tasks:

- fit a one-way ANOVA model and interpret the overall F test;
- carry out at least two follow-up contrasts among group means;
- fit a two-way ANOVA model with interaction and interpret the interaction term;
- fit an ANCOVA model and explain the meaning of adjusted group comparison;
- compare an ANCOVA model with and without interaction and explain which one you prefer.

Summary

In this chapter, we studied one-way ANOVA, two-way ANOVA, and ANCOVA within the general linear model framework. We emphasized that:

- ANOVA models are regression models with categorical predictors;
- two-way ANOVA introduces main effects and interaction;
- ANCOVA combines factor effects with continuous covariate adjustment;
- fitted means, adjusted means, and contrasts are often more meaningful than raw coding-specific coefficients;
- model comparison through general linear hypotheses unifies all these settings.

8 Diagnostics and Model Adequacy

In this chapter, we study how to assess whether a fitted linear regression model is adequate. After learning estimation, inference, ANOVA, and interpretation, we now turn to model checking. The main tools are residuals, diagnostic plots, and numerical measures that help identify nonlinearity, unequal variance, outliers, and influential observations.

Learning Objectives

By the end of this chapter, students should be able to:

- explain why model diagnostics are necessary after fitting a regression model;
- define raw, standardized, and studentized residuals;
- interpret residual plots for common types of model failure;
- distinguish between outliers in the response direction and influential observations in the design space;
- use leverage, Cook's distance, and related diagnostics;
- assess model adequacy using graphical and numerical tools.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on residual analysis
 - diagnostics for regression models
 - influence and unusual observations
- Montgomery, Peck, and Vining:
 - sections on residuals
 - diagnostic plots
 - outliers, leverage, and influence

Why Diagnostics Matter

A fitted regression model may look statistically significant and still be inappropriate. For example:

- the relationship may not be linear;
- the error variance may not be constant;

- the error distribution may be strongly non-normal;
- a small number of unusual observations may dominate the fit.

So regression analysis does not end when we obtain estimates and p -values. We must also ask whether the model assumptions are reasonable and whether the fit is being driven by problematic observations.

Review of the Linear Model Assumptions

Recall the classical linear model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

with assumptions such as

$$\mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

Under the normal linear model, we also assume

$$\varepsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

These assumptions support:

- unbiasedness of OLS;
- standard error formulas;
- t and F inference;
- prediction intervals.

Diagnostics help us investigate whether these assumptions seem plausible for the observed data.

8.1 Residuals

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\beta},$$

and the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

The i th residual is

$$e_i = Y_i - \hat{Y}_i.$$

A residual measures how far an observed response is from the fitted value at that observation. Residuals are the basic raw material of regression diagnostics.

! Residuals are not the true errors

Residuals are not the same as the true errors. The true model error is

$$\varepsilon_i = Y_i - \mathbb{E}[Y_i \mid \mathbf{x}_i],$$

whereas the residual is

$$e_i = Y_i - \hat{Y}_i.$$

Residuals are observable, but errors are not. Diagnostics therefore use residuals as proxies for model errors.

Properties of Residuals

From least squares geometry, we know that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

If the model contains an intercept, then

$$\sum_{i=1}^n e_i = 0.$$

Thus residuals are constrained and are not independent. This is one reason why it is useful to standardize them before interpretation.

Residual Variance and Leverage

Recall the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top.$$

Its diagonal entries h_{ii} are called **leverages**. Under the standard model,

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii}).$$

So residuals do not all have the same variance. Observations with high leverage tend to have smaller residual variance. This is why raw residuals alone can be misleading.

Standardized Residuals

A common adjustment is to divide each residual by its estimated standard deviation. The **standardized residual** is

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}.$$

This puts residuals on a comparable scale across observations. Large absolute values of r_i may indicate unusual observations in the response direction.

Studentized Residuals

A more refined version is the **studentized residual**. One version uses the variance estimate computed from the full model. Another version uses the variance estimate obtained after deleting the i th observation. The externally studentized residual is often written as

$$t_i = \frac{e_i}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}},$$

where $\hat{\sigma}_{(i)}^2$ is the residual variance estimate from the model fit without observation i . These are especially useful for outlier detection.

8.2 Diagnostic Plots

One of the most important diagnostic plots is the plot of residuals versus fitted values. This plot helps detect:

- nonlinearity;
- unequal variance;
- outliers;
- missing structure.

A good residual plot usually shows points randomly scattered around zero with roughly constant spread. Patterns are warning signs.

Interpreting Common Patterns

If the residual plot shows a curved pattern, that suggests the mean function may not be adequately linear. If the spread of residuals increases or decreases with the fitted values, that suggests heteroscedasticity, meaning nonconstant error variance. If a few points are isolated far

from the rest, they may be outliers or influential observations. Thus the residual plot is often the first and most important diagnostic tool.

Residuals Versus Individual Predictors

It is often helpful to plot residuals against each predictor separately. A residual-versus-predictor plot can reveal:

- nonlinearity in that predictor;
- different spread across ranges of the predictor;
- group-specific patterns;
- possible interactions not included in the model.

These plots are often more informative than a single omnibus diagnostic.

Normal Q-Q Plot

The normal Q-Q plot is used to assess whether the residual distribution is approximately normal. If the normality assumption is reasonable, the residual points should fall approximately along a straight line. Departures from linearity suggest:

- skewness;
- heavy tails;
- light tails;
- extreme outliers.

Normality matters most when sample size is small and exact t and F inference is important.

Histogram of Residuals

A histogram of residuals can also be useful, though it is usually less informative than a Q-Q plot. It may reveal:

- skewness;
- multimodality;
- heavy tails;
- extreme asymmetry.

However, the histogram depends strongly on bin choices, so it is usually used as a supplementary plot rather than the main normality diagnostic.

Scale-Location Plot

A scale-location plot often displays $\sqrt{|r_i|}$ against fitted values. This is another way to assess whether the variance is approximately constant across the fitted range. A roughly horizontal band is desirable. A systematic increase or decrease suggests heteroscedasticity.

Outliers

An **outlier** is an observation whose response value is unusual relative to the fitted model. In regression, an observation can be outlying in the response direction even if its predictor values are not unusual. Large residuals or large studentized residuals often indicate potential outliers. However, not every outlier is influential.

Leverage

Leverage measures how unusual an observation is in the predictor space. The leverage of observation i is h_{ii} . High leverage points are far from the center of the predictor cloud in a geometric sense. These points have greater potential to affect the fitted regression line or plane. A rough rule of thumb is that leverage values substantially larger than

$$\frac{2p}{n} \quad \text{or sometimes} \quad \frac{3p}{n}$$

may deserve attention, where p is the number of parameters including the intercept. These are only rough guidelines, not formal cutoffs.

Outlier Versus High-Leverage Point

It is important to distinguish between:

- a point with a large residual;
- a point with high leverage.

A point can have:

- low leverage and large residual;
- high leverage and small residual;
- both high leverage and large residual.

The last case is often the most concerning because such a point may strongly influence the fitted model. For additional coefficient-specific and partial-regression diagnostics, see [Optional linear-model extensions](#).

8.3 Influence

An observation is **influential** if removing it changes the fitted model noticeably. Influence depends on both:

- how unusual the response is;
- how unusual the predictor values are.

So influential points often combine high leverage with a sizable residual. Influence is not the same as being an outlier.

Cook's Distance

One of the most widely used influence measures is **Cook's distance**. Cook's distance for observation i measures how much the fitted values change when observation i is removed. A common formula is

$$D_i = \frac{e_i^2}{p\hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}.$$

Large values of D_i indicate potentially influential observations. Plots of Cook's distance help identify observations that deserve closer inspection.

Diagnosing Nonlinearity

If the mean function is nonlinear but we fit a linear model, residual plots may show curvature. Possible remedies include:

- adding polynomial terms;
- applying transformations;
- including interactions;
- using a different modelling framework.

Diagnostics should not be viewed only as fault-finding. They also guide model improvement.

Diagnosing Heteroscedasticity

If the error variance is not constant, residual plots may show a funnel shape or other changing spread. Possible remedies include:

- transforming the response;
- weighted least squares;
- modelling the variance structure explicitly;
- using heteroscedasticity-robust standard errors in some contexts.

At this stage, the main goal is to recognize the pattern and understand its consequences.

Diagnosing Non-Normality

If residuals are strongly non-normal, this may affect exact small-sample inference. Possible causes include:

- skewed responses;
- heavy-tailed errors;
- outliers;
- omitted structure.

Possible remedies include:

- transformation;
- alternative modelling assumptions;
- robust procedures;
- careful interpretation if sample size is large and inference is approximately stable.

Diagnostics Are Contextual

There is no single diagnostic that automatically declares a model valid or invalid. Instead, diagnostics require judgment. Students should ask:

- Is the pattern strong or mild?
- Is it scientifically meaningful?
- Does one point dominate the fit?
- Would conclusions change if the model were modified?
- Is the issue important for the goal of the analysis: explanation, prediction, or inference?

Worked Example With an Outlying Observation

Consider the data $x = (1, 2, 3, 4, 5, 6, 7, 8)$, and $y = (2, 4, 5, 8, 10, 11, 13, 25)$. The final observation may look unusual because the response jumps upward relative to the earlier trend. If we fit a simple linear regression, we should examine:

- the scatterplot with fitted line;
- residuals versus fitted values;
- studentized residuals;
- leverage;
- Cook's distance.

This is a good example for discussing the difference between an outlier and an influential point.

8.4 R Demonstration

Fit a simple model

```
x <- 1:8
y <- c(2, 4, 5, 8, 10, 11, 13, 25)

dat <- data.frame(x = x, y = y)
fit <- lm(y ~ x, data = dat)
summary(fit)
```

Call:

```
lm(formula = y ~ x, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.4762	-1.5179	-0.5595	1.1488	5.8333

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.3571	2.4532	-0.961	0.37374
x	2.6905	0.4858	5.538	0.00146 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

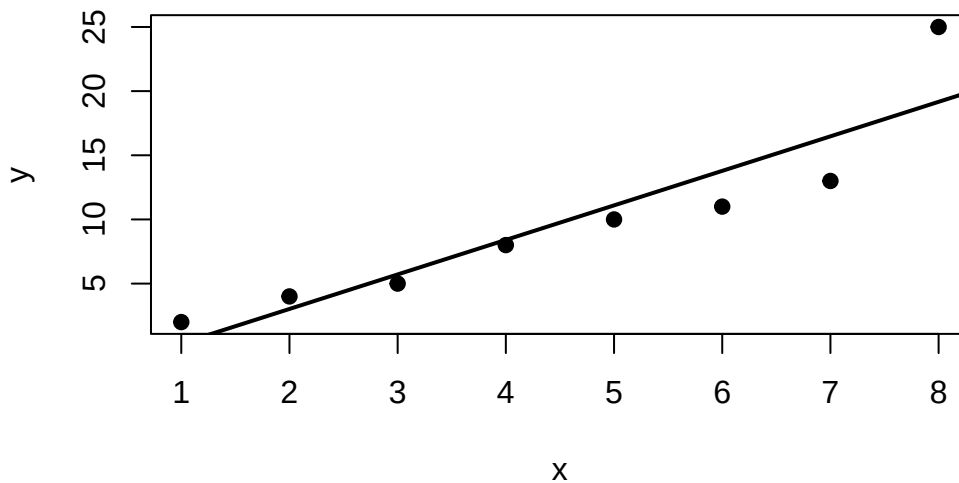
Residual standard error: 3.148 on 6 degrees of freedom

Multiple R-squared: 0.8364, Adjusted R-squared: 0.8091

F-statistic: 30.67 on 1 and 6 DF, p-value: 0.001462

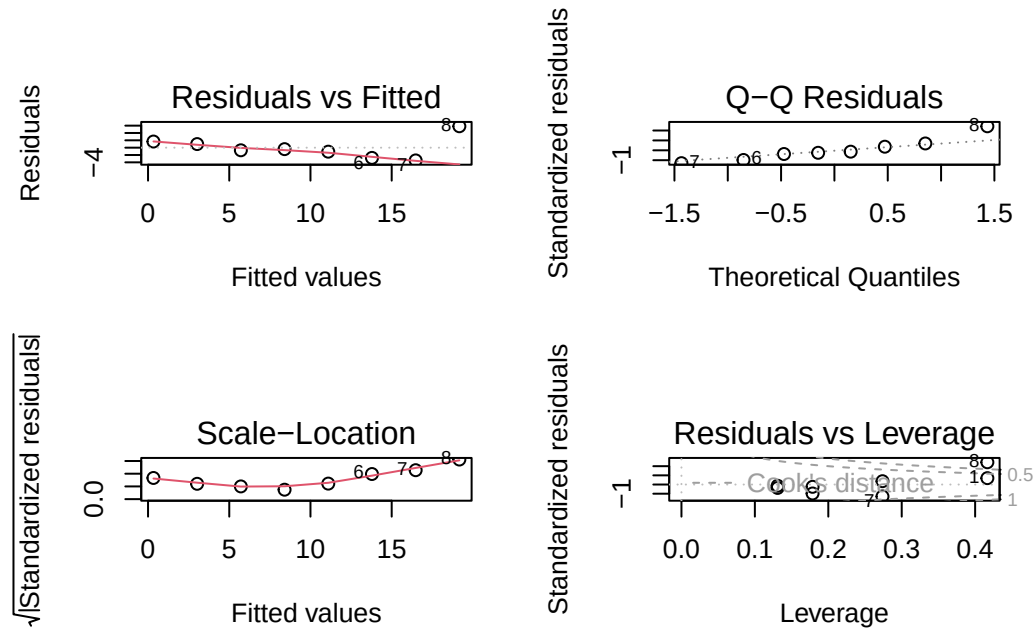
Scatterplot with fitted line

```
plot(dat$x, dat$y, pch = 19, xlab = "x", ylab = "y")  
abline(fit, lwd = 2)
```



Residual plots from base R

```
par(mfrow = c(2, 2))
plot(fit)
```



```
par(mfrow = c(1, 1))
```

Extract basic diagnostics numerically

```
data.frame(
  fitted = fitted(fit),
  residual = resid(fit),
  std_resid = rstandard(fit),
  stud_resid = rstudent(fit),
  leverage = hatvalues(fit),
  cooks_d = cooks.distance(fit)
)
```

	fitted	residual	std_resid	stud_resid	leverage	cooks_d
1	0.3333333	1.6666667	0.6930976	0.6596673	0.4166667	0.171565824
2	3.0238095	0.9761905	0.3638424	0.3358671	0.2738095	0.024957133
3	5.7142857	-0.7142857	-0.2503174	-0.2297101	0.1785714	0.006810742

```

4  8.4047619 -0.4047619 -0.1379056 -0.1260900 0.1309524 0.001432860
5 11.0952381 -1.0952381 -0.3731563 -0.3446665 0.1309524 0.010491110
6 13.7857143 -2.7857143 -0.9762380 -0.9716857 0.1785714 0.103591380
7 16.4761905 -3.4761905 -1.2956340 -1.3936654 0.2738095 0.316470107
8 19.1666667  5.8333333  2.4258417 15.9752413 0.4166667 2.101681345

```

Identify potentially unusual observations

```
which(abs(rstudent(fit)) > 2)
```

```

8
8

```

```
which(hatvalues(fit) > 2 * length(coef(fit)) / nrow(dat))
```

```
named integer(0)
```

```
which(cooks.distance(fit) > 4 / nrow(dat))
```

```

8
8

```

Example With Heteroscedasticity-Like Pattern

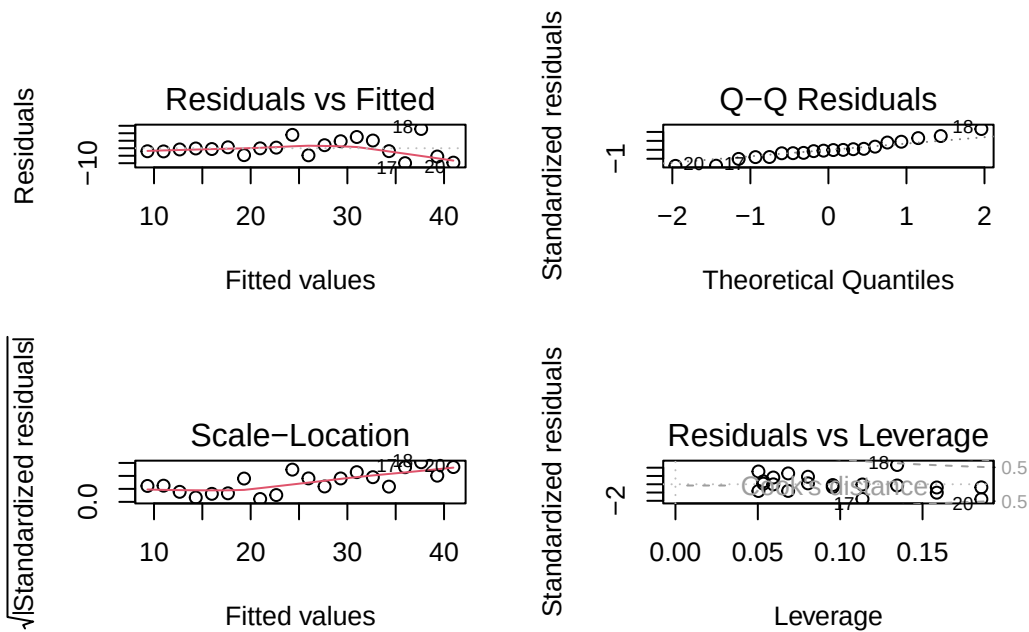
```

set.seed(8561)
x2 <- seq(1, 20, by = 1)
y2 <- 5 + 2 * x2 + rnorm(length(x2), sd = x2 / 3)

dat2 <- data.frame(x = x2, y = y2)
fit2 <- lm(y ~ x, data = dat2)

par(mfrow = c(2, 2))
plot(fit2)

```



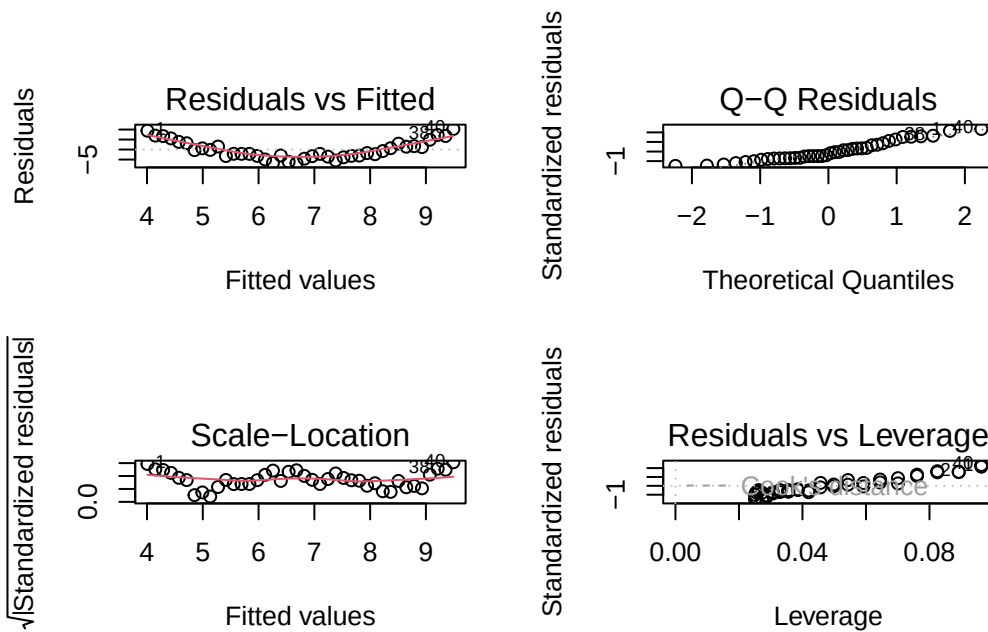
```
par(mfrow = c(1, 1))
```

Example With Curvature

```
set.seed(8561)
x3 <- seq(-3, 3, length.out = 40)
y3 <- 2 + x3 + 1.5 * x3^2 + rnorm(length(x3), sd = 1)

dat3 <- data.frame(x = x3, y = y3)
fit3 <- lm(y ~ x, data = dat3)

par(mfrow = c(2, 2))
plot(fit3)
```



```
par(mfrow = c(1, 1))
```

Comparing a Linear and Quadratic Fit

```
fit3_quad <- lm(y ~ x + I(x^2), data = dat3)
anova(fit3, fit3_quad)
```

Analysis of Variance Table

```
Model 1: y ~ x
Model 2: y ~ x + I(x^2)
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1     38 803.70
2     37  58.49  1    745.22 471.45 < 2.2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(fit3_quad)
```

Call:

```
lm(formula = y ~ x + I(x^2), data = dat3)

Residuals:
    Min       1Q   Median       3Q      Max
-2.66753 -0.78482  0.07729  0.64825  2.42384

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.91995     0.29834   6.435 1.62e-07 ***
x            0.91401     0.11194   8.165 8.45e-10 ***
I(x^2)       1.53156     0.07054  21.713 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.257 on 37 degrees of freedom
Multiple R-squared:  0.9357,    Adjusted R-squared:  0.9322
F-statistic: 269.1 on 2 and 37 DF,  p-value: < 2.2e-16
```

Interpreting Software Output

In practice, useful commands in R include:

- `plot(fit)` for the standard diagnostic panel;
- `resid(fit)` for residuals;
- `rstandard(fit)` for standardized residuals;
- `rstudent(fit)` for studentized residuals;
- `hatvalues(fit)` for leverages;
- `cooks.distance(fit)` for Cook's distances.

Students should learn to connect each numerical output to a concrete modelling question.

A Practical Diagnostic Workflow

A sensible basic workflow is:

- inspect the scatterplot and fitted line;
- examine the residuals-versus-fitted plot;
- check the normal Q-Q plot;
- inspect leverage and Cook's distance;
- investigate any unusual observations directly in the data;
- decide whether model revision is needed.

This sequence often works well for both simple and multiple regression.

What To Do After Finding a Problem

Finding a diagnostic issue does not mean we automatically delete observations. Instead, possible next steps include:

- checking for data entry or measurement errors;
- understanding whether the point represents a different regime;
- refitting with and without the point for sensitivity analysis;
- revising the model form;
- transforming variables;
- reporting the issue transparently.

Model criticism should be thoughtful, not mechanical.

In-Class Discussion Questions

1. Why are raw residuals not enough for identifying unusual observations?
2. Why can a high-leverage point have a small residual and still matter?
3. Why is Cook's distance more about influence than outlyingness?
4. What kinds of model failure are easiest to detect from a residual-versus-fitted plot?

8.5 Practice Problems

Conceptual

1. Explain the difference between an outlier, a high-leverage observation, and an influential observation.
2. Explain why residuals have unequal variance.
3. Explain why a normal Q-Q plot is useful even though residuals are not independent.

Computational

Suppose a regression model has $n = 25$ observations and $p = 4$ parameters.

1. Compute the rough leverage benchmark $2p/n$.
2. If one observation has $h_{ii} = 0.45$, explain whether this seems unusually large.
3. Suppose an observation has a large studentized residual but very small leverage. What kind of problem does this suggest?
4. Suppose another observation has large leverage but a very small residual. Why might it still deserve attention?

Model-Criticism Problem

A residual-versus-fitted plot shows a clear U-shape.

1. What model assumption is likely failing?
2. What changes to the model might you consider?

3. Why would simply reporting coefficient p -values be insufficient here?

Suggested Homework

Complete the following tasks:

- fit a regression model in R and produce the standard diagnostic plots;
- identify any observations with large studentized residuals, high leverage, or large Cook's distance;
- write a short interpretation of each diagnostic plot;
- modify a model to address one detected issue, such as curvature or unequal variance;
- compare the original and revised models.

Summary

In this chapter, we studied model diagnostics for linear regression. We focused on:

- residuals and standardized residuals;
- fitted-versus-residual plots;
- normal Q-Q plots;
- leverage and influence;
- Cook's distance and related ideas;
- practical judgment in assessing model adequacy.

These ideas are essential because regression analysis is not complete until the fitted model has been critically examined.

9 Transformations and Weighted Least Squares

In this chapter, we study what to do when the assumptions of the ordinary linear model are not adequate. Building on residual analysis and diagnostics from the preceding chapter, we now consider practical remedies for nonlinearity, nonconstant variance, and other violations of model assumptions. The emphasis is on understanding when transformations help, how weighted least squares works, and how to think systematically about model improvement.

Learning Objectives

By the end of this chapter, students should be able to:

- explain why transformations are used in regression analysis;
- distinguish between transforming the response and transforming predictors;
- interpret common response transformations such as logarithms and square roots;
- explain the basic idea of weighted least squares;
- derive the weighted least squares estimator;
- recognize situations where weighted least squares is appropriate;
- compare model improvement strategies based on diagnostics and scientific context.

Reading

Recommended reading for this chapter:

- Seber and Lee (2003) :
 - sections on transformations
 - weighted least squares
 - remedial measures for model inadequacy
- Montgomery et al. (2021) :
 - sections on transformations and model adequacy
 - weighted least squares
 - variance stabilization and practical remedies

Why Remedies Are Needed

Diagnostics often reveal that a fitted linear model is inadequate. Common problems include:

- curvature in the mean structure;
- nonconstant error variance;

- skewed response distributions;
- influential observations;
- omitted interactions or nonlinear effects.

When this happens, we should not stop at saying the model is flawed. We should also ask how the model might be improved. Some common remedies are:

- transforming the response;
- transforming predictors;
- adding polynomial or interaction terms;
- using weighted least squares;
- reconsidering the scope of the scientific question.

Review of the Ordinary Linear Model

The ordinary linear regression model is

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

The ordinary least squares estimator is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

provided that $\mathbf{X}$ has full column rank. When the variance is not constant, or when the mean structure is poorly represented by a linear form, OLS may still be computable, but its interpretation and inferential properties can become unsatisfactory.

9.1 Transformations in Regression

A transformation changes the scale on which a variable is analysed. For example:

- logarithm;
- square root;
- reciprocal;
- power transformation.

Transformations are often used for one or more of the following reasons:

- to improve linearity;
- to stabilize variance;
- to reduce skewness;
- to make the model more scientifically interpretable;
- to reduce the impact of extreme values.

Transformations should not be viewed as purely mechanical. They should be guided by diagnostics and by substantive understanding of the problem.

Transforming the Response

Suppose the original response is Y , but instead of modelling Y directly, we model $g(Y)$ for some transformation g . Then the model becomes

$$g(Y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i.$$

This can help if the original relationship between the mean and the predictors is nonlinear, or if the variability of Y changes with its level.

Log Transformation of the Response

A very common choice is the logarithm:

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

This is often useful when:

- the response is positive;
- the variance increases with the mean;
- the relationship is multiplicative rather than additive;
- the response distribution is right-skewed.

In this model, the interpretation of coefficients changes. If x increases by one unit, the conditional mean of $\log(Y)$ changes by β_1 . With conditional median zero errors, the median of Y is $\exp(\beta_0 + \beta_1 x)$, so its multiplicative change is exactly $\exp(\beta_1)$.

! Back-transformation changes the target

If $\varepsilon \mid x \sim N(0, \sigma^2)$, then

$$\text{median}(Y \mid x) = \exp(\beta_0 + \beta_1 x), \quad \mathbb{E}(Y \mid x) = \exp(\beta_0 + \beta_1 x + \sigma^2/2).$$

Exponentiating a fitted log response gives a fitted median under these assumptions, not the conditional mean on the original scale. The percentage change in the median is $100\{\exp(\beta_1) - 1\}\%$. The same mean ratio applies only when the error distribution is unchanged with x . A Gaussian model for $\log(Y)$ is also different from a Poisson GLM with a log link: the latter models $\log\{\mathbb{E}(Y \mid x)\}$ directly and allows zero counts.

Square-Root Transformation

Another common response transformation is the square root:

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

This is often used for count-like responses or responses whose variance tends to increase with the mean. Compared with the log transformation, the square-root transformation is milder and can still be used when the response includes zeros.

Reciprocal and Other Power Transformations

In some settings, transformations such as

$$\frac{1}{Y}, \quad Y^\lambda$$

or more general Box-Cox type power transformations may be useful. These transformations can sometimes help linearize relationships or stabilize variance, but they may make interpretation more difficult. So one should balance statistical convenience and interpretability.

Transforming Predictors

Instead of transforming the response, we may transform one or more predictors. For example, if the relationship between Y and x is curved, we might consider

- $\log(x)$;
- $\sqrt{x}$;
- x^2 or higher-order polynomial terms.

A model such as

$$Y_i = \beta_0 + \beta_1 \log(x_i) + \varepsilon_i$$

is still linear in the parameters, even though it is nonlinear in the original predictor. This is an important distinction: the model remains a linear model as long as it is linear in the coefficients.

Polynomial Terms as a Remedy

Suppose residual plots suggest curvature in the relationship between Y and x . A common remedy is to add a quadratic term:

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i.$$

This is still a linear model because it is linear in β_0 , β_1 , and β_2 . Polynomial terms often provide a more interpretable remedy than transforming the response, depending on the scientific context.

Choosing Between Transformations and Added Terms

Suppose diagnostics show nonlinearity. Then several remedies may be reasonable:

- transform the predictor;
- transform the response;
- add polynomial terms;
- include an interaction;
- restrict attention to a smaller range of the data.

There is often no single automatic answer. Choice depends on:

- what shape is suggested by diagnostics;
- which form is scientifically meaningful;
- how easy the resulting model is to explain;
- whether inference or prediction is the main goal.

Heteroscedasticity and Variance Stabilization

A common problem in regression is heteroscedasticity, meaning that $\text{Var}(\varepsilon_i)$ is not constant across observations. Residual plots may reveal a funnel shape or changing spread. Sometimes a response transformation can reduce this problem. For example:

- a log transformation often helps when the standard deviation is roughly proportional to the mean;
- a square-root transformation often helps for count-like data.

But in other cases, it is better to model the unequal variance directly. This leads to weighted least squares.

9.2 Weighted Least Squares

Suppose the observations still satisfy

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta,$$

but now the variance is

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{V},$$

where $\mathbf{V}$ is not the identity matrix. A particularly common case is when the errors are uncorrelated but have unequal variances:

$$\text{Var}(\varepsilon_i) = \sigma^2 v_i.$$

Then observations have different levels of reliability. Ordinary least squares treats all observations equally. Weighted least squares gives more weight to observations with smaller variance.

Basic Idea of Weights

If an observation has high variance, it contains less precise information about the mean. If an observation has low variance, it contains more precise information. So a sensible idea is to weight observations inversely to their variance. If

$$\text{Var}(\varepsilon_i) = \sigma^2 v_i,$$

then we often use weights

$$w_i = \frac{1}{v_i}.$$

Larger weights correspond to more precise observations.

Weighted Least Squares Criterion

In weighted least squares, we minimize

$$Q(\beta) = \sum_{i=1}^n w_i (Y_i - x_i^\top \beta)^2.$$

In matrix form, if

$$\mathbf{W} = \text{diag}(w_1, \dots, w_n),$$

then the criterion is

$$Q(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top \mathbf{W}(\mathbf{Y} - \mathbf{X}\beta).$$

This is the weighted analogue of the ordinary residual sum of squares.

Derivation of the Weighted Least Squares Estimator

Differentiate the weighted criterion with respect to β and set the derivative equal to zero. The weighted normal equations are

$$\mathbf{X}^\top \mathbf{W} \mathbf{X} \hat{\beta}_{WLS} = \mathbf{X}^\top \mathbf{W} \mathbf{Y}.$$

Assuming invertibility, the weighted least squares estimator is

$$\hat{\beta}_{WLS} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{Y}.$$

This has exactly the same structural form as OLS, but with $\mathbf{W}$ inserted to reflect unequal precision.

Transformation View of Weighted Least Squares

Weighted least squares can also be understood as transforming the model. If $\mathbf{W}^{1/2}$ is the diagonal matrix with entries $\sqrt{w_i}$, then multiplying the model by $\mathbf{W}^{1/2}$ gives

$$\mathbf{W}^{1/2} \mathbf{Y} = \mathbf{W}^{1/2} \mathbf{X} \beta + \mathbf{W}^{1/2} \varepsilon.$$

If the weights are chosen appropriately, the transformed errors have constant variance, and ordinary least squares on the transformed system becomes appropriate. This is a very useful conceptual link between WLS and OLS.

Interpreting Weighted Least Squares

Weighted least squares does not change the target mean model

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X} \beta.$$

Instead, it changes how observations are used in estimation. Observations with smaller variance have more influence on the fitted coefficients. Thus WLS is often best viewed as a remedy for unequal precision rather than a new mean model.

When Weighted Least Squares Is Appropriate

Weighted least squares is particularly useful when:

- variance is known up to a proportional constant;
- variance can be reasonably modelled as a function of the mean or a predictor;
- observations are averages based on different sample sizes;
- some observations are measured more precisely than others.

For example, if each response is an average based on m_i repeated measurements, then the variance may be proportional to $1/m_i$, so weights proportional to m_i are natural when the measurements contributing to each average are independent with a common variance. Correlated replicates require a different variance calculation. For estimated variance functions and feasible weighting, see [Optional GLS and covariance models](#).

Example of a Mean-Variance Relationship

Suppose residual plots suggest $\text{Var}(Y_i) \propto x_i^2$. Then it may be appropriate to use weights

$$w_i = \frac{1}{x_i^2}.$$

Alternatively, dividing the model through by x_i may also lead to a transformed model with approximately constant variance. This illustrates the close connection between variance modelling and transformation.

9.3 Response Transformation Versus WLS

Both response transformation and WLS can address heteroscedasticity, but they do so differently. A response transformation changes the scale of the response and often changes interpretation. Weighted least squares keeps the response on its original scale, but changes the estimation procedure. Which is better depends on the context. Students should compare:

- interpretability;
- adequacy of the residual plots after refitting;
- scientific plausibility of the variance model.

Practical Caveats

No remedy should be applied blindly. Questions to ask include:

- Does the transformed model make scientific sense?
- Does the transformed scale improve diagnostics meaningfully?
- Are the chosen weights justified?
- Does a more flexible mean model explain the apparent variance pattern?
- Would a different modelling framework be more appropriate?

Model repair is not merely technical. It should remain connected to the original scientific problem.

Worked Example With a Log Transformation

Suppose the response increases rapidly with the predictor and residual plots show larger spread for larger fitted values. A model for the original scale may look like

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

but the diagnostics suggest increasing variance and curvature. We might instead fit

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

If the transformed residual plots look more stable and more linear, then the log model may be preferable. Interpretation then becomes multiplicative rather than additive.

Worked Example With Weighted Least Squares

Suppose we observe responses with variance increasing as the predictor grows. If residual analysis suggests $\text{Var}(Y_i) \propto x_i^2$, then we may fit a weighted model with weights

$$w_i = \frac{1}{x_i^2}.$$

This gives lower weight to high-variance observations and can produce more stable coefficient estimates and more appropriate standard errors.

9.4 R Demonstration

Generate heteroscedastic data

```
set.seed(8561)
x <- seq(1, 20, by = 1)
y <- exp(1 + 0.12 * x + rnorm(length(x), sd = 0.25))

dat <- data.frame(x = x, y = y)

fit_raw <- lm(y ~ x, data = dat)
fit_log <- lm(log(y) ~ x, data = dat)
```

Compare the fitted models

```
summary(fit_raw)
```

```
Call:
lm(formula = y ~ x, data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-6.9915 -2.5592 -0.4927  1.3714 15.1442

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.4970     2.2582   0.220   0.828
x            1.0755     0.1885   5.705 2.07e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.861 on 18 degrees of freedom
Multiple R-squared:  0.6439,    Adjusted R-squared:  0.6241
F-statistic: 32.55 on 1 and 18 DF,  p-value: 2.073e-05
```

```
summary(fit_log)
```

```
Call:
lm(formula = log(y) ~ x, data = dat)

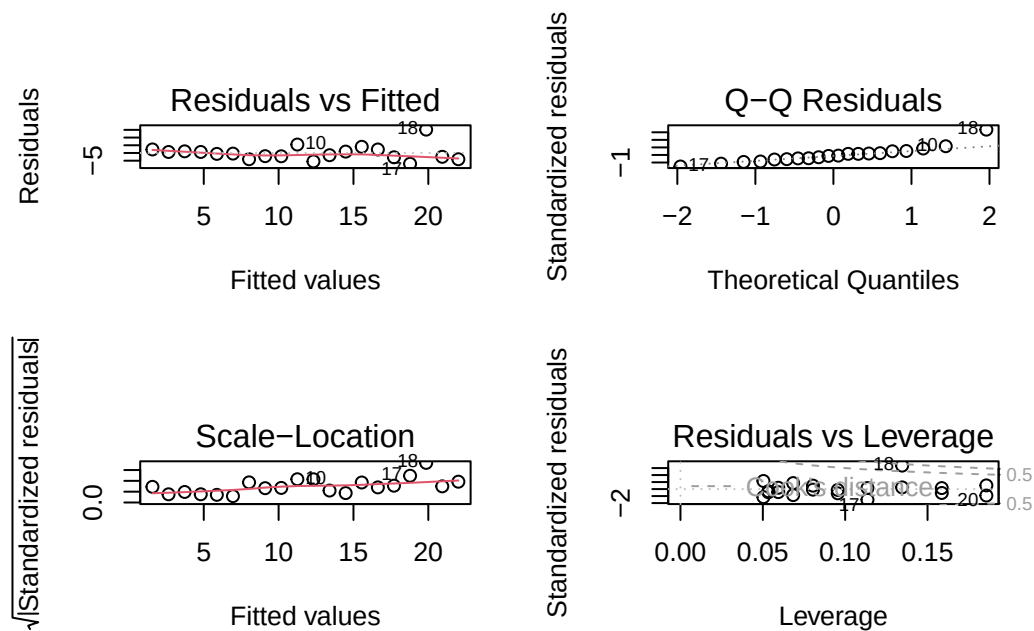
Residuals:
    Min       1Q   Median       3Q      Max
-0.52498 -0.20448  0.01606  0.12805  0.60913

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.1956     0.1449   8.249 1.58e-07 ***
x            0.1012     0.0121   8.365 1.29e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.312 on 18 degrees of freedom
Multiple R-squared:  0.7954,    Adjusted R-squared:  0.784
F-statistic: 69.97 on 1 and 18 DF,  p-value: 1.292e-07
```

Diagnostic plots for the untransformed model

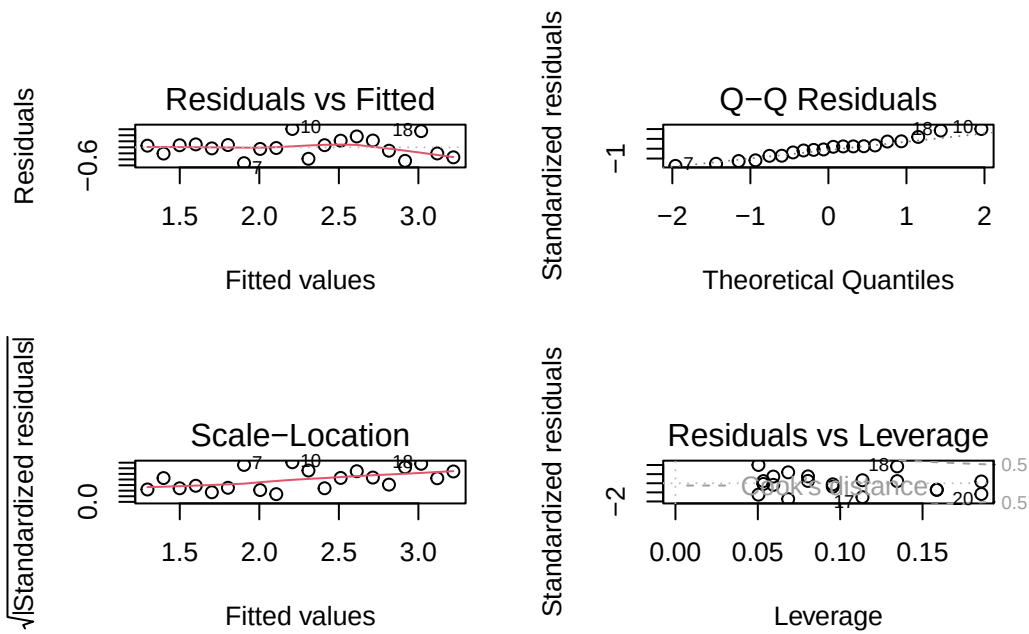
```
par(mfrow = c(2, 2))
plot(fit_raw)
```



```
par(mfrow = c(1, 1))
```

Diagnostic plots for the log-transformed model

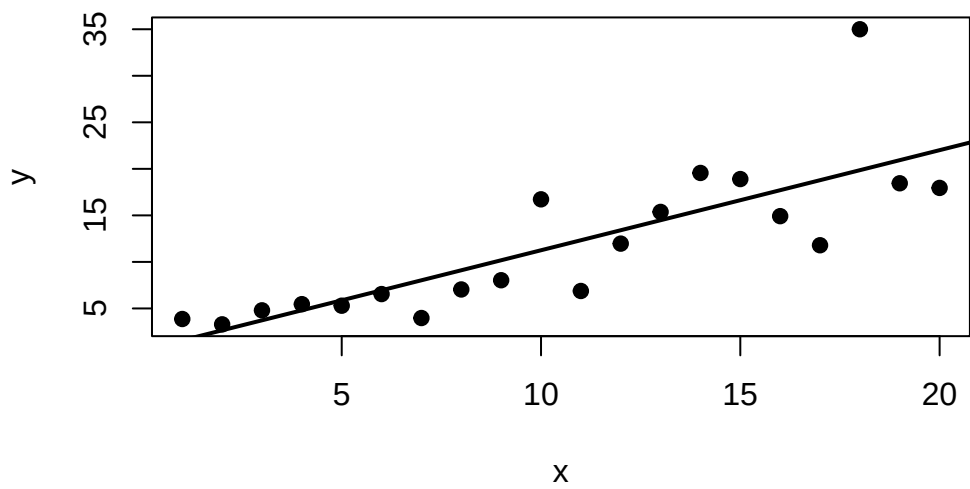
```
par(mfrow = c(2, 2))
plot(fit_log)
```



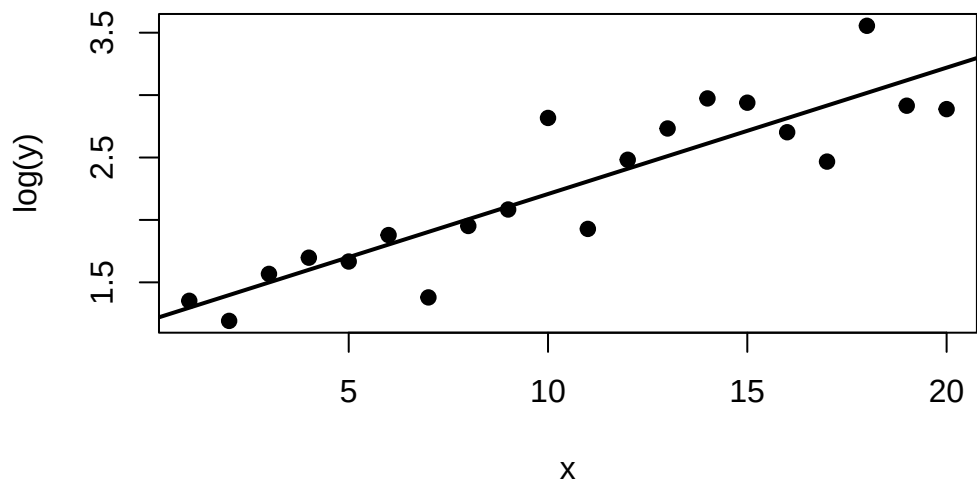
```
par(mfrow = c(1, 1))
```

Plot data on original and transformed scales

```
plot(dat$x, dat$y, pch = 19, xlab = "x", ylab = "y")
abline(fit_raw, lwd = 2)
```



```
plot(dat$x, log(dat$y), pch = 19, xlab = "x", ylab = "log(y)")
abline(fit_log, lwd = 2)
```



R Demonstration With Weighted Least Squares

Generate data with variance increasing in x

```
set.seed(8561)
x2 <- seq(1, 15, by = 1)
y2 <- 5 + 2 * x2 + rnorm(length(x2), sd = 0.6 * x2)

dat2 <- data.frame(x = x2, y = y2)

fit_ols <- lm(y ~ x, data = dat2)
fit_wls <- lm(y ~ x, data = dat2, weights = 1 / x^2)
```

Compare summaries

```
summary(fit_ols)
```

Call:

```
lm(formula = y ~ x, data = dat2)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.1677	-1.9056	0.6626	1.7668	12.3689

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.6129	3.2634	1.107	0.288
x	2.3824	0.3589	6.637	1.62e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.006 on 13 degrees of freedom

Multiple R-squared: 0.7722, Adjusted R-squared: 0.7546

F-statistic: 44.06 on 1 and 13 DF, p-value: 1.619e-05

```
summary(fit_wls)
```

Call:

```
lm(formula = y ~ x, data = dat2, weights = 1/x^2)
```

Weighted Residuals:

	Min	1Q	Median	3Q	Max
	-1.2815	-0.1740	0.1305	0.2625	1.3177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.2928	0.7033	7.526	4.34e-06 ***
x	2.1336	0.2283	9.346	3.92e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

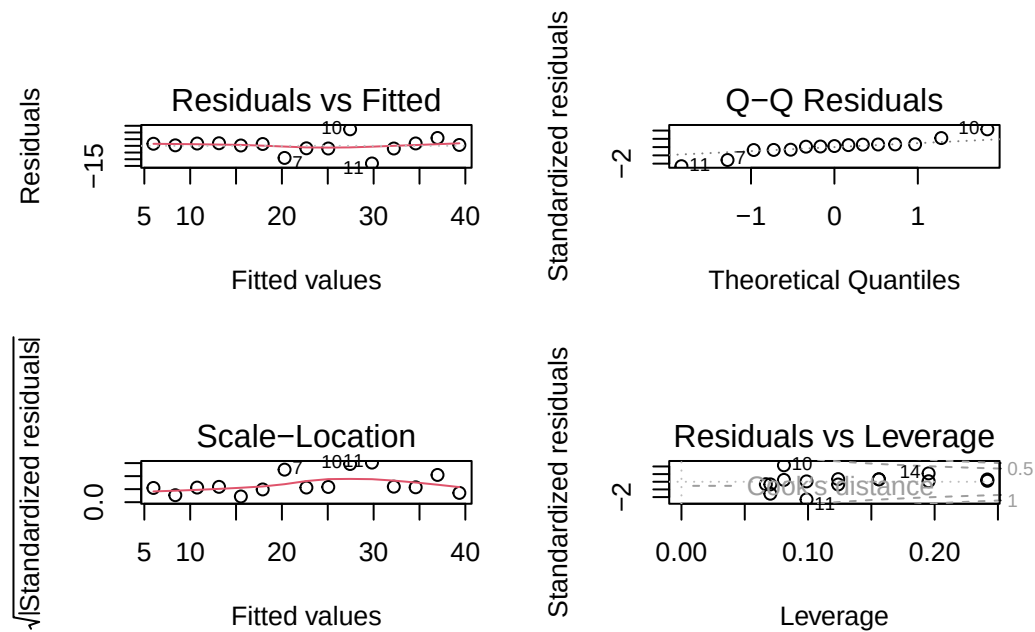
Residual standard error: 0.647 on 13 degrees of freedom

Multiple R-squared: 0.8705, Adjusted R-squared: 0.8605

F-statistic: 87.36 on 1 and 13 DF, p-value: 3.919e-07

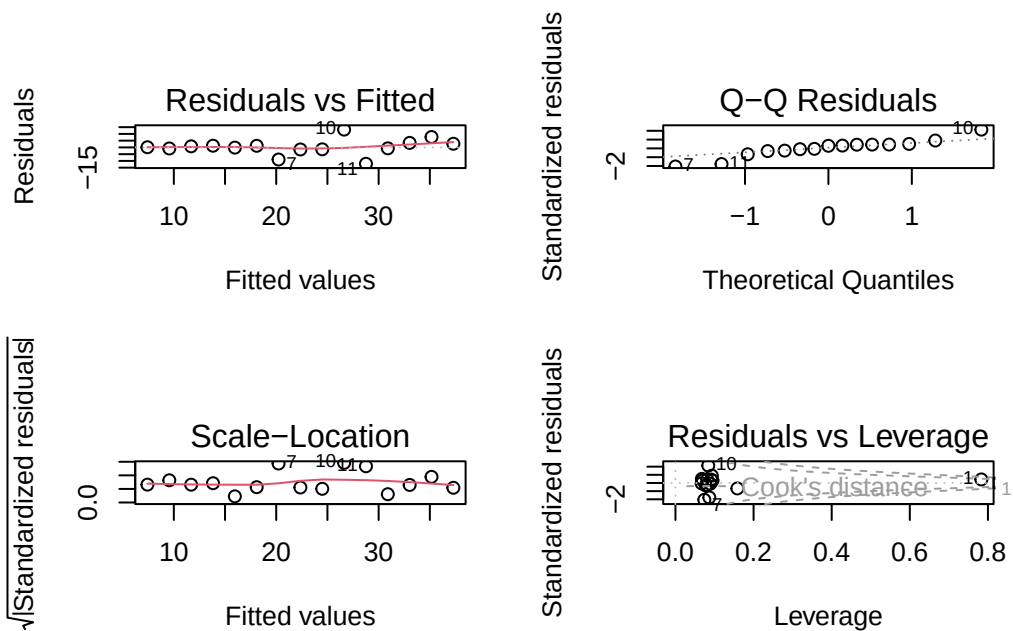
Compare diagnostic plots

```
par(mfrow = c(2, 2))
plot(fit_ols)
```



```
par(mfrow = c(1, 1))
```

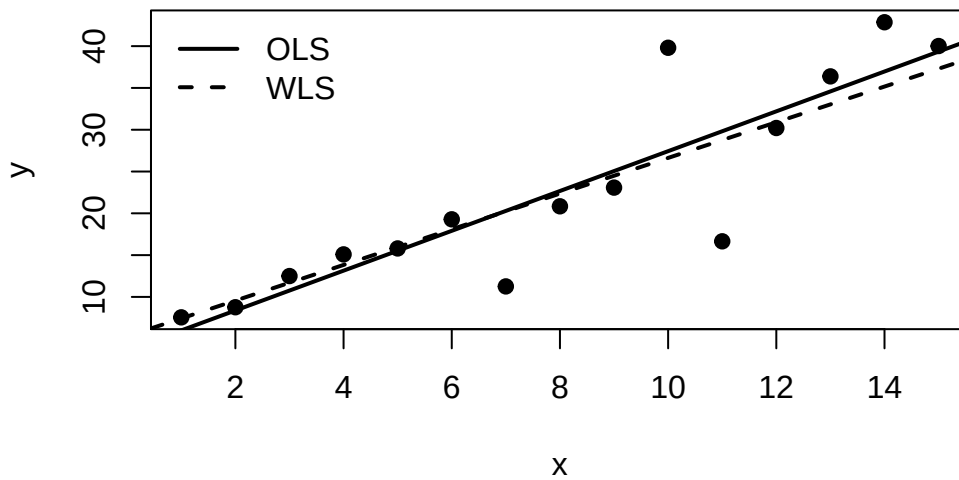
```
par(mfrow = c(2, 2))
plot(fit_wls)
```



```
par(mfrow = c(1, 1))
```

Plot fitted lines

```
plot(dat2$x, dat2$y, pch = 19, xlab = "x", ylab = "y")
abline(fit_ols, lwd = 2)
abline(fit_wls, lwd = 2, lty = 2)
legend("topleft",
      legend = c("OLS", "WLS"),
      lty = c(1, 2),
      lwd = 2,
      bty = "n")
```



Example With a Quadratic Remedy for Curvature

```
set.seed(8561)
x3 <- seq(-3, 3, length.out = 50)
y3 <- 3 + 2 * x3 + 1.8 * x3^2 + rnorm(length(x3), sd = 1)

dat3 <- data.frame(x = x3, y = y3)

fit_lin <- lm(y ~ x, data = dat3)
fit_quad <- lm(y ~ x + I(x^2), data = dat3)

summary(fit_lin)
```

Call:

```
lm(formula = y ~ x, data = dat3)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.006	-4.512	-1.085	3.409	11.147

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
(Intercept)			
x			
x^2			

```
(Intercept)  8.5027      0.7424  11.453 2.48e-15 ***
x            1.8405      0.4201   4.381 6.40e-05 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 5.25 on 48 degrees of freedom
```

```
Multiple R-squared:  0.2856,    Adjusted R-squared:  0.2707
```

```
F-statistic: 19.19 on 1 and 48 DF,  p-value: 6.398e-05
```

```
summary(fit_quad)
```

```
Call:
```

```
lm(formula = y ~ x + I(x^2), data = dat3)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-2.45522 -0.68482  0.03078  0.58492  2.37433
```

```
Coefficients:
```

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.89254     0.25022   11.56 2.43e-15 ***
x            1.84049     0.09437   19.50 < 2e-16 ***
I(x^2)       1.79673     0.05974   30.07 < 2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.179 on 47 degrees of freedom
```

```
Multiple R-squared:  0.9647,    Adjusted R-squared:  0.9632
```

```
F-statistic: 642.4 on 2 and 47 DF,  p-value: < 2.2e-16
```

```
anova(fit_lin, fit_quad)
```

```
Analysis of Variance Table
```

```
Model 1: y ~ x
```

```
Model 2: y ~ x + I(x^2)
```

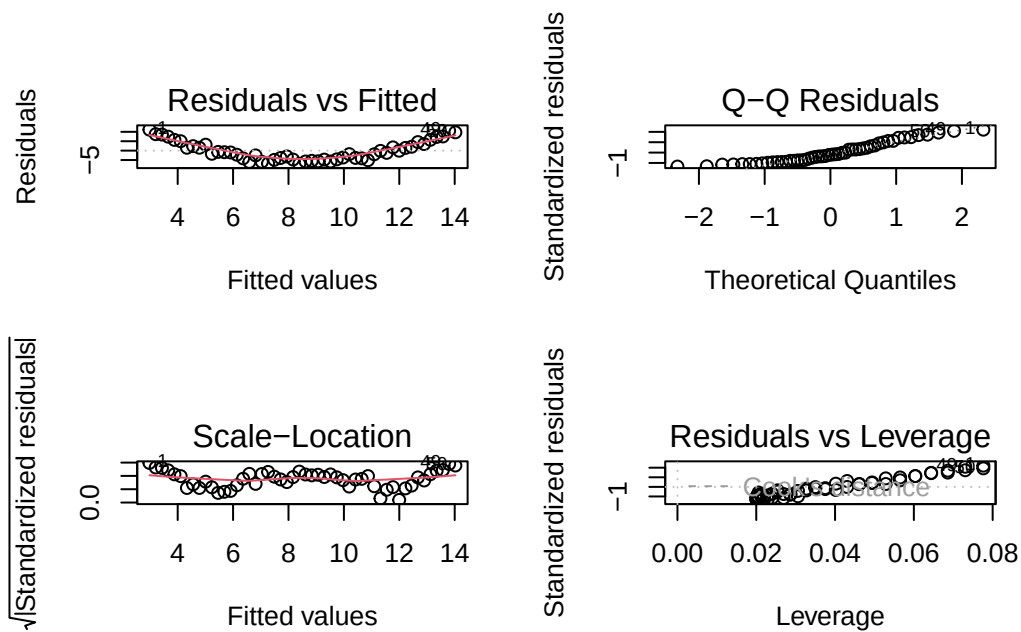
```
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     48 1322.81
2     47   65.35  1   1257.5 904.41 < 2.2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

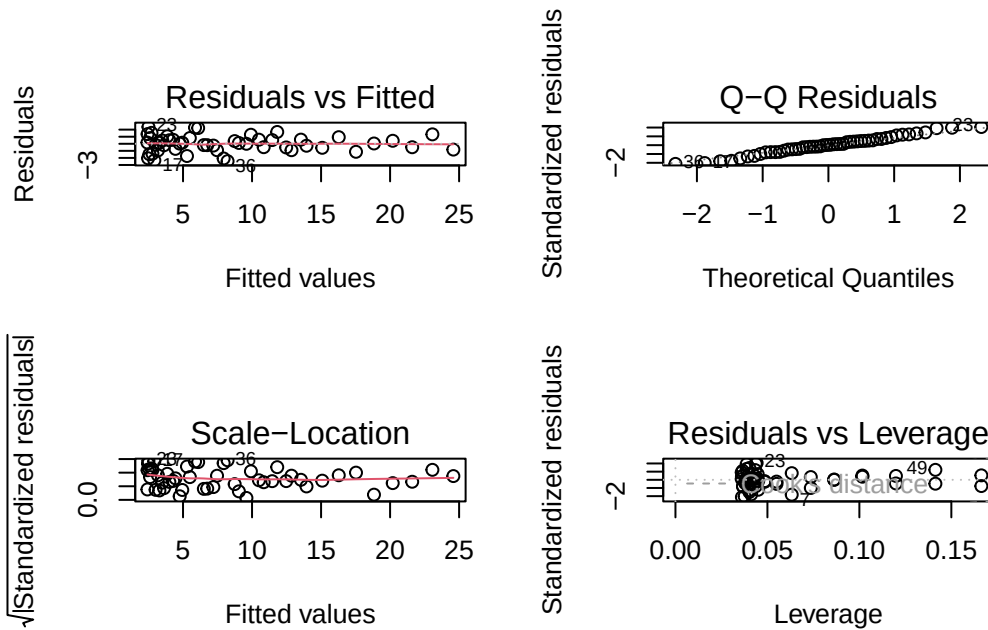
Compare diagnostic plots for linear and quadratic fits

```
par(mfrow = c(2, 2))  
plot(fit_lin)
```



```
par(mfrow = c(1, 1))
```

```
par(mfrow = c(2, 2))  
plot(fit_quad)
```



```
par(mfrow = c(1, 1))
```

Interpreting Software Output

Useful commands in R include:

- `lm(..., weights = ...)` for weighted least squares;
- `plot(fit)` for standard diagnostic plots;
- `anova(fit1, fit2)` for comparing nested mean models;
- `predict()` for fitted values on the chosen modelling scale.

Students should always keep track of the scale on which the model is fitted. This is especially important when the response is transformed.

A Practical Remedy Workflow

A useful workflow after diagnostics is:

- identify the main problem from residual analysis;
- decide whether the issue concerns the mean structure, the variance structure, or both;
- try a scientifically reasonable remedy;
- refit the model;
- compare diagnostics before and after the change;
- interpret the new model on the correct scale.

This encourages disciplined model improvement rather than ad hoc trial and error.

In-Class Discussion Questions

1. When is a response transformation preferable to adding polynomial terms?
2. How does the interpretation of coefficients change under a log transformation?
3. Why does weighted least squares downweight high-variance observations?
4. Why should model remedies be guided by both diagnostics and subject-matter knowledge?

9.5 Practice Problems

Conceptual

1. Explain the difference between transforming the response and transforming a predictor.
2. Explain why a log transformation may help when the variance grows with the mean.
3. Explain why weighted least squares can be viewed as ordinary least squares on a transformed system.

Computational

Suppose a regression model has

$$\text{Var}(\varepsilon_i) = \sigma^2 x_i^2.$$

1. What weights would be natural for weighted least squares?
2. Which observations receive the largest weights?
3. Why do these weights make sense?

Now suppose the fitted model is

$$\log(Y_i) = 1.5 + 0.2x_i.$$

1. What is the fitted log response when $x_i = 3$?
2. What is the fitted value on the original response scale if you exponentiate the fitted mean?
3. Why should interpretation on the original scale be made carefully?

Model-Improvement Problem

A residual-versus-fitted plot shows both a curved pattern and increasing spread.

1. What kinds of model inadequacy does this suggest?
2. Name two possible remedies.
3. How would you decide which remedy is more appropriate?

Suggested Homework

Complete the following tasks:

- fit a regression model that shows heteroscedasticity or curvature;
- apply one response transformation and assess whether the diagnostics improve;
- fit a weighted least squares model with a justified set of weights;
- compare OLS and WLS fits both numerically and graphically;
- write a short discussion explaining which remedy you prefer and why.

Summary

In this chapter, we studied practical remedies for model inadequacy in linear regression. We focused on:

- transforming the response or predictors;
- using polynomial terms to address curvature;
- stabilizing variance through transformation;
- using weighted least squares when error variances differ across observations;
- comparing alternative remedies using diagnostics and interpretation.

These ideas help students move from model criticism to model improvement.

10 Model Assessment and Prediction

A useful model must answer a stated question. Estimating an adjusted association, comparing treatments, and predicting a future response can favor different choices. This chapter keeps the main principles of collinearity and validation; automatic selection algorithms are in an optional appendix.

10.1 Stable estimates and scientific questions

Collinearity. When columns of X are nearly dependent, different coefficient combinations can produce very similar fitted values. The covariance $\sigma^2(X^\top X)^{-1}$ then explains why individual coefficients can be unstable even when fitted values are reasonably precise.

For a single nonconstant predictor column, $VIF_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing that column on the other predictors with an intercept. A large value is a diagnostic clue, not an automatic rule for deleting variables. Multi-column factors require additional care in interpreting column-specific VIFs.

Keep the question fixed. Removing a covariate changes the adjusted comparison. Retain terms required by the scientific question or design, and respect hierarchy when including interactions. Centering can clarify an intercept or reduce some polynomial-column correlations, but it does not create new information.

! Selection changes inference

Intervals and tests derived for a prespecified model do not automatically retain their interpretation after searching the data for a favorable model. A training/test split or cross-validation estimates predictive performance; it does not by itself validate post-selection coefficient intervals.

10.2 Comparing predictive performance

Training SSE and training R^2 reward added flexibility. Predictive evaluation instead asks how a fitted procedure behaves on observations it did not use for fitting.

A simple workflow. Specify a small set of candidate models, use the same observations and the same folds, fit each model only on the training portion, and evaluate it on the held-out portion. Any imputation, centering, feature construction, or variable search learned from data belongs inside the training step. If validation is used to choose a model, its winning validation error is itself optimistic; use an untouched test set or nested resampling for a final performance claim.

For continuous responses, one useful loss is $\text{MSE}_{test} = m^{-1} \sum_{i=1}^m (y_i - \hat{y}_i)^2$. For binary and count outcomes, the GLM chapter introduces response-appropriate interpretation and predictive checks.

i Respect the observational unit

Random row-wise folds are appropriate only when they match the intended prediction setting. With repeated measurements, keep a subject's observations together; with temporal prediction, evaluate on later periods. A large number of rows is not a substitute for independent subjects or future time periods.

AIC and model comparison. AIC balances likelihood fit and parameter count. It is useful when candidate likelihoods refer to the same response and comparable data, but it is not an effect estimate or a significance test. Comparing AIC values from a model for Y and one for $\log Y$ requires accounting for the transformation of the likelihood; do not compare the raw software values mechanically. BIC, C_p , and automatic searches are in [Automatic Model Selection \(Optional\)](#).

10.3 A small validation example

The following simulated curvature example uses the same five folds for two prespecified mean models. It illustrates predictive evaluation, not a guarantee that a quadratic model will always win.

```
set.seed(8561)
n <- 100
x <- runif(n, -2, 2)
dat <- data.frame(x = x, y = 2 + x + 0.7 * x^2 + rnorm(n, sd = 1.5))
fold <- sample(rep(1:5, length.out = n))
models <- list(linear = y ~ x, quadratic = y ~ x + I(x^2))
pred <- matrix(NA_real_, n, length(models),
               dimnames = list(NULL, names(models)))
for (k in 1:5) {
  for (j in seq_along(models)) {
    fit <- lm(models[[j]], data = dat[fold != k, ])
```

```

    pred[fold == k, j] <- predict(fit, newdata = dat[fold == k, ])
  }
}
cv_mse <- colMeans((dat$y - pred)^2)
cv_mse

```

```

linear quadratic
2.908542  2.567489

```

```

par(mfrow = c(1, 2))
for (j in seq_along(models)) {
  plot(dat$y, pred[, j], pch = 19, col = adjustcolor("steelblue", 0.6),
       xlab = "Observed response", ylab = "Held-out prediction",
       main = names(models)[j])
  abline(0, 1, lty = 2)
}
par(mfrow = c(1, 1))

```

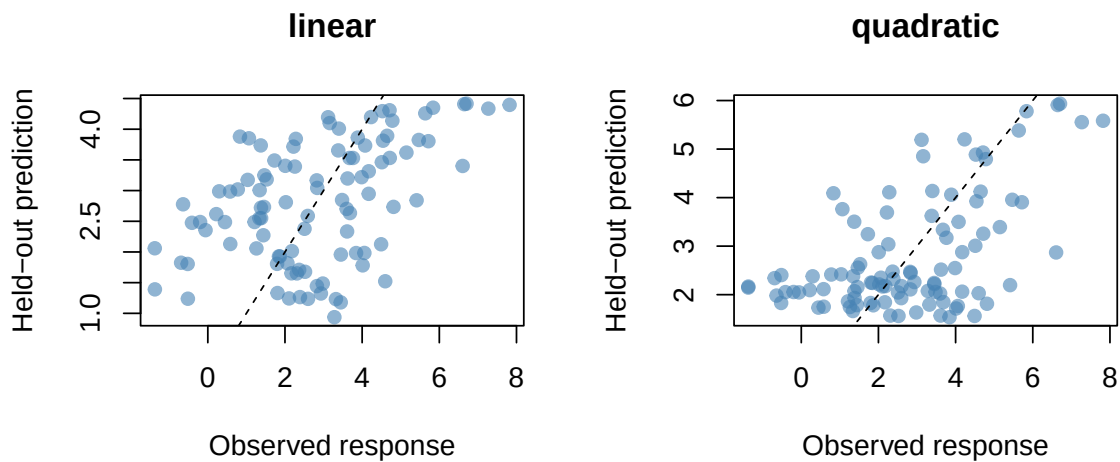


Figure 10.1: Predictions made for held-out observations. Every plotted prediction comes from a fit that excluded that observation's fold.

Report more than a winner. State the loss, the splitting rule, the population or future setting of interest, and how much performance differs. Investigate poor predictions. A single difference in estimated MSE does not establish a universal ranking of the methods.

10.4 Practice

1. Explain why adding a predictor can improve training fit without improving held-out predictions.
2. Explain why a confounder or design variable should not be removed solely because of its p-value or VIF.
3. Adapt the example to grouped data by assigning folds at the group level.
4. Write a short report distinguishing an inferential conclusion about a coefficient from a predictive conclusion about a fitted procedure.

11 Introduction to Generalized Linear Models

The same design matrix can represent continuous predictors, groups, interactions, and experimental comparisons. What changes when the response is a binary outcome or an event count? A generalized linear model (GLM) retains the linear predictor and changes the response distribution and its connection to the conditional mean.

Learning objectives. Choose a response family and link, interpret logistic and Poisson coefficients, obtain fitted probabilities or rates, compare nested models, and recognize when the basic model is inadequate. The core unit takes two lectures, followed by an integrated case in the final class.

11.1 From the normal linear model to GLMs

The normal linear model uses $\mathbb{E}(Y_i | x_i) = x_i^\top \beta$ and constant conditional variance. A binary outcome instead has a probability between zero and one, while a count is nonnegative and often becomes more variable as its mean increases.

Definition 11.1 (Components of a Generalized Linear Model). For conditionally independent observations, specify a response distribution, a linear predictor $\eta_i = x_i^\top \beta$, and a link function g satisfying

$$g(\mu_i) = \eta_i, \quad \mu_i = \mathbb{E}(Y_i | x_i).$$

The chosen family also specifies how the conditional variance depends on the mean and, when applicable, a dispersion parameter.

Response	Conditional distribution	Link	Conditional mean
Continuous	Gaussian	identity	$\mu_i = x_i^\top \beta$
Binary	Bernoulli	logit	$\pi_i = \{1 + \exp(-x_i^\top \beta)\}^{-1}$
Count	Poisson	log	$\mu_i = \exp(x_i^\top \beta)$

What stays the same? Building X , representing factors and interactions, defining a comparison, and checking the observational unit remain essential. The Gaussian identity-link GLM recovers the ordinary normal linear model.

What changes? The mean can be nonlinear in x , the variance need not be constant, and estimation generally uses likelihood rather than minimizing unweighted SSE. The link transforms the conditional mean; it does not require transforming each observed response. This matters when the data contain zeros.

! GLM and GLS address different extensions

GLM means **generalized linear model** and concerns the response family and mean link. GLS means **generalized least squares** and concerns a covariance model for linear-mean regression. A standard independent-observation GLM does not automatically account for repeated observations on the same subject. Correlated-error GLS is in an [optional appendix](#).

11.2 Logistic regression: probabilities and odds

For $Y_i | x_i \sim \text{Bernoulli}(\pi_i)$, the conditional variance is $\pi_i(1 - \pi_i)$. Logistic regression specifies

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = x_i^\top \beta, \quad \pi_i = \frac{\exp(x_i^\top \beta)}{1 + \exp(x_i^\top \beta)}.$$

Interpretation. In a model without an interaction involving x_j , a one-unit increase in x_j , holding other predictors fixed, multiplies the odds by $\exp(\beta_j)$. It does not multiply the probability by that number, and it does not give a constant probability difference. An interaction makes the corresponding odds ratio depend on the interacting predictor.

Example 11.1 (The Same Odds Ratio Can Imply Different Probability Changes). An odds ratio of 2 changes a baseline probability of 0.10 to $2(0.10)/(1 - 0.10 + 2(0.10)) \approx 0.182$. At a baseline probability of 0.50 it gives $2(0.50)/(1 - 0.50 + 2(0.50)) \approx 0.667$. Report predicted probabilities for meaningful predictor values when communicating the result.

Simulated example. Imagine a binary completion outcome, a centered baseline score, and a group indicator. The following data are simulated to illustrate the model; they are not observations from a real study.

```

set.seed(8561)
n <- 240
score <- runif(n, -2, 2)
group <- factor(sample(c("Control", "Treatment"), n, replace = TRUE))
prob <- plogis(-0.5 + 0.8 * score + 0.6 * (group == "Treatment"))
binary <- data.frame(score, group,
                     completed = rbinom(n, size = 1, prob = prob))
fit_logit <- glm(completed ~ score + group,
                 family = binomial(link = "logit"),
                 data = binary)
summary(fit_logit)

```

Call:

```
glm(formula = completed ~ score + group, family = binomial(link = "logit"),
    data = binary)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.2905	0.1932	-1.504	0.133
score	0.6982	0.1262	5.531	3.19e-08 ***
groupTreatment	0.2101	0.2818	0.746	0.456

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 330.31 on 239 degrees of freedom
Residual deviance: 295.04 on 237 degrees of freedom
AIC: 301.04

Number of Fisher Scoring iterations: 4

Coefficient intervals and fitted probabilities. The intervals below are approximate Wald intervals. Exponentiating the intercept gives baseline odds at score zero in the reference group; exponentiating the remaining coefficients gives the corresponding conditional odds ratios for this additive model.

```

b <- coef(fit_logit)
se <- sqrt(diag(vcov(fit_logit)))
exp(cbind(estimate = b, lower = b - qnorm(0.975) * se,
          upper = b + qnorm(0.975) * se))

```

	estimate	lower	upper
(Intercept)	0.747878	0.5121390	1.092128
score	2.010082	1.5694918	2.574354
groupTreatment	1.233769	0.7102261	2.143242

```
new_binary <- expand.grid(score = seq(-2, 2, length.out = 100),
                          group = levels(binary$group))
link_fit <- predict(fit_logit, newdata = new_binary,
                   type = "link", se.fit = TRUE)
new_binary$probability <- plogis(link_fit$fit)
new_binary$lower <- plogis(link_fit$fit - qnorm(0.975) * link_fit$se.fit)
new_binary$upper <- plogis(link_fit$fit + qnorm(0.975) * link_fit$se.fit)
```

```
cols <- c(Control = "steelblue4", Treatment = "darkorange3")
plot(NA, xlim = range(new_binary$score), ylim = c(0, 1),
     xlab = "Centered baseline score", ylab = "Completion probability")
for (g in levels(binary$group)) {
  z <- new_binary[new_binary$group == g, ]
  polygon(c(z$score, rev(z$score)), c(z$lower, rev(z$upper)),
          border = NA, col = adjustcolor(cols[g], 0.13))
  lines(z$score, z$probability, col = cols[g], lwd = 2)
}
legend("topleft", legend = names(cols), col = cols, lwd = 2, bty = "n")
```

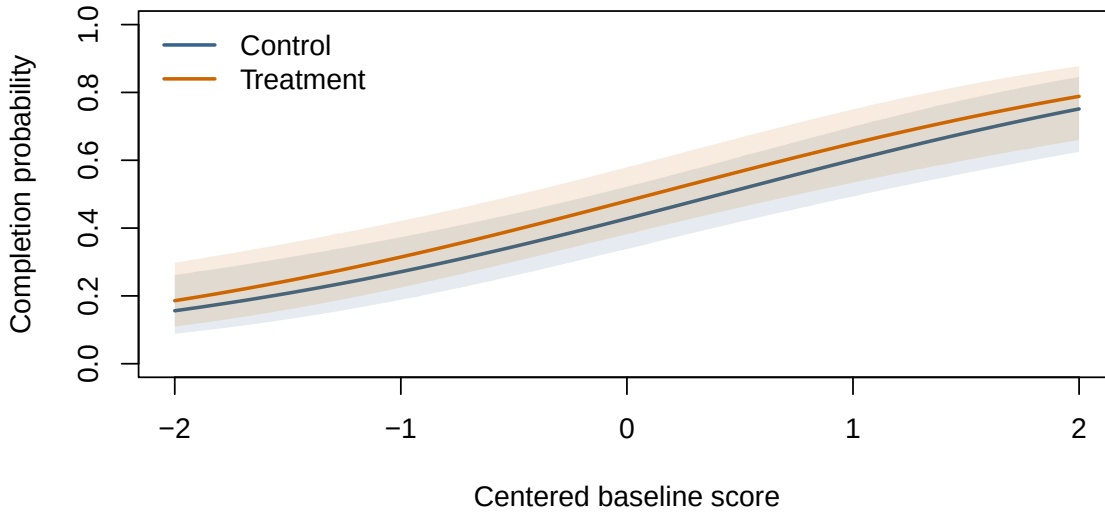


Figure 11.1: Fitted completion probabilities with approximate pointwise 95% confidence bands. Group differences on the probability scale vary with the baseline score.

i Confidence for a probability is not prediction of an individual outcome

These bands quantify uncertainty in the fitted conditional probability. An individual future outcome is still either zero or one. The Gaussian prediction-interval formula from Chapter 3 does not transfer directly to a Bernoulli outcome. The bands are pointwise, not simultaneous bands over the entire score range.

11.3 Poisson regression: counts, rates, and exposure

For a Poisson outcome, $\mathbb{E}(Y_i | x_i) = \text{Var}(Y_i | x_i) = \mu_i$. With a log link, $\log \mu_i = x_i^\top \beta$, ensuring positive fitted means.

Counts versus rates. Units may be observed for different lengths of time. If $t_i > 0$ is the known exposure, model the rate λ_i using

$$\mu_i = t_i \lambda_i, \quad \log \mu_i = \log t_i + x_i^\top \beta.$$

The term $\log t_i$ is an **offset** with coefficient fixed at one. Under this rate model, doubling exposure doubles the expected count while leaving the rate unchanged. For an additive log-rate model, $\exp(\beta_j)$ is a conditional rate ratio for a one-unit increase in x_j .

! A log link does not mean taking log of the observed count

A Poisson GLM models $\log\{E(Y_i | x_i)\}$. It can use zero counts directly. Replacing the response with $\log(Y_i + 1)$ creates a different model with a different target and is not an equivalent implementation.

Simulated example. Suppose sites have different exposure times, a baseline workload, and two service protocols. Here exposure is measured in days, so the model's rate is events per day.

```
set.seed(8561)
n <- 240
workload <- runif(n, -1.5, 1.5)
protocol <- factor(sample(c("A", "B"), n, replace = TRUE))
exposure <- runif(n, 1, 8)
rate <- exp(-0.1 + 0.35 * workload + 0.4 * (protocol == "B"))
counts <- data.frame(workload, protocol, exposure,
                     events = rpois(n, lambda = exposure * rate))
fit_pois <- glm(events ~ workload + protocol + offset(log(exposure)),
               family = poisson(link = "log"), data = counts)
summary(fit_pois)
```

Call:

```
glm(formula = events ~ workload + protocol + offset(log(exposure)),
    family = poisson(link = "log"), data = counts)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.07682	0.04370	-1.758	0.0787 .
workload	0.32626	0.03278	9.954	< 2e-16 ***
protocolB	0.38246	0.05723	6.683	2.34e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 366.12 on 239 degrees of freedom

Residual deviance: 225.30 on 237 degrees of freedom
AIC: 993.2

Number of Fisher Scoring iterations: 4

For a one-day exposure, a fitted expected count is also a fitted daily rate. Use the original exposure for expected counts over the original observation periods.

```
new_counts <- expand.grid(workload = seq(-1.5, 1.5, length.out = 100),  
                          protocol = levels(counts$protocol), exposure = 1)  
new_counts$rate <- predict(fit_pois, newdata = new_counts, type = "response")  
c(rate_ratio_protocol_B = exp(coef(fit_pois)["protocolB"]))
```

```
rate_ratio_protocol_B.protocolB  
1.465891
```

```
cols <- c(A = "steelblue4", B = "darkorange3")  
plot(counts$workload, counts$events / counts$exposure, pch = 19,  
      col = adjustcolor(cols[as.character(counts$protocol)], 0.35),  
      xlab = "Centered workload", ylab = "Events per day")  
for (g in levels(counts$protocol)) {  
  z <- new_counts[new_counts$protocol == g, ]  
  lines(z$workload, z$rate, col = cols[g], lwd = 2)  
}  
legend("topleft", legend = names(cols), col = cols, lwd = 2, bty = "n")
```

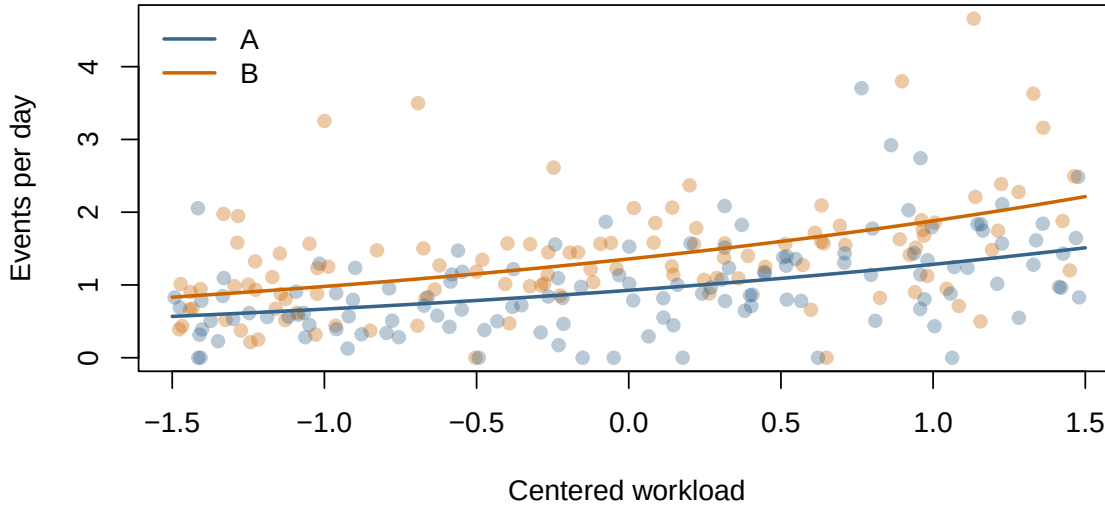


Figure 11.2: Observed events per exposure day and fitted conditional rates. Different exposure times make the observed rates unequally variable; the model fits counts with a log-exposure offset.

11.4 Inference and model checks

From sums of squares to likelihood. OLS compares residual sums of squares. In a GLM, the likelihood measures how well the fitted conditional distribution accounts for the observations. Deviance measures lack of fit relative to a saturated model; for the binomial and Poisson models here, a difference in deviances equals twice the difference in maximized log likelihoods.

Theorem 11.1 (Approximate Likelihood-Ratio Test). *For nested, correctly specified binomial or Poisson GLMs fitted to the same responses and observations, let the full model add r identifiable parameters. Under the reduced model and standard regularity conditions, including an interior true parameter and finite, regular maximum-likelihood estimates,*

$$D_R - D_F = 2(\ell_F - \ell_R) \stackrel{\text{approx}}{\sim} \chi_r^2.$$

This is an asymptotic approximation, not the exact finite-sample F test of the normal linear model. Separation, sparse data, or boundary problems can invalidate the usual approximation.

```
fit_logit_reduced <- glm(completed ~ score,
                        family = binomial(), data = binary)
anova(fit_logit_reduced, fit_logit, test = "LRT")
```

Analysis of Deviance Table

Model 1: completed ~ score

Model 2: completed ~ score + group

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	238	295.60			
2	237	295.04	1	0.55759	0.4552

```
fit_pois_reduced <- glm(events ~ workload + offset(log(exposure)),
                        family = poisson(), data = counts)
anova(fit_pois_reduced, fit_pois, test = "LRT")
```

Analysis of Deviance Table

Model 1: events ~ workload + offset(log(exposure))

Model 2: events ~ workload + protocol + offset(log(exposure))

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	238	270.44			
2	237	225.30	1	45.147	1.828e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

In each comparison, state the group or protocol question before reading the p-value. Keep the response, observations, exposure definition, and relevant likelihood assumptions comparable. For estimated-dispersion quasi models, the Poisson/binomial chi-square rule is not automatically appropriate; see [R's analysis-of-deviance documentation](#).

Check the model in the units of the question.

Question	What to inspect
Is the observation unit independent?	Repeated subjects, sites, clustering, and time order; a different family alone does not remove dependence.
Is the mean structure reasonable?	Curvature, interactions, fitted probabilities or rates across the observed predictor range.

Question	What to inspect
Is the logistic fit regular?	Convergence warnings, sparse predictor patterns, very large coefficients/SEs, and separation.
Is a Poisson variance plausible?	Pearson residuals, unusually variable counts conditional on predictors, and unexplained heterogeneity.
Will predictions transfer?	Held-out loss and observed-versus-predicted summaries in a split that matches future use.

For a Poisson fit, the Pearson statistic divided by residual degrees of freedom is a useful rough dispersion diagnostic:

```
pearson_dispersion <- sum(residuals(fit_pois, type = "pearson")^2) /
  df.residual(fit_pois)
pearson_dispersion
```

```
[1] 0.9579286
```

A value substantially above one can signal overdispersion, missing mean structure, dependence, or other misspecification. It is not a stand-alone proof of a particular alternative model, especially with sparse counts. Recognizing when the basic model is inadequate is required; negative-binomial, quasi-likelihood, zero-inflated, and mixed-model remedies are beyond this introductory unit.

A simple held-out probability check. The Brier score is the mean squared difference between a predicted probability and a binary outcome. The following split evaluates a prespecified logistic model and an intercept-only baseline. Both are fitted on the training set.

```
set.seed(8561)
train <- sample(seq_len(nrow(binary)), floor(0.7 * nrow(binary)))
fit_train <- glm(completed ~ score + group,
  family = binomial(), data = binary[train, ])
test <- binary[-train, ]
p_hat <- predict(fit_train, newdata = test, type = "response")
p_baseline <- mean(binary$completed[train])
c(model_Brier = mean((test$completed - p_hat)^2),
  baseline_Brier = mean((test$completed - p_baseline)^2))
```

```
model_Brier baseline_Brier
0.2199555    0.2484056
```

A single small split is noisy. Use it to understand the evaluation procedure, not to assert that one model universally dominates another. A classification accuracy at one threshold does not summarize the quality of predicted probabilities.

For the likelihood representation, the role of WLS in fitting, and further theory, see [GLM Likelihood and Computation \(Optional\)](#).

11.5 Practice and integrated analysis

1. Explain which GLM component changes when moving from Gaussian OLS to logistic regression, and which parts of the design matrix can remain the same.
2. Convert an odds ratio of 1.5 into a new probability for baseline probabilities 0.10 and 0.60. Explain why the probability differences differ.
3. Explain why exposure belongs in the Poisson example as an offset rather than an ordinary predictor with an estimated coefficient under the proportional-exposure model.
4. State the null hypothesis tested by each R model comparison. Explain why the reference distribution differs from Chapter 4's F test.
5. Write a one-page analysis: define the observational unit, choose a response family/link, report a probability or rate comparison with uncertainty, and identify one model limitation. Separate uncertainty about a conditional mean from variation in a future observation.

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John Wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part I

Assessments

Assignment 1

This assignment focuses on the foundations of the linear model:

- matrix representation,
- least squares estimation,
- projection
- basic inference.

Instructions

- Submit both your source file (`.qmd` or `.Rmd`) and the rendered HTML or PDF file.
- Name your source file as `A1_LastName_FirstName.qmd` or `A1_LastName_FirstName.Rmd`.
- Clearly label every question and subpart.
- Type all mathematical work using LaTeX.
- Include all R code used to obtain your answers.
- Your document must render from beginning to end without error.
- Do not submit screenshots of code, R output, or mathematical derivations.
- Round numerical answers to three decimal places when appropriate.
- For True/False questions, enter only T or F.
- For short-answer questions, one or two clear sentences are sufficient unless a derivation is requested.

The total value of the assignment is [40].

Question 1: Matrix Representation of the Linear Model [8]

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, 5,$$

with the following observed data:

i	x_i	Y_i
1	0	2
2	1	4
3	2	5
4	3	8
5	4	9

(a) [2]

Which of the following is the correct design matrix $\mathbf{X}$?

☐ **A.**

$$\begin{pmatrix} 0 & 2 \\ 1 & 4 \\ 2 & 5 \\ 3 & 8 \\ 4 & 9 \end{pmatrix}$$

☐ **B.**

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

☐ **C.**

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$$

☐ **D.**

$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}$$

(b) [2]

Fill in the following table.

Quantity	Dimension / Rank
$\mathbf{Y}$	---
$\mathbf{X}$	---
β	---
$\text{rank}(\mathbf{X})$	---

(c) [2]

Which pair is correct?

☐ **A.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

☐ **B.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 5 \\ 5 & 30 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

☐ **C.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 77 \\ 28 \end{pmatrix}$$

☐ **D.**

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 77 \end{pmatrix}$$

(d) [2]

Assume

$$\mathbb{E}[\varepsilon] = \mathbf{0}, \quad \text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_5.$$

For each statement, enter T or F.

Statement	T/F
$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\beta$	---
$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$	---
$\mathbb{E}[\mathbf{Y}] = \mathbf{0}$	---
The five errors all have variance σ^2	---

Question 2: Least Squares and Projection [12]

Continue using the data and design matrix from Question 1.

(a) [4]

Starting from

$$S(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

derive the normal equations

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{Y},$$

and then state the OLS estimator when $\mathbf{X}$ has full column rank.

Keep your derivation concise.

(b) [3]

Using the quantities from Question 1, compute

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

Fill in the blanks:

$$\hat{\beta}_0 = \underline{\hspace{2cm}}, \quad \hat{\beta}_1 = \underline{\hspace{2cm}}.$$

Hence,

$$\hat{Y} = \underline{\hspace{2cm}} + \underline{\hspace{2cm}}x.$$

(c) [2]

For the least squares residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

which statement must be true?

- ☐ **A.** $\mathbf{X}\mathbf{e} = \mathbf{0}$
- ☐ **B.** $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$
- ☐ **C.** $\mathbf{e}^\top \mathbf{e} = 0$
- ☐ **D.** $\mathbf{H}\mathbf{e} = \mathbf{e}$

In one sentence, explain the geometric meaning of your choice.

Your explanation:

(d) [3]

For the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top,$$

enter T or F.

Statement	T/F
$\mathbf{H}^\top = \mathbf{H}$	---
$\mathbf{H}^2 = \mathbf{H}$	---
$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$	---
$\mathbf{H}$ is an orthogonal projection matrix	---

Question 3: Consequences and Geometry of Least Squares [12]

Continue using the data and fitted regression model from Questions 1 and 2.

Recall that

$$\hat{\beta}_0 = 2.0, \quad \hat{\beta}_1 = 1.8.$$

(a) Fitted Values and Residuals [3]

Compute the fitted values

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

and the residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

Fill in the blanks:

$$\hat{\mathbf{Y}} = \begin{pmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \\ \rule{1cm}{0.4pt} \end{pmatrix}.$$

(b) Consequences of Including an Intercept [3]

For each statement, enter T or F.

Statement	T/F
The residuals satisfy $\sum_{i=1}^n e_i = 0$.	---
The fitted values satisfy $\hat{Y} = \bar{Y}$.	---
The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$.	---
These properties necessarily hold for every regression model without an intercept.	---

(c) Residual-Maker Matrix [3]

Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

For each statement, enter T or F.

Statement	T/F
$\mathbf{e} = \mathbf{MY}$	---
$\mathbf{M}^\top = \mathbf{M}$	---
$\mathbf{M}^2 = \mathbf{M}$	---
$\mathbf{HM} = \mathbf{0}$	---

(d) Rank and Uniqueness [3]

Suppose now that a design matrix $\mathbf{X}$ does **not** have full column rank.

For each statement, enter T or F.

Statement	T/F
$(\mathbf{X}^\top \mathbf{X})^{-1}$ necessarily exists.	---
The least squares coefficient vector $\hat{\beta}$ may not be unique.	---
The fitted vector $\hat{\mathbf{Y}}$ is still unique.	---
The residual vector $\mathbf{e}$ is still unique.	---

Question 4: Linear Regression in R [8]

For this question, use the built-in R dataset `cars`.

The variables are:

- `speed`: speed of the car in miles per hour;
- `dist`: stopping distance in feet.

Consider the model

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \varepsilon_i.$$

Load the data using

```
data(cars)
```

(a) [4]

Fit the model using `lm()` and complete the table.

```
fit <- lm(dist ~ speed, data = cars)
summary(fit)
```

Call:

```
lm(formula = dist ~ speed, data = cars)
```

Residuals:

Min	1Q	Median	3Q	Max
-29.069	-9.525	-2.272	9.215	43.201

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-17.5791	6.7584	-2.601	0.0123 *
speed	3.9324	0.4155	9.464	1.49e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.38 on 48 degrees of freedom

Multiple R-squared: 0.6511, Adjusted R-squared: 0.6438

F-statistic: 89.57 on 1 and 48 DF, p-value: 1.49e-12

Quantity	Your answer
$\hat{\beta}_0$	---
$\hat{\beta}_1$	---

Which interpretation of $\hat{\beta}_1$ is correct?

- ☐ **A.** For each 1 mph increase in speed, the stopping distance increases by exactly 3.9324 feet for every car.
- ☐ **B.** For each 1 mph increase in speed, the fitted mean stopping distance increases by approximately 3.9324 feet.
- ☐ **C.** The average speed increases by 3.9324 mph for every extra foot of stopping distance.
- ☐ **D.** A car travelling at 0 mph has a stopping distance of 3.9324 feet.

(b) [4]

Construct **Y** and **X** manually and reproduce the OLS coefficients using matrix operations.

Complete the following code:

```
Y <- matrix(cars$dist, ncol = 1)

X <- cbind(
  1,
  cars$speed
)

# Replace NULL with your matrix expression
beta_hat <- NULL

beta_hat
```

NULL

Then report

$$\hat{\beta} = \begin{pmatrix} \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} \end{pmatrix}.$$

Grading Summary

Question	Topic	Value
Question 1	Matrix representation	8
Question 2	Least squares and projection	12
Question 3	Distribution and inference	12
Question 4	Regression in R	8
Total		40

Assignment 2

Question: Distribution and Inference [12]

Suppose a multiple linear regression model has $n = 20$ observations and $p = 3$ regression parameters. Assume

$$\mathbf{Y} \sim N_{20}(\mathbf{X}\beta, \sigma^2 \mathbf{I}_{20}).$$

Suppose

$$\hat{\beta} = \begin{pmatrix} 2.0 \\ 1.5 \\ -0.5 \end{pmatrix},$$

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 0.20 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.08 \end{pmatrix},$$

and

$$\text{SSE} = 34.$$

Use

$$t_{0.975, 17} = 2.110.$$

(a) [3]

Choose the correct distributions.

For $\hat{\beta}$:

- ☐ **A.** $N_3(\mathbf{0}, \sigma^2 \mathbf{I}_3)$
- ☐ **B.** $N_3(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$
- ☐ **C.** t_{17}
- ☐ **D.** χ_{17}^2

For SSE/σ^2 :

- ☐ **A.** $N(0, 1)$
- ☐ **B.** t_{17}
- ☐ **C.** χ_{17}^2
- ☐ **D.** $F_{3,17}$

(b) [2]

Compute

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p}.$$

$$\hat{\sigma}^2 = \underline{\hspace{2cm}}.$$

(c) [4]

For testing

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_1 : \beta_1 \neq 0,$$

where β_1 is the second entry of β , fill in the following:

$$\text{SE}(\hat{\beta}_1) = \underline{\hspace{2cm}},$$

$$t = \underline{\hspace{2cm}},$$

and choose the decision:

- ☐ **A.** Reject H_0
- ☐ **B.** Fail to reject H_0

(d) [3]

Which of the following is the 95% confidence interval for β_1 ?

- ☐ **A.** (0.8329, 2.1671)
- ☐ **B.** (1.1838, 1.8162)
- ☐ **C.** (−0.6671, 0.6671)
- ☐ **D.** (0, 3.0000)

A Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.

B Linear-Model Extensions (Optional)

Optional appendix

The main chapters use identifiable mean comparisons and practical diagnostics. This appendix develops the Gauss–Markov result, rank deficiency, and additional diagnostics; it is not a prerequisite for the GLM chapter.

Efficiency of least squares. The following result supplements the moment calculations in Chapter 3.

Theorem B.1 (Gauss–Markov: What “Best” Means). *Let X be fixed with full column rank, $\mathbb{E}(\varepsilon | X) = 0$, and $\text{Var}(\varepsilon | X) = \sigma^2 I_n$. Among estimators linear in Y and unbiased for every β , OLS has the smallest covariance in the positive-semidefinite ordering. Thus every fixed linear combination of its coefficients has variance no larger than that of any competing linear unbiased estimator. Normality is not required.*

Proof idea. Write a competing estimator as AY , with $AX = I_p$. If $B = (X^\top X)^{-1} X^\top$, then $A = B + D$ with $DX = 0$. The cross terms vanish, so

$$\text{Var}(AY | X) - \text{Var}(\hat{\beta} | X) = \sigma^2 DD^\top \succeq 0.$$

Scope. “Best” refers to this linear unbiased class and these covariance assumptions. It does not claim optimality among all biased or nonlinear estimators, nor that OLS keeps this efficiency property under arbitrary unequal variances or correlation.

B.1 Estimability

So far, we have mostly assumed that $\mathbf{X}$ has full column rank. But in some important cases, the design matrix is rank-deficient. Then the parameter vector itself is not uniquely identifiable. In such settings, we ask a more refined question: Which linear functions of β can still be estimated uniquely from the model? This leads to the concept of **estimability**. **Why Estimability Is Needed** Suppose two different parameter vectors, say β and β^* , produce the same mean vector:

$$\mathbf{X}\beta = \mathbf{X}\beta^*.$$

Then the data cannot distinguish between these two parameter vectors. So a particular coefficient may not be uniquely meaningful. However, some combinations of coefficients may still be uniquely determined by the mean structure. Those combinations are estimable. **Definition of Estimability** A linear function $a^\top \beta$ is **estimable** if there exists a vector t such that

$$a^\top = t^\top \mathbf{X}.$$

Equivalently, a must lie in the row space of $\mathbf{X}$. This condition ensures that the target quantity depends only on the mean vector $\mathbf{X}\beta$, and not on the particular parameterization used. **Interpretation of Estimability** Estimability means that the quantity is determined by the model's observable mean structure. If a linear function is not estimable, then different parameter vectors producing the same fitted mean can give different values of that function.

So no unbiased linear estimator can uniquely recover it. **Example With a Factor Model** Suppose we write a one-way mean model as

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij},$$

for groups $i = 1, \dots, g$. If we include all group indicators together with an intercept, then the parameters are not uniquely identified, because adding a constant to all τ_i and subtracting it from μ gives the same mean structure. In this case:

- μ alone is not uniquely defined;
- τ_i alone is not uniquely defined;
- but differences such as $\tau_i - \tau_j$ are estimable.

This is a classic example. **Estimable Functions in Rank-Deficient Models** Even when the coefficient vector is not unique, fitted values are still unique, provided we project onto the column space of $\mathbf{X}$. Likewise, every estimable linear function has a unique value determined by the model.

So regression analysis in rank-deficient settings often focuses on estimable functions rather than on individual raw coefficients. **## Least Squares in Rank-Deficient Models**

When $\mathbf{X}$ is rank-deficient, the normal equations do not yield a unique coefficient vector. Different generalized inverse solutions may produce different coefficient vectors. However:

- the fitted values are unique;
- the residual sum of squares is unique;
- estimable linear functions have unique least squares estimates.

So the inferential target should be framed carefully. **Generalized Inverse View** A generalized inverse of $\mathbf{X}^\top \mathbf{X}$ can be used to write one least squares solution. This leads to expressions similar to the full-rank case, but students should remember that not every coefficient itself is uniquely meaningful. What matters most is whether the function of interest is estimable. **Software and Estimability** Modern software often handles rank deficiency automatically. It may:

- drop aliased columns;
- report coefficients as not estimable;
- use a default parameterization that makes the fit identifiable.

Students should not interpret every reported coefficient mechanically. They should understand the underlying estimable structure. **Example of rank deficiency**

```
X_bad <- cbind(1, diag(3))
X_bad <- rbind(X_bad, X_bad)
colnames(X_bad) <- c("Intercept", "G1", "G2", "G3")

qr(X_bad)$rank
```

```
[1] 3
```

```
ncol(X_bad)
```

```
[1] 4
```

```
X_bad
```

```
      Intercept G1 G2 G3
[1,]          1  1  0  0
[2,]          1  0  1  0
[3,]          1  0  0  1
[4,]          1  1  0  0
[5,]          1  0  1  0
[6,]          1  0  0  1
```

B.2 DFFITS and DFBETAS

Other influence measures include:

- **DFFITS**, which measures the effect of deleting an observation on its fitted value;
- **DFBETAS**, which measure the effect of deleting an observation on each estimated coefficient.

These are useful when we want to know not only whether a point is influential, but how it changes the model.

For an introductory treatment, Cook's distance and leverage usually provide a strong starting point. ## Added-Variable and Partial Residual Plots

When there are several predictors, ordinary residual plots may not fully reveal whether one variable needs a nonlinear term or whether its effect remains after adjustment. Useful advanced plots include:

- **added-variable plots**, which assess the contribution of one predictor after adjusting for others;
- **partial residual plots**, which help visualize possible nonlinearity for a specific predictor.

These are especially useful in multiple regression, though they are often introduced after students become comfortable with basic residual plots.

C Automatic Model Selection (Optional)

Optional appendix

The main course compares a small set of scientifically justified models and checks predictive performance. Automatic searches and additional selection criteria are supplementary. A search-selected model does not inherit the usual fixed-model interpretation of its p -values and confidence intervals.

C.1 Forward Selection

In **forward selection**, we begin with a small model, often the intercept-only model, and add predictors one at a time. At each step, we add the variable that gives the best improvement according to some criterion, such as:

- partial F test;
- smallest p -value;
- largest drop in AIC;
- largest increase in adjusted R^2 .

This continues until no candidate addition meets the chosen rule. **Backward Elimination** In **backward elimination**, we begin with a larger model and remove predictors one at a time. At each step, we remove the least useful variable according to a chosen rule. This continues until all remaining variables satisfy the stopping criterion. Backward elimination requires that the initial model be estimable, and it may be sensitive to collinearity and hierarchical structure. **Stepwise Selection**

Stepwise selection combines forward and backward ideas. A variable may enter at one stage and later be removed if its contribution becomes weak after other variables enter the model. This procedure is popular in software because it is automated, but it should be used cautiously. Automatic procedures can be unstable and may encourage overly mechanical modelling. **Problems With Automatic Selection**

Automatic variable selection can create several difficulties:

- it ignores model uncertainty;

- repeated searching inflates the chance of false discoveries;
- selected coefficients and p -values may look more certain than they really are;
- different but similar datasets may lead to different selected models;
- scientific structure may be lost if the procedure is used blindly.

So automated methods should be treated as exploratory tools, not as final arbiters of truth.

C.2 Additional comparison criteria

Criteria for Comparing Models

Several criteria are commonly used in model comparison.

Adjusted R Squared

Adjusted R^2 is

$$R_{\text{adj}}^2 = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)}.$$

Unlike ordinary R^2 , adjusted R^2 can decrease when unnecessary predictors are added. It rewards fit but penalizes unnecessary complexity in a simple way.

AIC

The Akaike information criterion is typically written as

$$\text{AIC} = -2 \log L + 2k,$$

where L is the fitted likelihood and k is the number of estimated parameters. Smaller AIC indicates a better tradeoff between fit and complexity. AIC is often more prediction-oriented than strict parsimony-oriented.

BIC

The Bayesian information criterion is

$$\text{BIC} = -2 \log L + k \log n.$$

Because the penalty term is stronger than that of AIC when n is moderate or large, BIC tends to favor smaller models.

Mallows' Cp

Another classical criterion is Mallows' C_p , which compares model bias and variance relative to a fuller model. It is less frequently emphasized in introductory software workflows today, but it remains conceptually important in regression theory.

C.3 An executable selection example

Generate data with multicollinearity

```
set.seed(8561)
n <- 80
x1 <- rnorm(n)
x2 <- 0.92 * x1 + rnorm(n, sd = 0.25)
x3 <- rnorm(n)
y <- 3 + 2 * x1 - 1.5 * x2 + 1.2 * x3 + rnorm(n, sd = 1)

dat <- data.frame(y = y, x1 = x1, x2 = x2, x3 = x3)

fit_full <- lm(y ~ x1 + x2 + x3, data = dat)
summary(fit_full)
```

Call:

```
lm(formula = y ~ x1 + x2 + x3, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.53882	-0.85013	-0.08111	0.85767	2.24530

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.0750	0.1281	24.008	< 2e-16 ***
x1	2.0710	0.4768	4.343	4.28e-05 ***
x2	-1.6962	0.5124	-3.310	0.00143 **
x3	1.1336	0.1277	8.880	2.27e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.117 on 76 degrees of freedom

Multiple R-squared: 0.5981, Adjusted R-squared: 0.5822

F-statistic: 37.7 on 3 and 76 DF, p-value: 4.966e-15

Compare with simpler models

```
fit_small <- lm(y ~ x1 + x3, data = dat)
fit_alt <- lm(y ~ x2 + x3, data = dat)

summary(fit_small)
```

Call:

```
lm(formula = y ~ x1 + x3, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.7697	-0.6645	-0.1148	0.9949	2.3101

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1177	0.1354	23.02	< 2e-16 ***
x1	0.5343	0.1159	4.61	1.57e-05 ***
x3	1.1421	0.1356	8.42	1.58e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.187 on 77 degrees of freedom

Multiple R-squared: 0.5401, Adjusted R-squared: 0.5282

F-statistic: 45.22 on 2 and 77 DF, p-value: 1.027e-13

```
summary(fit_alt)
```

Call:

```
lm(formula = y ~ x2 + x3, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.7455	-0.7881	-0.1158	1.0147	2.5958

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1185	0.1417	22.003	< 2e-16 ***
x2	0.4702	0.1301	3.615	0.000534 ***

```
x3          1.1415      0.1417    8.057 7.91e-12 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 1.24 on 77 degrees of freedom

Multiple R-squared: 0.4983, Adjusted R-squared: 0.4853

F-statistic: 38.24 on 2 and 77 DF, p-value: 2.927e-12

```
anova(fit_small, fit_full)
```

Analysis of Variance Table

Model 1: $y \sim x1 + x3$

Model 2: $y \sim x1 + x2 + x3$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	77	108.498				
2	76	94.824	1	13.673	10.959	0.001427 **

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Use AIC-based stepwise selection

```
fit_null <- lm(y ~ 1, data = dat)
fit_scope <- lm(y ~ x1 + x2 + x3, data = dat)

step_forward <- step(fit_null,
                     scope = list(lower = fit_null, upper = fit_scope),
                     direction = "forward",
                     trace = 0)

step_backward <- step(fit_scope, direction = "backward", trace = 0)

summary(step_forward)
```

Call:

```
lm(formula = y ~ x3 + x1 + x2, data = dat)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.53882	-0.85013	-0.08111	0.85767	2.24530

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.0750	0.1281	24.008	< 2e-16 ***
x3	1.1336	0.1277	8.880	2.27e-13 ***
x1	2.0710	0.4768	4.343	4.28e-05 ***
x2	-1.6962	0.5124	-3.310	0.00143 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.117 on 76 degrees of freedom

Multiple R-squared: 0.5981, Adjusted R-squared: 0.5822

F-statistic: 37.7 on 3 and 76 DF, p-value: 4.966e-15

```
summary(step_backward)
```

Call:

```
lm(formula = y ~ x1 + x2 + x3, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.53882	-0.85013	-0.08111	0.85767	2.24530

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.0750	0.1281	24.008	< 2e-16 ***
x1	2.0710	0.4768	4.343	4.28e-05 ***
x2	-1.6962	0.5124	-3.310	0.00143 **
x3	1.1336	0.1277	8.880	2.27e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.117 on 76 degrees of freedom

Multiple R-squared: 0.5981, Adjusted R-squared: 0.5822

F-statistic: 37.7 on 3 and 76 DF, p-value: 4.966e-15

Compare AIC, BIC, and adjusted R squared

```
model_summary <- function(model) {  
  c(  

```

```

    AIC = AIC(model),
    BIC = BIC(model),
    adj_R2 = summary(model)$adj.r.squared
  )
}

rbind(
  small = model_summary(fit_small),
  alt = model_summary(fit_alt),
  full = model_summary(fit_full),
  step_forward = model_summary(step_forward),
  step_backward = model_summary(step_backward)
)

```

	AIC	BIC	adj_R2
small	259.4064	268.9345	0.5281786
alt	266.3670	275.8951	0.4852878
full	250.6302	262.5403	0.5822139
step_forward	250.6302	262.5403	0.5822139
step_backward	250.6302	262.5403	0.5822139

Interpretation. Compare the selected variable sets, but do not treat this demonstration as evidence that an automatic search is scientifically optimal. If evaluating a selection procedure by resampling, run the entire selection procedure inside each training fold. Ordinary prediction validation does not repair post-selection confidence intervals.

D Generalized Least Squares and Correlated Errors (Optional)

In this optional appendix, we study what happens when the error terms in a linear model are no longer independent with common variance. This leads to generalized least squares, a natural extension of ordinary least squares that accounts for nonidentity covariance structure. The goal is to help students understand how the linear model changes when observations are correlated or have unequal precision in a more general matrix form.

i Optional appendix: GLS is different from GLM

GLS retains the mean model $E(Y) = X\beta$ and changes the error covariance. A generalized linear model instead specifies a response family and a link for its conditional mean. This appendix is supplementary to the main course.

Learning Objectives

By the end of this chapter, students should be able to:

- explain why ordinary least squares is not always appropriate when errors are correlated or heteroscedastic;
- state the generalized least squares covariance assumption for the error vector in a linear model context;
- derive the generalized least squares estimator;
- explain how generalized least squares reduces to ordinary least squares and weighted least squares in special cases;
- interpret the transformed-model view of generalized least squares;
- recognize practical examples of correlated errors and structured covariance models.

Reading

Recommended reading for this chapter:

- Seber and Lee:
 - sections on generalized least squares
 - correlated observations and covariance structure
 - extensions of least squares methods

- Montgomery, Peck, and Vining:
 - sections on departures from ordinary linear model assumptions
 - weighted and generalized least squares ideas
 - practical modelling considerations for dependence and unequal variance

Why Ordinary Least Squares Can Fail

So far, much of the course has relied on the classical assumption

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

This means:

- all observations have the same error variance;
- errors are uncorrelated.

But many real datasets do not satisfy this assumption. Examples include:

- repeated observations on the same unit;
- measurements taken over time;
- clustered or grouped data;
- observations with known unequal precision;
- spatial or serial dependence.

In such cases, ordinary least squares may still be unbiased for the mean model under suitable assumptions, but it is not generally the most efficient linear unbiased estimator, and its usual spherical-error standard errors need not be valid. Unbiasedness still requires a correctly specified conditional mean.

Review of the Linear Model

Recall the linear model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon,$$

with

$$\mathbb{E}[\varepsilon] = \mathbf{0}.$$

Under ordinary least squares, we assumed

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}_n.$$

This gave the estimator

$$\hat{\beta}_{OLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Now we allow a more general covariance structure.

D.1 A More General Covariance Model

Suppose instead that

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{V},$$

where $\mathbf{V}$ is a known positive definite matrix. Then

$$\text{Var}(\mathbf{Y}) = \sigma^2 \mathbf{V}.$$

The matrix $\mathbf{V}$ may represent:

- unequal variances;
- correlations among observations;
- both at once.

This is the setting of generalized least squares.

Why the Covariance Matrix Matters

The effect of correlation depends on the quantity being estimated. For equal marginal variance σ^2 and correlation ρ , $\text{Var}\{(Y_1 + Y_2)/2\} = \sigma^2(1 + \rho)/2$, whereas $\text{Var}(Y_1 - Y_2) = 2\sigma^2(1 - \rho)$. Positive correlation increases uncertainty for the pair average but decreases it for the within-pair difference. If one observation has much larger variance than another, it should typically receive less weight in estimation.

Thus the covariance structure affects how much information each part of the data contributes. Ordinary least squares ignores this structure. Generalized least squares incorporates it directly.

D.2 The Generalized Least Squares Criterion

In generalized least squares, we minimize the quadratic form

$$Q(\beta) = (\mathbf{Y} - \mathbf{X}\beta)^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\beta).$$

This is the natural analogue of the ordinary least squares criterion

$$(\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta),$$

but now weighted by the inverse covariance structure. The matrix $\mathbf{V}^{-1}$ downweights directions in the data that are more variable or more redundant.

Derivation of the Generalized Least Squares Estimator

Differentiate the generalized least squares criterion with respect to β and set the derivative equal to zero. This gives the generalized normal equations

$$\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \hat{\beta}_{GLS} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}.$$

Assuming invertibility, the generalized least squares estimator is

$$\hat{\beta}_{GLS} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}.$$

This is the central formula for the week.

Interpretation of the GLS Estimator

The GLS estimator has the same structural form as OLS and WLS, but with the covariance matrix inserted. It can be interpreted as an estimator that gives appropriate weight to observations according to the joint covariance structure. Compared with OLS:

- observations with larger variance are effectively downweighted;
- correlated observations are not treated as if they were independent;
- the estimator uses the information in the data more efficiently when $\mathbf{V}$ is correctly specified.

Special Case: Ordinary Least Squares

If $\mathbf{V} = \mathbf{I}_n$, then GLS becomes

$$\hat{\beta}_{GLS} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

which is exactly the ordinary least squares estimator. So OLS is a special case of GLS.

Special Case: Weighted Least Squares

If the covariance matrix is diagonal, $\mathbf{V} = \text{diag}(v_1, \dots, v_n)$, then

$$\mathbf{V}^{-1} = \text{diag}(1/v_1, \dots, 1/v_n).$$

This is the weighted least squares setting from the preceding chapter, with weights proportional to inverse variances. So WLS is also a special case of GLS.

D.3 The Transformed-Model View

A very important way to understand GLS is through a transformation. Because $\mathbf{V}$ is positive definite, there exists a matrix $\mathbf{A}$ such that

$$\mathbf{A}^\top \mathbf{A} = \mathbf{V}^{-1}.$$

For example, we may take $\mathbf{A} = \mathbf{V}^{-1/2}$. Multiplying the model by $\mathbf{A}$ gives

$$\mathbf{A}\mathbf{Y} = \mathbf{A}\mathbf{X}\beta + \mathbf{A}\epsilon.$$

Now the transformed error has variance

$$\text{Var}(\mathbf{A}\epsilon) = \mathbf{A}(\sigma^2 \mathbf{V})\mathbf{A}^\top = \sigma^2 \mathbf{I}_n.$$

So GLS can be viewed as ordinary least squares applied to a transformed model with spherical errors. This is one of the most important conceptual ideas in the course.

Distribution of the GLS Estimator

If the covariance structure is correctly specified and

$$\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{V}),$$

then the GLS estimator is normally distributed:

$$\hat{\beta}_{GLS} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}).$$

This parallels the OLS distribution theory, but with $\mathbf{V}^{-1}$ appearing throughout.

Gauss-Markov Interpretation

Under the covariance model

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{V},$$

the generalized least squares estimator is the best linear unbiased estimator among all linear unbiased estimators. So GLS is the natural Gauss-Markov extension of OLS to nonidentity covariance structures. This is sometimes phrased by saying that OLS is BLUE under spherical errors, while GLS is BLUE under the more general covariance model.

Estimating Sigma Squared Under GLS

In the transformed model, residual sums of squares can be defined using the GLS criterion. The generalized residual sum of squares is

$$SSE_{GLS} = (\mathbf{Y} - \mathbf{X}\hat{\beta}_{GLS})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\beta}_{GLS}).$$

Under the normal model with known $\mathbf{V}$, this plays the same role as the ordinary SSE after transformation. An estimator of σ^2 is then

$$\hat{\sigma}_{GLS}^2 = \frac{SSE_{GLS}}{n - p}.$$

Why Known $\mathbf{V}$ Is Rare

In practice, the covariance matrix $\mathbf{V}$ is usually not known exactly. Instead, we may have:

- scientific knowledge suggesting a covariance pattern;
- repeated-measures structure suggesting dependence;
- known weights from design considerations;
- an estimated covariance model.

When $\mathbf{V}$ is replaced by an estimate, the resulting estimator is often called **feasible GLS**.

Feasible Weighted Least Squares

In many applications, the variances are not known exactly. Instead, we estimate the variance pattern from the data and then use estimated weights. This is often called **feasible weighted least squares**. Typical workflow:

- fit an initial OLS model;
- inspect residuals to understand how variance changes;
- propose a variance model;
- compute estimated weights;
- refit using WLS.

This approach is practical, though it introduces additional modelling decisions.

! Known versus estimated covariance

Exact normal-model t and F distributions apply to the transformed model when the relative covariance is known and the remaining assumptions hold. Plugging in estimated weights or covariance parameters does not automatically preserve those exact distributions.

Feasible Generalized Least Squares

The basic idea of feasible GLS is:

- propose a covariance model for $\mathbf{V}$;
- estimate the unknown covariance parameters from the data;
- plug the estimated covariance matrix into the GLS formula.

This is practical, but it introduces an extra modelling step and relies on the covariance model being approximately correct.

D.4 Examples of Covariance Structures

Several structured covariance matrices appear often in applications.

Unequal Variances Only

If observations are independent but have different variances, then $\mathbf{V}$ is diagonal. This is the weighted least squares case.

Compound Symmetry

In grouped or repeated-measures settings, one may assume a common variance and a common correlation within group. This gives a covariance pattern often called compound symmetry.

Autoregressive Structure

For time-ordered observations, nearby errors may be more strongly correlated than distant ones. A simple model is the AR(1) structure, where correlations decay geometrically with lag.

Block-Diagonal Structure

If observations are grouped into independent clusters, then $\mathbf{V}$ may be block diagonal, with each block describing within-cluster dependence. These examples help students see that covariance structure is part of model formulation, not just a technical afterthought.

Correlated Errors in Time-Ordered Data

Suppose we observe data over time:

$$Y_t = \beta_0 + \beta_1 x_t + \varepsilon_t.$$

If the errors satisfy

$$\text{Corr}(\varepsilon_t, \varepsilon_{t-1}) \neq 0,$$

then ordinary least squares may still estimate the mean trend, but standard errors based on independence are not trustworthy. This is one of the classic motivations for generalized least squares.

Clustered and Repeated Observations

Suppose several observations come from the same individual, school, hospital, or geographic region. Then responses within the same cluster may be correlated. In this case, the assumption of independent errors is not appropriate, and a structured covariance model or a mixed-model approach may be better suited. GLS provides an important stepping stone toward these more advanced frameworks.

Relationship to Robust Standard Errors

One alternative to full covariance modelling is to keep the OLS mean estimator but adjust the standard errors to be robust to heteroscedasticity or dependence. This is a different strategy from GLS.

- GLS changes the estimator itself using a covariance model;
- robust standard errors keep the estimator but correct inference approximately.

Both are useful, but they serve different purposes. This distinction is valuable for students to understand even if robust methods are treated only briefly.

When GLS Is Worth Using

GLS is especially attractive when:

- there is a defensible covariance model;
- the dependence or heteroscedasticity is substantial;
- efficient estimation matters;
- the transformed model remains interpretable.

If the covariance model is poorly specified, the gains from GLS may be limited, and robustness or alternative modelling strategies may be preferable.

Diagnostics for Correlated Errors

Symptoms that may suggest correlated errors include:

- residuals that display runs or systematic temporal patterns;
- residual plots against time showing clustering;
- repeated-measures structure built into the design;
- known grouping or spatial dependence in the data collection process.

The key lesson is that covariance structure should be motivated both by diagnostics and by how the data were collected.

Worked Example With Known Unequal Precision

Suppose each response is an average of m_i repeated measurements. Then the variance of the average may satisfy

$$\text{Var}(Y_i) = \frac{\sigma^2}{m_i}.$$

In that case, a natural choice is

$$v_i = \frac{1}{m_i}, \quad w_i = m_i.$$

So observations based on more repeated measurements get more weight. This is a very intuitive example of GLS through weighted least squares.

Worked Example With Correlated Pairs

Suppose observations come in natural pairs, and within each pair the errors are positively correlated. Then the covariance matrix has a block structure, with a 2×2 covariance block for each pair. GLS accounts for covariance when estimating the target mean or contrast. For positive within-pair correlation, averaging and differencing have different information consequences; compare their variances above.

R Demonstration With Weighted Least Squares as GLS

Simulate data with unequal variances

```

set.seed(8561)
n <- 40
x <- seq(1, 20, length.out = n)
v <- 0.3 + 0.08 * x^2
y <- 5 + 1.2 * x + rnorm(n, sd = sqrt(v))

dat <- data.frame(y = y, x = x, v = v)

fit_ols <- lm(y ~ x, data = dat)
fit_wls <- lm(y ~ x, data = dat, weights = 1 / v)

summary(fit_ols)

```

Call:

```
lm(formula = y ~ x, data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.636	-2.163	0.190	1.948	9.834

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.4316	1.4586	3.724	0.000635 ***
x	1.1426	0.1225	9.331	2.26e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.356 on 38 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6882

F-statistic: 87.07 on 1 and 38 DF, p-value: 2.257e-11

```
summary(fit_wls)
```

Call:

```
lm(formula = y ~ x, data = dat, weights = 1/v)
```

Weighted Residuals:

Min	1Q	Median	3Q	Max
-2.65080	-0.75591	0.07032	0.67369	2.38994

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.44884	0.47201	11.54	5.46e-14 ***
x	1.14032	0.08545	13.34	6.42e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

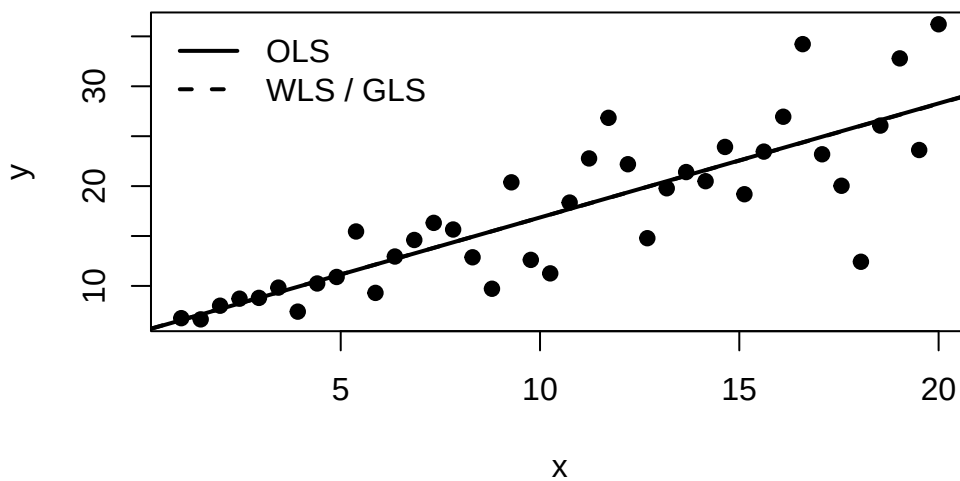
Residual standard error: 1.239 on 38 degrees of freedom

Multiple R-squared: 0.8241, Adjusted R-squared: 0.8195

F-statistic: 178.1 on 1 and 38 DF, p-value: 6.418e-16

Compare fitted lines

```
plot(dat$x, dat$y, pch = 19, xlab = "x", ylab = "y")
abline(fit_ols, lwd = 2)
abline(fit_wls, lwd = 2, lty = 2)
legend("topleft",
      legend = c("OLS", "WLS / GLS"),
      lty = c(1, 2),
      lwd = 2,
      bty = "n")
```



D.5 R Demonstration With a Hand-Built GLS Computation

Construct a covariance matrix and compute GLS directly

```
set.seed(8561)
n2 <- 8
x2 <- seq(1, n2)
X <- cbind(1, x2)

rho <- 0.6
V <- outer(1:n2, 1:n2, function(i, j) rho^abs(i - j))
beta_true <- c(2, 1.5)

eps <- t(chol(V)) %*% rnorm(n2)
y2 <- as.vector(X %*% beta_true + eps)

Y <- matrix(y2, ncol = 1)

beta_gls <- solve(t(X) %*% solve(V) %*% X) %*% t(X) %*% solve(V) %*% Y
beta_ols <- solve(t(X) %*% X) %*% t(X) %*% Y

beta_gls
```

```
      [,1]
3.336799
x2 1.299329
```

```
beta_ols
```

```
      [,1]
3.379372
x2 1.309102
```

Transform the model and verify the GLS view

```
A <- solve(t(chol(V))) # chol(V) is upper triangular: V = R'R
Y_tilde <- A %*% Y
X_tilde <- A %*% X

beta_transformed <- solve(t(X_tilde) %*% X_tilde) %*% t(X_tilde) %*% Y_tilde
beta_transformed
```

```
[,1]  
3.336799  
x2 1.299329
```

```
max(abs(beta_transformed - beta_gls)) # numerical agreement with direct GLS
```

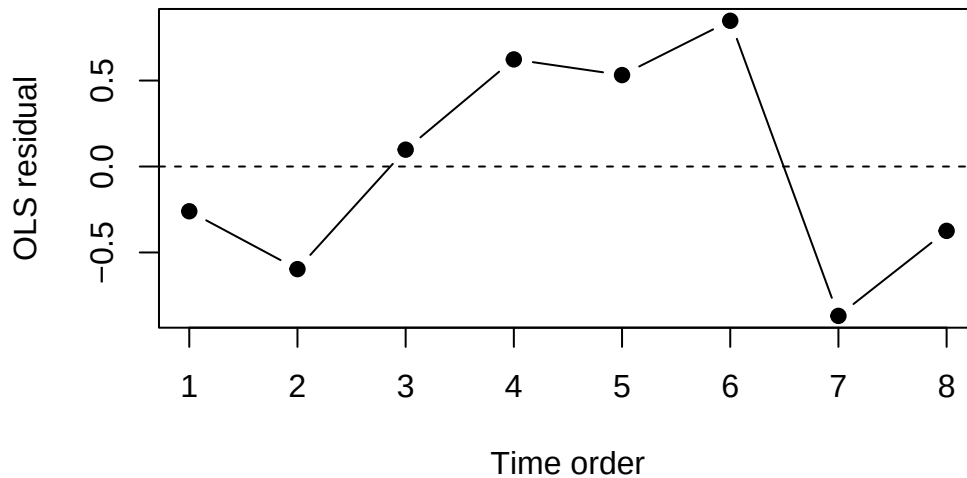
```
[1] 9.769963e-15
```

```
max(abs(A %*% V %*% t(A) - diag(n2))) # whitening check
```

```
[1] 1.804112e-16
```

Compare residual patterns

```
fit_ols2 <- lm(y2 ~ x2)  
  
plot(x2, resid(fit_ols2), type = "b", pch = 19,  
      xlab = "Time order", ylab = "OLS residual")  
abline(h = 0, lty = 2)
```



Simple example of grouped covariance intuition

```

set.seed(8561)
group <- rep(1:10, each = 3)
x3 <- rnorm(length(group))
u <- rnorm(10, sd = 1.2)
eps3 <- rep(u, each = 3) + rnorm(length(group), sd = 0.5)
y3 <- 4 + 2 * x3 + eps3

dat3 <- data.frame(y = y3, x = x3, group = factor(group))
fit3 <- lm(y ~ x, data = dat3)
summary(fit3)

```

Call:

```
lm(formula = y ~ x, data = dat3)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.3937	-0.8770	-0.0909	0.9381	2.9870

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.7464	0.2929	12.789	3.27e-13 ***
x	2.1451	0.2450	8.757	1.66e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.603 on 28 degrees of freedom

Multiple R-squared: 0.7325, Adjusted R-squared: 0.723

F-statistic: 76.68 on 1 and 28 DF, p-value: 1.656e-09

Interpreting Software Output

In base R, ordinary `lm()` directly handles OLS and weighted least squares through the `weights=` argument. More general correlated-error models often require additional packages in practice, but the matrix formula is already enough to understand the core idea of GLS. Students should focus on:

- what covariance structure is being assumed;
- why that structure is plausible;
- how the weighting or transformation changes the estimator;
- what practical goal is being improved.

A Practical Workflow for GLS Thinking

A sensible workflow is:

- ask whether independence and equal variance are plausible from the design of the study;
- inspect residual patterns for signs of heteroscedasticity or dependence;
- propose a covariance structure or weighting scheme when justified;
- compare OLS and GLS-style fits;
- interpret the results in light of both efficiency and model credibility.

In-Class Discussion Questions

1. Why is weighted least squares a special case of generalized least squares?
2. Why can correlated observations carry less information than independent observations?
3. What is the conceptual benefit of the transformed-model view of GLS?
4. When might robust standard errors be preferred to specifying a full covariance model?

D.6 Practice Problems

Conceptual

1. Explain why OLS is not generally efficient when the covariance matrix is not proportional to the identity.
2. Explain how GLS uses the inverse covariance matrix to weight information in the data.
3. Explain the difference between modelling the covariance structure and simply correcting standard errors.

Computational

Suppose

$$\text{Var}(\varepsilon) = \sigma^2 \mathbf{V}, \quad \mathbf{V} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}.$$

1. Write down $\mathbf{V}^{-1}$.
2. Which observation receives more weight in GLS?
3. Explain why.

Now suppose

$$\hat{\beta}_{GLS} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}.$$

1. Show how this reduces to OLS when $\mathbf{V} = \mathbf{I}_n$.
2. Show how this reduces to WLS when $\mathbf{V}$ is diagonal.

Modelling Problem

You have repeated observations on each subject over time.

1. Why is the independence assumption questionable?
2. What kinds of covariance structure might be reasonable?
3. Why might GLS be more appropriate than OLS in this setting?

Suggested Homework

Complete the following tasks:

- fit an OLS model and a weighted least squares model for data with unequal variance;
- compare coefficient estimates, standard errors, and residual plots;
- compute a GLS estimator by hand for a small example with a known covariance matrix;
- explain the transformed-model view in your own words;
- write a short discussion of when a covariance model is scientifically credible enough to justify GLS.

Summary

In this chapter, we studied generalized least squares as an extension of ordinary least squares to models with nonidentity covariance structure. We emphasized that:

- OLS assumes spherical errors;
- GLS allows unequal variances and correlated errors;
- WLS is a special case of GLS;
- GLS can be understood as OLS on a transformed model;
- covariance structure is part of model formulation and should be justified by both design and diagnostics.

E GLM Likelihood and Computation (Optional)

i Optional appendix

The main GLM chapter emphasizes response distributions, links, interpretation, and practical model checks. This appendix connects those ideas to exponential-family likelihoods and iteratively reweighted least squares. The general regularity theory is outside the core syllabus.

E.1 Exponential-family structure

For independent observations conditional on the predictors, a standard exponential-dispersion-family representation is

$$f(y_i; \theta_i, \phi) = \exp \left\{ \frac{y_i \theta_i - b(\theta_i)}{a_i(\phi)} + c(y_i, \phi) \right\}.$$

Then $\mu_i = b'(\theta_i)$ and $\text{Var}(Y_i | x_i) = a_i(\phi)b''(\theta_i)$. For unit-weight observations it is common to write this variance as $\phi V(\mu_i)$. A link specifies $g(\mu_i) = x_i^\top \beta$; a canonical link makes the linear predictor equal to the natural parameter θ_i .

Family	Mean	Variance, unit weight	Canonical link
Gaussian	μ	σ^2	identity
Bernoulli	π	$\pi(1 - \pi)$	logit
Poisson	μ	μ	log

For grouped binomial observations, the trial count must be included; the mean and variance for a count differ from those for a proportion.

E.2 Why weighted least squares reappears

At a current fit, let $\eta_i = x_i^\top \beta + o_i$, where o_i is a known offset, and define the working response and weights

$$z_i = \eta_i + (y_i - \mu_i)g'(\mu_i), \quad w_i = \{\text{Var}(Y_i | x_i)[g'(\mu_i)]^2\}^{-1}.$$

A Fisher-scoring step solves weighted least squares for $z - o$:

$$\beta^{new} = (X^\top W X)^{-1} X^\top W (z - o).$$

In computation, solve the weighted system with a factorization rather than explicitly forming this inverse. Recompute the working mean and weights until convergence. These are local, iteration-dependent weights; they are different from fixed precision weights in the Gaussian WLS model.

For Bernoulli logistic regression, $w_i = \pi_i(1 - \pi_i)$; for Poisson regression with a log link, $w_i = \mu_i$. For a Gaussian identity-link model with constant variance, the update reduces to ordinary least squares.

Why convergence matters. A full-rank design alone does not guarantee a finite logistic-regression MLE. Complete or quasi-complete separation can drive coefficients toward infinity. Inspect fitting warnings and the data pattern before interpreting standard errors or Wald tests.

R's `glm.fit()` uses iteratively reweighted least squares; its working weights should not be confused with the user-supplied prior weights. See the [R glm documentation](#).

E.3 Likelihood-based inference

A local quadratic approximation to the log likelihood gives an approximate covariance based on Fisher information. Under suitable regularity conditions, this yields asymptotic normal inference for estimable coefficients. For nested models and an interior null parameter, twice the maximized log-likelihood difference has an asymptotic chi-square reference distribution with degrees of freedom equal to the number of independent restrictions.

These approximations can fail or be inaccurate near boundaries, under separation, with sparse data, or with an incorrectly specified sampling model. They are not the exact finite-sample t and F results established for the normal linear model.

Beyond the core. Profile likelihood, bias-reduced logistic estimation, negative-binomial models, quasi-likelihood, and clustered/generalized mixed models require additional assumptions

and methods. Recognizing the limitation of a basic fit is a core skill; developing all of these remedies is supplementary.

F Quantile Regression (Optional)

Optional self-study appendix

This reading extends the regression framework to conditional quantiles. It is supplementary to the main lecture sequence. Read Chapters 2, 5, and 8 first: least squares, coefficient interpretation, and diagnostics are the useful prerequisites. The R examples use the `quantreg` package; install it once with `install.packages("quantreg")` if needed.

F.1 When the mean is not the whole question

OLS describes a conditional mean. In some studies, the scientific question concerns a typical response, a low outcome, or an unusually high outcome. For example, average delivery time and the 90th percentile of delivery time answer different service-planning questions. A covariate may change the spread of the response even when its relationship with the mean is simple.

Definition F.1 (Conditional Quantile). For $0 < \tau < 1$, define

$$Q_\tau(Y \mid X = x) = \inf\{q : P(Y \leq q \mid X = x) \geq \tau\}.$$

A linear quantile model specifies $Q_\tau(Y \mid X = x) = x^\top \beta_\tau$. The coefficients may vary with τ . At $\tau = 0.5$, the target is a conditional median.

Interpretation. Without an interaction involving x_j , $\beta_{\tau,j}$ is the change in the modeled conditional τ -quantile for a one-unit change in x_j , holding the other predictors fixed. This compares conditional distributions. It does not track a particular person who remains at the same percentile, and it is not automatically a causal effect.

Use all observations to estimate a conditional quantile

Fitting the 90th percentile does not mean selecting the largest 10% of responses and fitting OLS to them. Quantile regression uses the full dataset with a loss function that gives different penalties to positive and negative residuals. A marginal percentile of all responses is also different from a percentile conditional on the predictors.

Why a different loss? Least squares penalizes a residual $u = y - x^\top b$ by u^2 . Quantile regression instead uses the check loss, also called pinball loss:

$$\rho_\tau(u) = u\{\tau - \mathbf{1}(u < 0)\} = \begin{cases} \tau u, & u \geq 0, \\ (1 - \tau)(-u), & u < 0, \end{cases}$$

$$\hat{\beta}_\tau \in \arg \min_b \sum_{i=1}^n \rho_\tau(y_i - x_i^\top b).$$

At $\tau = 0.5$, minimizing check loss is equivalent to minimizing the sum of absolute residuals. At $\tau = 0.9$, underpredicting the response costs nine times as much per unit as overpredicting it. The fitted target consequently moves toward the upper tail.

Proposition F.1 (Why Check Loss Targets a Quantile). *If $\mathbb{E}(|Y|) < \infty$, the minimizers of $\mathbb{E}\{\rho_\tau(Y - a)\}$ are characterized by $F(a-) \leq \tau \leq F(a)$; the conditional-quantile definition above selects the lower endpoint when the minimizer is not unique. Where F is continuous, the derivative with respect to a is $F(a) - \tau$. It is nonpositive below a minimizing quantile and nonnegative above it. The same argument applies to the conditional distribution at a fixed x .*

Scope of the linear model. The population argument allows a different quantile at every x . A linear quantile regression restricts these values to one linear function. If the true conditional quantile is nonlinear, the fitted line is a best approximation under the chosen loss, rather than an exact description at every predictor value.

```
check_loss <- function(u, tau) u * (tau - (u < 0))
u <- seq(-3, 3, length.out = 301)
taus <- c(0.1, 0.5, 0.9)
cols <- c("steelblue4", "gray35", "darkorange3")
loss <- sapply(taus, function(tau) check_loss(u, tau))
matplot(u, loss, type = "l", lty = 1, lwd = 2, col = cols,
        xlab = "Residual: observed - predicted", ylab = "Check loss")
abline(v = 0, lty = 3, col = "gray70")
legend("top", legend = paste("tau =", taus), col = cols,
      lty = 1, lwd = 2, bty = "n", horiz = TRUE)
```

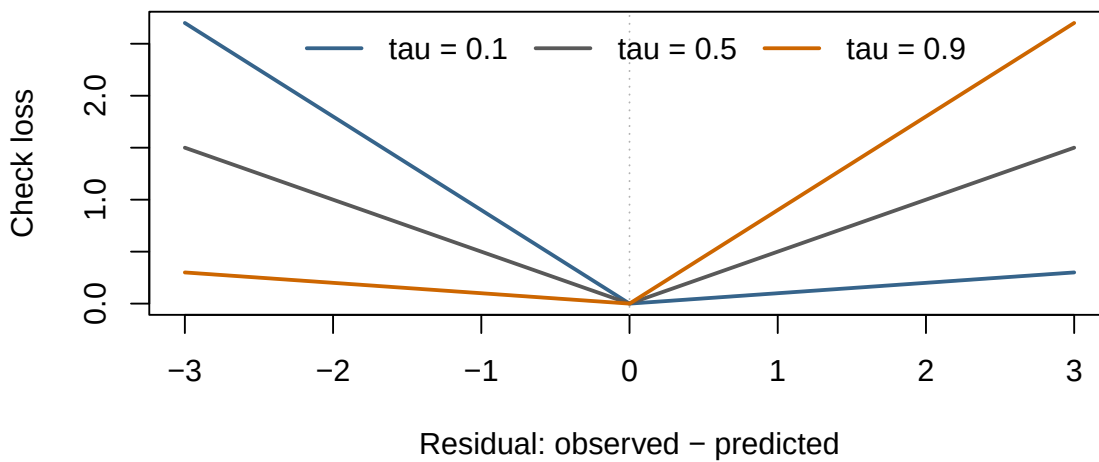


Figure F.1: Check loss for three quantiles. A positive residual means the prediction is too low. The upper-quantile loss penalizes that side more strongly.

F.2 A mean model can miss changing spread

Consider the simulated model $Y = 5 + 2x + (0.5 + 0.8x)Z$, with $0 \leq x \leq 4$ and $Z \sim N(0, 1)$ independent of x . Its conditional mean and median are both $5 + 2x$, while

$$Q_{\tau}(Y \mid x) = 5 + 2x + (0.5 + 0.8x)\Phi^{-1}(\tau).$$

The conditional-quantile slopes are $2 + 0.8\Phi^{-1}(\tau)$. Different slopes here reflect changing spread, even though every observation follows the same data-generating model. Quantile regression need not be reserved for skewed distributions.

```
set.seed(8561)
n <- 350
x <- runif(n, 0, 4)
dat_qr <- data.frame(x, y = 5 + 2 * x +
                     (0.5 + 0.8 * x) * rnorm(n))
fit_mean <- lm(y ~ x, data = dat_qr)
fit_quantiles <- quantreg::rq(y ~ x, tau = taus, data = dat_qr)
round(coef(fit_quantiles), 3)
```

	tau= 0.1	tau= 0.5	tau= 0.9
(Intercept)	4.509	5.078	5.850
x	0.768	1.816	2.965

```
data.frame(tau = taus, true_slope = 2 + 0.8 * qnorm(taus),
           estimated_slope = unname(coef(fit_quantiles)["x", ]))
```

	tau	true_slope	estimated_slope
1	0.1	0.9747587	0.7676385
2	0.5	2.0000000	1.8156789
3	0.9	3.0252413	2.9651924

The matrix returned by `coef()` contains one column per requested quantile. R's [rq\(\) documentation](#) describes its formula interface and the `tau` argument.

```
new_x <- data.frame(x = seq(0, 4, length.out = 200))
qhat <- predict(fit_quantiles, newdata = new_x)
plot(dat_qr$x, dat_qr$y, pch = 16, cex = 0.65,
     col = adjustcolor("gray40", 0.35), xlab = "x", ylab = "Response")
matlines(new_x$x, qhat, col = cols, lty = 1, lwd = 2)
abline(fit_mean, lty = 2, lwd = 2)
legend("topleft", legend = c(paste("Quantile", taus), "OLS mean"),
     col = c(cols, "black"), lty = c(1, 1, 1, 2), lwd = 2, bty = "n")
```

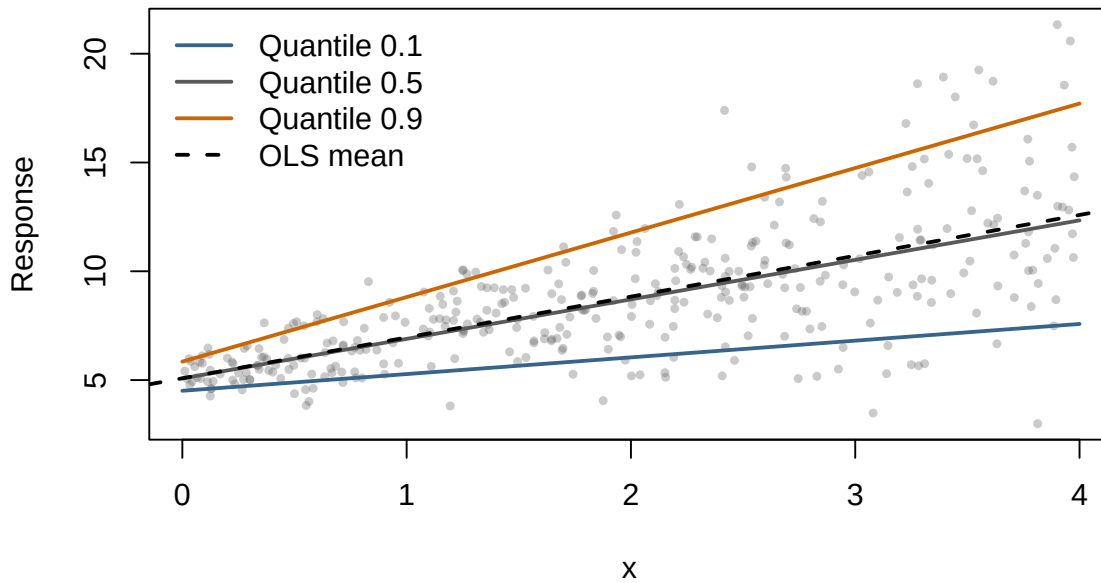


Figure F.2: Fitted conditional quantiles and the OLS mean in simulated data with increasing spread. The growing distance between the lower and upper quantiles reveals variation that a mean line alone does not describe.

Read the graph. Compare the predicted 10th, 50th, and 90th percentiles at a fixed x . The gap between the 10th and 90th percentiles describes an interval containing the central 80% of a continuous conditional distribution when the quantile model is correct. The fitted gap is not a confidence interval for the mean or for a regression coefficient, and its predictive coverage is not guaranteed in a finite sample.

F.3 Uncertainty and practical checks

Quantile regression avoids a normal-error or constant-variance requirement for defining its target, but it is not assumption-free. Reliable estimation and inference still depend on the quantile specification, the design, the sampling structure, and sufficient information near the quantile of interest. Very extreme quantiles are difficult to estimate from small datasets.

A pairs-bootstrap illustration. These simulated observations are independent draws, so resampling whole (x_i, y_i) pairs is appropriate. The code requests a bootstrap standard error

for the median-regression slope and forms an approximate normal interval. The replicate count is kept moderate for the notes; increase it and check Monte Carlo stability in a substantive analysis.

```
set.seed(8561)
fit_median <- quantreg::rq(y ~ x, tau = 0.5, data = dat_qr)
median_summary <- summary(fit_median, se = "boot", R = 499,
                           bsmethod = "xy")
b <- coef(fit_median)["x"]
se_b <- median_summary$coefficients["x", "Std. Error"]
c(slope = unname(b), bootstrap_SE = se_b,
  lower = unname(b - qnorm(0.975) * se_b),
  upper = unname(b + qnorm(0.975) * se_b))
```

slope	bootstrap_SE	lower	upper
1.815679	0.136845	1.547468	2.083890

This is not the exact finite-sample OLS t interval from Chapter 3. For dependent or clustered observations, an independent-pairs bootstrap generally fails to preserve the sampling structure. The package's [summary.rq\(\)](#) [documentation](#) describes alternative inference methods.

! Two fitted quantiles need a joint comparison

If one slope is significant at $\tau = 0.1$ and another is not significant at $\tau = 0.9$, that does not establish that the slopes differ. They are estimated from the same observations and are dependent. A difference such as $\beta_{0.9,j} - \beta_{0.1,j}$ requires joint inference that retains this dependence, for example refitting both quantiles in each common bootstrap sample.

Before interpreting a fit: check the functional form, influential predictor values, the amount of information in the tails, and whether separately fitted quantiles cross. Linear loss is less sensitive to very large response residuals than squared loss, but it does not eliminate leverage problems. Crossing fitted quantiles cannot describe a valid ordered distribution at that x ; investigate misspecification or use methods that enforce ordering rather than silently relabeling the curves.

Evaluate the stated target. For held-out data, compare candidate predictions at the same τ using mean check loss. A coverage diagnostic compares the fraction of responses below the predicted quantile with τ , including checks across meaningful predictor ranges. Good overall coverage can hide poor conditional coverage. Use splits that match future prediction, as in Chapter 10.

F.4 Optional application and self-study

i Optional application: household expenditure

The `engel` dataset bundled with `quantreg` contains income and food expenditure. Use `data("engel", package = "quantreg")`, plot the variables, and fit the 0.1, 0.5, and 0.9 conditional quantiles of `foodexp` given `income`. Explain how a household-expenditure quantile differs from mean expenditure. Treat the fitted relationships as descriptive associations, and avoid extrapolating beyond the observed incomes.

1. Use the loss definition to explain why the 90th percentile is penalized more for underprediction.
2. Derive the true intercept and slope at each quantile in the simulated example. Explain why the median and mean agree here.
3. Predict the 10th and 90th percentiles at $x = 1$ and $x = 3$. Explain what their widening gap means.
4. Design a bootstrap for comparing two quantile slopes. State which observations must stay together if the data are clustered.

Suggested reading. Start with the [quantreg package documentation](#), then the vignette available through `vignette("rq", package = "quantreg")`. Roger Koenker's *Quantile Regression* develops the estimation and inference theory in greater depth.

G Mixed-Effects Models (Optional)

Optional self-study appendix

This reading extends linear models to grouped and repeated observations. It is supplementary to the main lecture sequence. Chapters 1, 3, and 7 supply the useful prerequisites: covariance, inference, and the experimental unit. The R examples use `lme4`; install it once with `install.packages("lme4")` if needed. The covariance connection to GLS is developed here, so the GLS appendix is not a prerequisite.

G.1 Why group membership changes the analysis

Measurements from the same person, classroom, clinic, or production batch often share influences. Treating all rows as independent can give misleading standard errors. Repeated measurements provide information about within-unit change, but they do not create additional independently sampled or randomized units.

A linear mixed-effects model combines shared regression coefficients with group-specific deviations. **Fixed effects** describe common coefficients, possibly including measured group characteristics. **Random effects** represent latent group deviations through a distribution. These terms describe components of the model; they do not mean that a predictor must be nonrandom or that treatment assignment was randomized.

Definition G.1 (A Random-Intercept Model). For observation i in group j , let

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_j + \varepsilon_{ij},$$
$$b_j \stackrel{\text{iid}}{\sim} N(0, \tau^2), \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

Condition on the observed predictors. Assume that the group effects and residual errors are mutually independent, with the displayed distributions unchanged after conditioning on those predictors. Conditional on b_j , the group has intercept $\beta_0 + b_j$. After integrating over b_j , its mean is $\beta_0 + \beta_1 x_{ij}$.

For two distinct observations in the same group, the shared b_j gives covariance τ^2 ; their marginal variances are $\tau^2 + \sigma^2$. Therefore the within-group correlation in this simple model is

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

This intraclass correlation describes the fraction of conditional-on-predictors variation attributable to a shared group intercept. It is not a universal correlation formula: random slopes or additional residual correlation change it.

Connection to the course. In matrix notation,

$$Y = X\beta + Zb + \varepsilon, \quad b \sim N(0, G), \quad \varepsilon \sim N(0, R), \\ \text{Var}(Y \mid X, Z) = ZGZ^\top + R,$$

with b and ε independent and the design conditioned on. For a known covariance $V = ZGZ^\top + R$, estimation of β has the GLS form. In practice, the covariance parameters are estimated too, so the exact OLS t/F theory does not transfer unchanged.

! A random intercept does not repair every dependence problem

The basic model assumes independent groups and independent residual errors conditional on the random effects. Serial correlation, household relationships across groups, or an omitted random slope may require a different covariance structure. A random-effects model also does not remove confounding: correlation between group effects and predictors can invalidate its interpretation unless modeled appropriately.

G.2 Partial pooling and shrinkage

Three possible analyses of group means illustrate what the random-effects assumption adds:

Approach	What is shared?	Main tradeoff
Complete pooling	One mean for every group	Ignores genuine group differences.
Separate group means	Each group has its own unrestricted mean	Small groups can have very noisy estimates.
Partial pooling	Groups share a distribution of deviations	Stabilizes estimates using an explicit exchangeability assumption.

Proposition G.1 (Shrinkage in the Intercept-Only Model). *For $Y_{ij} = \mu + b_j + \varepsilon_{ij}$ under the Gaussian assumptions above, suppose μ , $\tau^2 > 0$, and $\sigma^2 > 0$ are known. If group j has n_j observations, then*

$$\mathbb{E}(\mu + b_j \mid \bar{Y}_j) = \mu + w_j(\bar{Y}_j - \mu), \quad w_j = \frac{\tau^2}{\tau^2 + \sigma^2/n_j}.$$

The conditional group mean is a weighted average of the observed group mean and the common mean. Small groups have less precise sample means and receive more shrinkage toward the common mean. Larger between-group variation reduces shrinkage.

Why the weight appears. The sample mean contains a group signal b_j with variance τ^2 and averaged residual noise with variance σ^2/n_j . Conditioning a jointly normal pair yields the signal-to-total-variance weight. Plugging estimated parameters into this expression gives an empirical best linear unbiased prediction (empirical BLUP) in this simple model. More general random-slope models have a matrix version rather than this scalar weight.

```
set.seed(8561)
n_group <- c(3, 4, 5, 6, 8, 10, 12, 16, 20, 24, 30, 40)
group <- factor(rep(LETTERS[seq_along(n_group)], n_group))
b_true <- rnorm(length(n_group), sd = 2)
dat_group <- data.frame(group,
  y = 12 + b_true[as.integer(group)] + rnorm(length(group), sd = 3))
fit_pool <- lme4::lmer(y ~ 1 + (1 | group), data = dat_group,
  REML = TRUE)
mu_hat <- unname(lme4::fixef(fit_pool)[1])
tau2_hat <- as.numeric(lme4::VarCorr(fit_pool)$group[1, 1])
sigma2_hat <- sigma(fit_pool)^2
raw_mean <- tapply(dat_group$y, dat_group$group, mean)
shrunk_mean <- mu_hat + lme4::ranef(fit_pool)$group[, "(Intercept)"]
w_hat <- tau2_hat / (tau2_hat + sigma2_hat / n_group)
round(data.frame(n = n_group, raw_mean, mixed_mean = shrunk_mean,
  weight_on_group_mean = w_hat), 3)
```

	n	raw_mean	mixed_mean	weight_on_group_mean
A	3	16.277	14.780	0.647
B	4	9.330	10.117	0.709
C	5	15.146	14.380	0.753
D	6	12.371	12.300	0.786
E	8	12.053	12.051	0.830
F	10	11.601	11.663	0.859
G	12	7.901	8.399	0.880

H	16	11.816	11.837	0.907
I	20	12.033	12.033	0.924
J	24	16.297	16.025	0.936
K	30	8.738	8.909	0.948
L	40	11.989	11.991	0.961

```
c(ICC = tau2_hat / (tau2_hat + sigma2_hat))
```

```
ICC
0.3790268
```

The syntax `(1 | group)` requests a random intercept for each group. The shared intercept is returned by `fixef()`, while `ranef()` returns estimated group deviations. These are different quantities; adding them gives the fitted group mean in this intercept-only example. See the [lmer\(\) reference](#) for model syntax.

```
j <- seq_along(n_group)
plot(raw_mean, j, pch = 1, xlim = range(c(raw_mean, shrunk_mean)),
     ylim = c(0.5, length(j) + 2), yaxt = "n", xlab = "Group mean",
     ylab = "Group and sample size")
axis(2, at = j, labels = paste0(levels(group), " (", n_group, ")"),
     las = 1, cex.axis = 0.75)
segments(raw_mean, j, shrunk_mean, j, col = "gray60", lwd = 1.5)
points(shrunk_mean, j, pch = 16, col = "steelblue4")
segments(mu_hat, 0.5, mu_hat, length(j) + 0.35,
     lty = 2, col = "darkorange3")
legend("top", legend = c("Separate means", "Partial pooling"),
     pch = c(1, 16), col = c("black", "steelblue4"),
     horiz = TRUE, cex = 0.85, bty = "n")
```

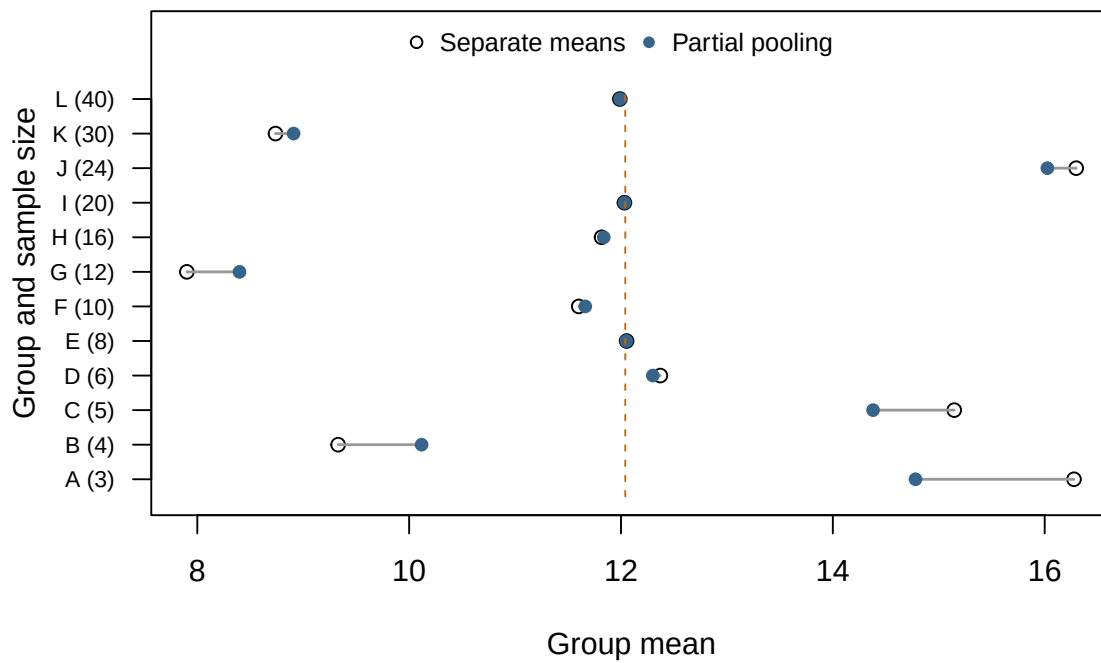


Figure G.1: Unpooled sample means and partially pooled group means in simulated data. The dashed line is the fitted common mean. The amount of shrinkage reflects group sample size and the estimated variance components.

What shrinkage does not establish. It is not evidence that a small group's true mean equals the common mean. It reflects uncertainty and the assumed distribution of group effects. Point predictions of group deviations are also not confidence intervals or a justified ranking of all groups.

G.3 Optional application: repeated measurements and random slopes

i Optional application: reaction times during sleep restriction

The public `sleepstudy` data bundled with `lme4` contain repeated reaction-time measurements for 18 subjects. `Reaction` is the average reaction time for a subject on a recorded day, measured in milliseconds. Days 0–1 are adaptation/training and day 2 is baseline. We use baseline and later measurements, with `day = Days - 2`. The independent grouping unit is the subject. This linear-trend example is not an estimate of a randomized treatment contrast. See the [dataset documentation](#) for the study and variables.

A random intercept shifts a subject's line vertically. A **random slope** also allows the subject's rate of change to vary:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{0j} + b_{1j} t_{ij} + \varepsilon_{ij}, \quad \begin{pmatrix} b_{0j} \\ b_{1j} \end{pmatrix} \sim N_2(0, G).$$

Here β_1 describes the average linear trend across the modeled subject population, while $\beta_1 + b_{1j}$ is the subject-specific slope. Allowing covariance between b_{0j} and b_{1j} recognizes that baseline level and rate of change may be related.

A covariance model as well as a mean model. For two measurements on the same subject, at distinct occasions t and s , the covariance contributed by the random effects is $g_{00} + (t + s)g_{01} + tsg_{11}$. Thus correlations can depend on the observation times. One constant random-intercept ICC no longer summarizes the whole dependence structure.

```
data("sleepstudy", package = "lme4")
study <- subset(sleepstudy, Days >= 2)
study$day <- study$Days - 2
fit_sleep <- lme4::lmer(Reaction ~ day + (day | Subject),
                        data = study, REML = TRUE)
lme4::fixef(fit_sleep)
```

```
(Intercept)      day
  267.96742    11.43543
```

```
lme4::VarCorr(fit_sleep)
```

Groups	Name	Std.Dev.	Corr
Subject	(Intercept)	30.9573	
	day	6.7659	0.178
Residual		25.5264	

```
lme4::isSingular(fit_sleep)
```

```
[1] FALSE
```

In (day | Subject), the random intercept is included along with the random slope, and their covariance is estimated. This is a model for continuous Gaussian responses. Binary or count outcomes with random effects lead to **generalized linear mixed models**, which add a link and response family; they require further study beyond this appendix.

```
subjects <- levels(study$Subject)[1:4]
plot_grid <- expand.grid(day = seq(0, 7, length.out = 40),
                        Subject = subjects)
plot_grid$conditional <- predict(fit_sleep, newdata = plot_grid)
plot_grid$population <- predict(fit_sleep, newdata = plot_grid,
                              re.form = NA)
op <- par(mfrow = c(2, 2), mar = c(3.5, 4, 2, 0.5),
         mgp = c(2.4, 0.7, 0))
for (subject in subjects) {
  d <- study[study$Subject == subject, ]
  g <- plot_grid[plot_grid$Subject == subject, ]
  plot(d$day, d$Reaction, pch = 16, ylim = range(study$Reaction),
       xlab = "Days since baseline", ylab = "Reaction time (ms)",
       main = paste("Subject", subject))
  lines(g$day, g$conditional, col = "steelblue4", lwd = 2)
  lines(g$day, g$population, lty = 2, lwd = 2)
}
par(op)
```

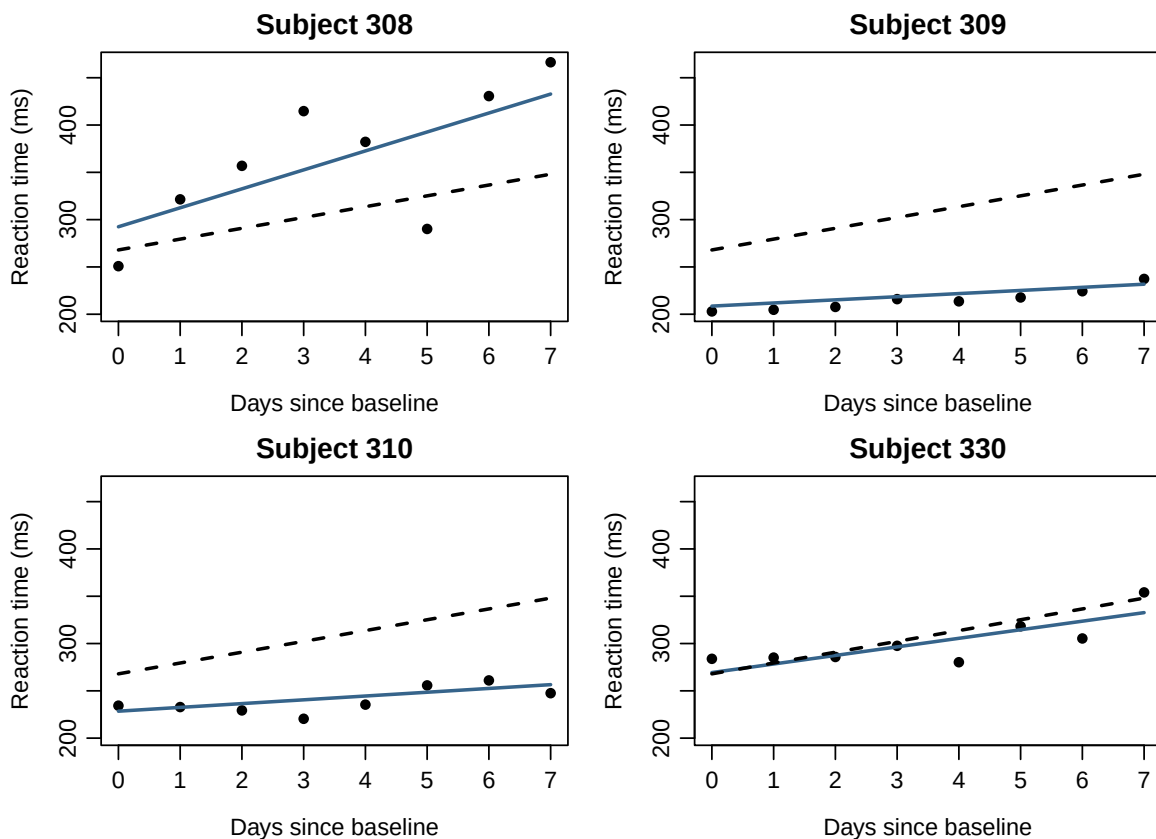


Figure G.2: Four subjects from the sleepstudy data. Blue lines include each subject's fitted random effects; dashed lines use only the common fixed effects. A common linear trend does not require identical subject trajectories.

Prediction target. For a previously observed subject, predictions can use that subject's estimated random effects. For a new subject with no response data yet, those deviations are unknown; setting them to zero gives the population mean in this Gaussian identity-link model. A prediction interval for the new subject must still include between-subject and residual variation, together with parameter uncertainty. A fixed-effect line alone is not such an interval. With a nonlinear GLMM link, setting random effects to zero generally does not equal averaging the response over their distribution.

G.4 Inference, model checks, and further reading

ML and REML. Maximum likelihood estimates regression and covariance parameters through the marginal model. Restricted maximum likelihood (REML) accounts for the estimation

of fixed effects when fitting covariance parameters. When comparing different fixed-effect structures by likelihood, use ML fits on the same observations and keep the specified random-effects structure comparable; the REML criterion is not directly comparable across different fixed-effect designs.

The following comparison asks whether the average linear trend is zero while allowing the same subject-specific random intercept and slope structure in both models. Its likelihood-ratio reference is asymptotic, not Chapter 4's exact F distribution.

```
fit_sleep_ml <- update(fit_sleep, REML = FALSE)
fit_no_trend_ml <- lme4::lmer(Reaction ~ 1 + (day | Subject),
                             data = study, REML = FALSE)
anova(fit_no_trend_ml, fit_sleep_ml, refit = FALSE)
```

Data: study

Models:

fit_no_trend_ml: Reaction ~ 1 + (day | Subject)

fit_sleep_ml: Reaction ~ day + (day | Subject)

	npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
fit_no_trend_ml	5	1444.4	1459.3	-717.21	1434.4			
fit_sleep_ml	6	1425.2	1443.0	-706.58	1413.2	21.266	1	3.997e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

! Variance components require special care

Testing whether a random-effect variance is zero places the null parameter on a boundary. The usual chi-square rule based only on the parameter-count difference need not apply. Small numbers of independent groups also limit large-sample approximations. Methods such as parametric bootstrap or appropriate small-sample corrections need additional justification. Do not use the ordinary OLS F test simply because the model contains a linear predictor.

Check what the design can support. Examine conditional residuals versus fitted values, residual patterns across time and groups, convergence messages, and near-singularity. A singular fit means the fitted random-effects covariance is on or near a lower-dimensional boundary, such as an estimated variance near zero or a redundant intercept/slope combination. It is not synonymous with an optimizer failing to converge. Review the design and model complexity before changing optimizer settings. See `isSingular()`.

```

op <- par(mfrow = c(1, 2), mar = c(4, 4, 2, 0.5),
          mgp = c(2.4, 0.7, 0))
plot(fitted(fit_sleep), resid(fit_sleep), pch = 16,
     col = adjustcolor("steelblue4", 0.5),
     xlab = "Conditional fitted value", ylab = "Conditional residual")
abline(h = 0, lty = 2)
qqnorm(resid(fit_sleep), pch = 16, cex = 0.65,
       main = "Residual Q-Q plot")
qqline(resid(fit_sleep), col = "darkorange3")
par(op)

```

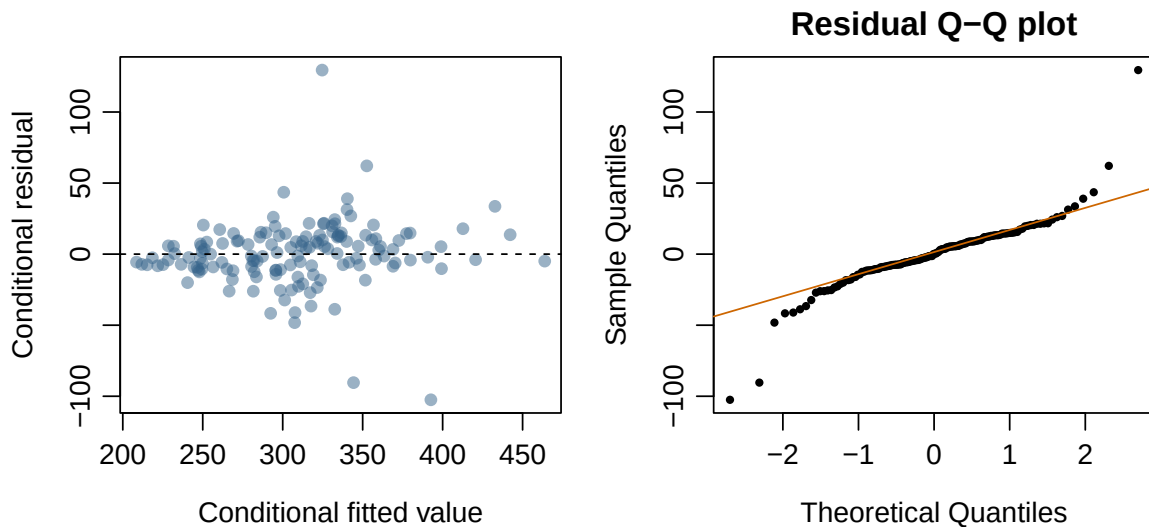


Figure G.3: Basic conditional-residual checks for the fitted sleepstudy model. These plots can reveal mean or residual-distribution problems, but they do not prove that the full dependence structure is correct.

Plots of estimated random effects can be informative, but they display shrunken estimates with unequal uncertainty, not directly observed independent normal draws. With few groups, distributional checks are particularly limited. For prediction to new subjects, evaluate with subject-level holdouts; random row-wise splitting can let the same subject appear in both training and test data.

Self-study questions.

1. Derive the covariance and ICC for the random-intercept model. Explain why increasing observations within a group is different from adding new independent groups.

2. In the simulation, verify that $\text{mu_hat} + \text{w_hat} * (\text{raw_mean} - \text{mu_hat})$ matches the mixed-model group predictions. Explain the role of n_j .
3. Write down the fixed and random portions of the `sleepstudy` model. Explain what changes if `(day | Subject)` is replaced with `(1 | Subject)`.
4. Explain why a new subject's prediction interval is wider than an interval that conditions on known subject effects, and why a row-wise validation split may answer the wrong question.
5. State the null hypothesis in the ML comparison and explain why testing a random-effect variance of zero is a different inferential problem.

Suggested reading. The authors' [Fitting Linear Mixed-Effects Models Using lme4](#) develops the model and computation. The package's [inference guide](#) discusses likelihood, bootstrap, and small-sample approaches. Continue to generalized mixed models only after distinguishing conditional and marginal prediction targets in the Gaussian case.